Safe prompts, safe fields:
the order-1 Wasserstein conformal region for linear flow matching

Distilled from the proof-of-concept note of this folder
companion to *Pipeline-Agnostic Meta-Learning: an Optimal Transport Approach*

Abstract

A prompt-conditioned flow-matching model turns each prompt into a time-dependent velocity field. We restrict the field of a new prompt to the order-1 Wasserstein conformal region built from the fields of approved prompts. For velocity fields that are linear in their parameters, a class that contains the first-order linearisations of conditional networks, we define every object involved. The region contains the field of a new approved prompt with probability at least $1-\beta$. Projecting a field onto the region is a second-order cone program; in the region's own metric, its solution is a mixture of the prompt's field and the approved fields, and for a model affine in the prompt it is the field of a mixed prompt. When fields are compared through their noise-to-output maps, every field of the region generates outputs within order-1 Wasserstein distance $(M+1)\varepsilon$ of the approved outputs, with no Grönwall factor. This map metric is a Mahalanobis distance for one-step samplers, and multi-step samplers pay their straightness defects. We summarise the toy evidence of the proof of concept, then test the method on a two-dimensional prompt-conditioned toy. Coverage holds and time behaves as predicted, but on a diverse family of approved prompts the region as written is too wide to exclude small unsafe modes, and its certificate is vacuous. Scoring a request against the nearest approved field repairs detection; the region becomes a nearest-neighbour conformal detector, which is not convex. A truncated map distance keeps the score immune to the rare trajectories that a sampler sends far away, and its certificate, which applies Markov's inequality to a bounded variable, becomes rigorous once each deployed field is verified on fresh noise. Under two white-box attacks on the prompt embedding, localised field-space regions keep the unsafe mass at a few percent, whereas Euclidean prompt-space regions admit prompts with $P(A)$ from 0.496 to 1.000; a whitened prompt-space region also resists, but only because the approved embeddings of the toy are degenerate. Finally, every mechanism is applied to a pretrained text-to-image model, Sana 600M, with a safe set of winter scenes only. The model fills a season left unspecified with a snow-free scene, so such requests are unsafe too. Every region keeps its coverage, and what it excludes depends on its space more than on its weights: the source slice of the fields excludes every summer request and almost no safe unfamiliar one, because the season changes the mean output; field regions in an image-feature space exclude about half of the unsafe requests but, when they enforce, keep the requested scene; a whitened prompt region refuses every safe rewording; nearest-anchor weights repair the field regions only with many anchors, and then refuse safe scenes that no anchor shows; and the plug-in certificates are vacuous. Coverage depends on the decision alone, so a prompt-side decision can be paired with a field-side repair: deciding with the source slice, backing off towards the approved prompt nearest in the field distance, and stopping once the deployed outputs are as close to that prompt's as approved outputs are to their nearest anchor's, leaves 0.06 of the summer images and 0.02 of the season-unspecified ones unsafe, against 0.45 and 0.44 for the source slice alone, and keeps the requested scene.

1 Problem and summary of the proof of concept

A prompt-conditioned flow-matching model maps a prompt $\mathbf{e}$ (a text embedding, a class, a goal) to a velocity field $v : \mathbb{R}^n \times [0,1] \to \mathbb{R}^n$; integrating the field from Gaussian noise generates the outputs of the prompt. The fields of approved prompts $\mathbf{e}_1, \dots, \mathbf{e}_M$ generate safe outputs. For a new prompt, which may be unsafe or adversarial, we want to (i) decide whether its field is compatible with the safe fields, (ii) replace it by a compatible field otherwise, and (iii) know what compatibility implies for the generated outputs. On a space of weights, the manuscript answers (i) and (ii) with the order-1 score $s_{d,1}(\boldsymbol{\theta}; \mathcal{D}) = \frac{1}{M(M+1)} \sum_i d(\boldsymbol{\theta}, \boldsymbol{\theta}_i)$, the order-1 Wasserstein distance between the empirical law of the historical weights before and after the candidate is added, and with a split-conformal radius ε such that $\bar{\Theta} = \{\boldsymbol{\theta} : s_{d,1}(\boldsymbol{\theta}; \mathcal{D}) \le \varepsilon\}$ contains the weights of a new task with probability at least $1-\beta$. This note builds the region on a space of fields and answers (iii).

What the proof of concept established. The note `flow_matching_safe_set_POC.tex` (the POC below) contains the following facts, which are kept here in a reorganised form.

1. The closed form of $s_{d,1}$ only uses the triangle inequality, so it holds on any pseudometric space of fields, and split-conformal coverage holds for any score (Lemma 5.3, Proposition 5.4). The substantive choice is the distance between fields.
2. Where in space and time two fields are compared decides what the region sees. Costs evaluated along the safe fields' trajectories (*off-policy*: an L^2 distance with a fixed Gram matrix, the forward path divergence) are convex in the parameters but nearly blind to a small unsafe mode: a mode of mass m costs at most $\log \frac{1}{1-m} \approx m$ in forward divergence. Costs evaluated along the candidate's own trajectories (*on-policy*: the reverse divergence) charge about $m \log(m/q)$, q the safe probability of the unsafe region, but are not convex (Appendix A).
3. For a velocity linear in its parameters, L^2 distances between fields are Mahalanobis distances between parameters, the region is the manuscript's Euclidean region after whitening, and constrained training is a second-order cone program.
4. Region membership certifies a total-variation bound to the safe mixture (from path divergences; valid but loose), a rare-event bound (from the reverse divergence; informative), and closeness of ODE trajectories through Grönwall's lemma (valid but vacuous: an a-priori bound on the factor is about 10^{15} in the toy; Remark 2.14).
5. On a two-dimensional toy, coverage is nominal; on-policy costs exclude contaminated fields that off-policy costs admit; the certificates hold on every split; and constrained few-shot fitting beats least squares but not tuned shrinkage (Section 8).
6. The prompt-conditioned results (POC Section 8) and the experiment plan (POC Section 10) are untested there; Section 9 tests them.

What this note adds. Section 2 collects the mathematical background: measures and L^2 spaces of fields, projections in Hilbert spaces, conditional expectations, flows, couplings, convexity and exchangeability. Section 3 defines linear flow matching, reflow (which straightens its fields by re-pairing noise and outputs), and the prompt-conditioned version. Sections 4–7 build the theory: distances between fields and the role of time (Section 4); the order-1 region of the safe fields and its coverage (Section 5); the constrained field of a new prompt, which mixes its field with the safe fields and, for a model affine in the prompt, is the field of a mixed prompt (Section 6); and a Wasserstein certificate on the outputs of every field of the region (Section 7). Section 8 summarises the evidence of the POC, Section 9 implements the method on a prompt-conditioned toy and follows one request through it step by step, with figures of its fields, of the regions and of the certificates, then criticises the method and improves it; Section 10 compares regions built on the prompts (with a Euclidean and a whitened distance) and on the fields, with static and adaptive weights, and tests reflow, pull-back metrics and denser anchors as remedies; Section 11 applies every mechanism to a pretrained text-to-image model whose safe prompts are winter scenes, analyses, method by method, what works and what does not, and then combines prompt-side decisions with field-side repairs, with the conformal guarantees that such combinations keep (Section 11.8); and Section 12 lists what remains to be tested. The version of 14 October 2026 was audited (`safe_prompt_fields_audit.tex`); this revision corrects it: a probe on which the 1-rectified sampler diverges, a stronger attack, a whitened prompt baseline and a verified deployment were added, and the claims they contradict were withdrawn.

2 Mathematical preliminaries

This section fixes the objects that the note manipulates and states every fact that a later proof uses. Textbook results are quoted with a reference; the others are proved, and later proofs cite the items they use.

Notation. $[\![M]\!] = \{1, \dots, M\}$, $\Delta^{M-1} = \{w \in \mathbb{R}^M : w_i \geq 0, \sum_i w_i = 1\}$ is the probability simplex, and $\mathbf{1}_b$ is the b-th unit vector. On $\mathbb{R}^{n'}$, $\langle \mathbf{a}, \mathbf{b} \rangle = \mathbf{a}^\top \mathbf{b}$ and $\|\cdot\|$ are the Euclidean inner product and norm. For a matrix A, $\|A\|_F = (\sum_{j,l} A_{jl}^2)^{1/2}$ is the Frobenius norm, $\|A\mathbf{a}\| \leq \|A\|_F \|\mathbf{a}\|$ by the Cauchy–Schwarz inequality applied to each row, and $A \otimes A'$ is the Kronecker product. $H \succeq 0$ ($H \succ 0$) means that H is symmetric positive semidefinite (definite). For $B \subseteq \mathbb{R}^{n'}$ and $r > 0$, $\operatorname{dist}(\mathbf{y}, B) = \inf_{\mathbf{y}' \in B} \|\mathbf{y} - \mathbf{y}'\|$ and $B^{+r} = \{\mathbf{y} : \operatorname{dist}(\mathbf{y}, B) < r\}$. $\arg\min$ is the set of minimisers, identified with its element when it has exactly one. $\mathcal{U}[a, b]$, $\mathcal{N}(\mathbf{c}, \Sigma)$ and $\delta_{\mathbf{x}}$ are the uniform, Gaussian and Dirac laws, $\mathbb{1}_E$ is the indicator of E, and $\mu_0 = \mathcal{N}(0, I_n)$ throughout.

Measures and integrals.

Definition 2.1 (Borel maps and laws)**.** *The spaces below are Borel subsets of Euclidean spaces, such as $\mathbb{R}^{n'}$, $\mathbb{R}^n \times [0, 1]$ or $\mathbb{R}^{n \times D}$, with their Borel σ-algebras. A map between two of them is* Borel *if the preimage of every Borel set is Borel, and $\mathcal{P}(\mathcal{X})$ is the set of Borel probability measures on $\mathcal{X}$. Random variables are measurable maps from a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ into these spaces, and the* law *of X is $\mathbb{P} \circ X^{-1}$. The* push-forward *of $\mu \in \mathcal{P}(\mathcal{X})$ by a Borel map T is $T_{\#}\mu = \mu \circ T^{-1}$, the law of $T(X)$ when $X \sim \mu$. The* product *$\mu \otimes \nu$ is the law of (X, X') for independent $X \sim \mu$ and $X' \sim \nu$. For $w \in \Delta^{M-1}$, the* mixture *$\sum_i w_i \mu_i$ is the measure $B \mapsto \sum_i w_i \mu_i(B)$, which is the law of X_I when $X_i \sim \mu_i$ and $I \sim w$ is independent of*

$(X_1, \dots, X_M)$. *A law* μ *on* $\mathbb{R}^{n'}$ *has a* finite p-th moment *if* $\int \|\mathbf{x}\|^p d\mu < \infty$. *A property holds* μ-almost everywhere *(a.e.), or* almost surely *(a.s.) under* $\mathbb{P}$*, if it holds outside a set of measure zero.*

Remark 2.2 (Standard tools). (i) Continuous maps are Borel, and so are sums, products and compositions of Borel maps. (ii) *Change of variables*: $\int f d(T_{\#}\mu) = \int f \circ T d\mu$ for every Borel $f \geq 0$ and every f integrable under $T_{\#}\mu$; in particular $\mathbb{E}\, f(X) = \int f d\mu$ for $X \sim \mu$ (Williams 1991, Sec. 6.12). Also $(S \circ T)_{\#}\mu = S_{\#}(T_{\#}\mu)$. (iii) *Fubini–Tonelli*: the integral against $\mu \otimes \nu$ of a nonnegative or integrable function is the iterated integral, in either order (Brezis 2011, Thms. 4.4–4.5). For instance $\int f d(\mu \otimes \delta_t) = \int f(\mathbf{x}, t)\mu(d\mathbf{x})$, and $\mathbb{E}\, g(I, X) = \sum_i w_i\, \mathbb{E}\, g(i, X)$ when $I \sim w$ and X are independent. (iv) *Dominated convergence* (Brezis 2011, Thm. 4.2); the *Cauchy–Schwarz inequality* $\mathbb{E}\,|\langle X, X'\rangle| \leq (\mathbb{E}\,\|X\|^2)^{1/2}(\mathbb{E}\,\|X'\|^2)^{1/2}$; and *Markov's inequality* $\mathbb{P}(Z \geq r) \leq \mathbb{E}\, Z/r$ for $Z \geq 0$ and $r > 0$ (Williams 1991, Secs. 6.4 and 6.8). (v) For every $B \subseteq \mathbb{R}^{n'}$, $|\operatorname{dist}(\mathbf{y}, B) - \operatorname{dist}(\mathbf{y}', B)| \leq \|\mathbf{y} - \mathbf{y}'\|$, so B^{+r} is open, hence Borel.

Pseudometrics, Hilbert spaces and projections.

Definition 2.3 (Pseudometrics, seminorms, inner products)**.** *An* extended pseudometric *on a set* V *is a map* $d : V \times V \to [0, +\infty]$ *with* $d(x, x) = 0$*,* $d(x, y) = d(y, x)$ *and* $d(x, z) \leq d(x, y) + d(y, z)$*; it is a* pseudometric *if it is finite, and a* metric *if moreover* $d(x, y) = 0$ *only when* $x = y$*. A map* $f : V \to \mathbb{R}$ *is* L-Lipschitz *if* $|f(x) - f(y)| \leq L d(x, y)$*; for a pseudometric* d *and* $y \in V$*,* $d(\cdot, y)$ *is* 1*-Lipschitz by the triangle inequality. A* seminorm *on a real vector space* V *is a map* $\|\cdot\| : V \to [0, \infty)$ *with* $\|cx\| = |c|\|x\|$ *for* $c \in \mathbb{R}$ *and* $\|x + y\| \leq \|x\| + \|y\|$*; its* kernel $\{x : \|x\| = 0\}$ *is a subspace, and it is a* norm *if its kernel is* $\{0\}$*. An* inner product *is a symmetric bilinear form with* $\langle x, x\rangle > 0$ *for* $x \neq 0$*; with the norm* $\|x\| = \langle x, x\rangle^{1/2}$*,* V *is a* Hilbert space *if every Cauchy sequence converges.*

Lemma 2.4 (Elementary properties)**.** *(i) A seminorm induces the pseudometric* $(x, y) \mapsto \|x - y\|$*, which is a metric if and only if the kernel is* $\{0\}$*. (ii) If* d *is an extended pseudometric on* V *and* $g : U \to V$*, then* $(u, u') \mapsto d(g(u), g(u'))$ *is an extended pseudometric on* U*. (iii) If* d *is a metric and* $r > 0$*, then* $\min(d, r)$ *is a metric with values in* $[0, r]$*. (iv) If* (d_ω) *are pseudometrics on* V*, indexed by the points* ω *of a probability space, with* $\omega \mapsto d_\omega(x, y)$ *measurable for all* x, y*, then* $(x, y) \mapsto \int d_\omega(x, y)\mathbb{P}(d\omega)$ *is an extended pseudometric. (v) In an inner-product space,* $|\langle x, y\rangle| \leq \|x\|\|y\|$ *(Cauchy–Schwarz), and* $\|x+y\|^2 = \|x\|^2+\|y\|^2$ *when* $\langle x, y\rangle = 0$ *(Pythagoras). A finite-dimensional inner-product space is a Hilbert space, and a finite-dimensional subspace of a Hilbert space is closed.*

Proof: (i)–(ii) The axioms transfer directly, and $\|x-y\| = 0$ means that $x-y$ is in the kernel. (iii) $\min(\cdot, r)$ is nondecreasing and $\min(a + b, r) \leq \min(a, r) + \min(b, r)$ for $a, b \geq 0$. (iv) The three axioms hold for every ω and are preserved by integration. (v) Both follow from bilinearity (Brezis 2011, Sec. 5.1); finite-dimensional normed spaces are complete, and a complete subset of a metric space is closed. □

Lemma 2.5 (Mahalanobis seminorms and Gram matrices)**.** *For* $H \in \mathbb{R}^{D\times D}$ *with* $H \succeq 0$*, let* $\|\mathbf{a}\|_H = (\mathbf{a}^\top H\mathbf{a})^{1/2}$ *and* $\langle \mathbf{a}, \mathbf{b}\rangle_H = \mathbf{a}^\top H\mathbf{b}$*. (i) If* $H = CC^\top$ *(for instance* $C = H^{1/2}$*, or the Cholesky factor when* $H \succ 0$*), then* $\|\mathbf{a}\|_H = \|C^\top\mathbf{a}\|$*. Hence* $\|\cdot\|_H$ *is a seminorm with kernel* $\ker H$*; it is a norm, and* $(\mathbb{R}^D, \langle\cdot,\cdot\rangle_H)$ *is a Hilbert space, if and only if* $H \succ 0$*. (ii) If* P *is the orthogonal projector onto the range of* H*, then* $\|\mathbf{a}\|_H = \|P\mathbf{a}\|_H$ *for all* $\mathbf{a}$*, and* $\|\cdot\|_H$ *is a norm on that range. (iii) Let* $\mathsf{L} : \mathbb{R}^D \to \mathcal{H}$ *be linear,* $\mathcal{H}$ *a Hilbert space. Its* Gram matrix $G = (\langle \mathsf{L}\mathbf{1}_j, \mathsf{L}\mathbf{1}_l\rangle)_{j,l}$ *satisfies* $G \succeq 0$*,* $\langle \mathsf{L}\mathbf{a}, \mathsf{L}\mathbf{b}\rangle = \mathbf{a}^\top G\mathbf{b}$*,* $\|\mathsf{L}\mathbf{a}\| = \|\mathbf{a}\|_G$ *and* $\ker G = \ker \mathsf{L}$*: a linear map pulls the geometry of* $\mathcal{H}$ *back to a Mahalanobis seminorm on* $\mathbb{R}^D$*.*

Proof: (i) $\mathbf{a}^\top CC^\top\mathbf{a} = \|C^\top\mathbf{a}\|^2$, so $\|\cdot\|_H$ is the Euclidean norm composed with a linear map, a seminorm. With $C = H^{1/2}$, $\|\mathbf{a}\|_H = 0$ if and only if $H^{1/2}\mathbf{a} = 0$, that is, $H\mathbf{a} = 0$. The form $\langle\cdot,\cdot\rangle_H$ is symmetric and bilinear, and it is an inner product exactly when this kernel is trivial; completeness is Lemma 2.4(v). (ii) H is symmetric with range $(\ker H)^\perp$, so $H = PHP$ and $\mathbf{a}^\top H\mathbf{a} = (P\mathbf{a})^\top H(P\mathbf{a})$; and $\ker H \cap \operatorname{range}(H) = \{0\}$. (iii) By bilinearity, $\langle \mathsf{L}\mathbf{a}, \mathsf{L}\mathbf{b}\rangle = \sum_{j,l} a_j b_l \langle \mathsf{L}\mathbf{1}_j, \mathsf{L}\mathbf{1}_l\rangle = \mathbf{a}^\top G\mathbf{b}$. Hence $\mathbf{a}^\top G\mathbf{a} = \|\mathsf{L}\mathbf{a}\|^2 \geq 0$, and by (i) $\mathbf{a}^\top G\mathbf{a} = 0$ if and only if $G\mathbf{a} = 0$. □

Theorem 2.6 (Projection onto a closed convex set)**.** *Let* $\mathcal{C}$ *be a non-empty closed convex subset of a Hilbert space* $\mathcal{H}$*. Every* $x \in \mathcal{H}$ *has a unique nearest point* $P_{\mathcal{C}}x$ *in* $\mathcal{C}$*, characterised by* $P_{\mathcal{C}}x \in \mathcal{C}$ *and* $\langle x - P_{\mathcal{C}}x, y - P_{\mathcal{C}}x\rangle \leq 0$ *for all* $y \in \mathcal{C}$ *(Brezis 2011, Thm. 5.2). Consequently: (i)* $\|x-y\|^2 \geq \|x-P_{\mathcal{C}}x\|^2+\|P_{\mathcal{C}}x-y\|^2$ *for all* $y \in \mathcal{C}$*; (ii)* $\|P_{\mathcal{C}}x-P_{\mathcal{C}}x'\|^2 \leq \langle x-x', P_{\mathcal{C}}x-P_{\mathcal{C}}x'\rangle$*, hence* $\|P_{\mathcal{C}}x - P_{\mathcal{C}}x'\| \leq \|x - x'\|$ *(Brezis 2011, Prop. 5.3); (iii) if* $\mathcal{C}$ *is a closed subspace,* $P_{\mathcal{C}}$ *is linear and characterised by* $P_{\mathcal{C}}x \in \mathcal{C}$ *and* $x - P_{\mathcal{C}}x \perp \mathcal{C}$ *(the* orthogonal projection*) (Brezis 2011, Cor. 5.4), and* $\|x\|^2 = \|P_{\mathcal{C}}x\|^2 + \|x - P_{\mathcal{C}}x\|^2$*.*

Proof of (i)–(iii): (i) $\|x - y\|^2 = \|x - P_{\mathcal{C}}x\|^2 + \|P_{\mathcal{C}}x - y\|^2 - 2\langle x - P_{\mathcal{C}}x, y - P_{\mathcal{C}}x\rangle$. (ii) Add the characterisation at x with $y = P_{\mathcal{C}}x'$ and at x' with $y = P_{\mathcal{C}}x$, then apply Cauchy–Schwarz. (iii) Pythagoras, since $P_{\mathcal{C}}x \perp x - P_{\mathcal{C}}x$. □

Definition 2.7 (L^p spaces, linear growth)**.** *Let* $\mu \in \mathcal{P}(\mathcal{X})$ *and* $p \in \{1, 2\}$*. For a Borel map* $F : \mathcal{X} \to \mathbb{R}^{n'}$*,* $\|F\|_{L^p(\mu)} = (\int \|F\|^p d\mu)^{1/p} \in [0, +\infty]$*, and* $L^p(\mu; \mathbb{R}^{n'})$ *is the set of Borel* F *with* $\|F\|_{L^p(\mu)} < \infty$*, two of which are identified when they agree* μ*-a.e.; on* $L^2(\mu; \mathbb{R}^{n'})$*,* $\langle F, F'\rangle_{L^2(\mu)} = \int\langle F, F'\rangle d\mu$*. Matrix-valued maps are treated alike with* $\|\cdot\|_F$*, and the same definitions apply to random variables on* $(\Omega, \mathcal{F}, \mathbb{P})$*. A Borel map* F *on* $\mathbb{R}^n$ *or on* $\mathbb{R}^n \times [0, 1]$ *has* linear growth *if* $\|F(\mathbf{x})\| \leq C(1 + \|\mathbf{x}\|)$*, respectively* $\|F(\mathbf{x}, t)\| \leq C(1 + \|\mathbf{x}\|)$*, for some* C *and all arguments.*

Proposition 2.8 (Facts on L^p spaces)**.** *(i)* $\|F + F'\|_{L^p(\mu)} \leq \|F\|_{L^p(\mu)} + \|F'\|_{L^p(\mu)}$ *for all Borel* F, F' *(Minkowski). Hence* $(F, F') \mapsto \|F - F'\|_{L^p(\mu)}$ *is an extended pseudometric on Borel maps, which vanishes exactly when* $F = F'$ μ*-a.e., and* $\|\cdot\|_{L^p(\mu)}$ *is a norm on* $L^p(\mu; \mathbb{R}^{n'})$*. (ii)* $\|F\|_{L^1(\mu)} \leq \|F\|_{L^2(\mu)}$*. (iii)* $L^2(\mu; \mathbb{R}^{n'})$ *is a Hilbert space. (iv)*

$\|F \circ T\|_{L^p(\mu)} = \|F\|_{L^p(T_\#\mu)}$ *for every Borel* T. *(v) If* F *has linear growth with constant* C *and* $m_2 = \int \|\mathbf{x}\|^2 d\mu < \infty$, *where* $\mathbf{x}$ *is the* $\mathbb{R}^n$*-coordinate, then* $\|F\|^2_{L^2(\mu)} \le 2C^2(1+m_2)$. *For* μ_0, $m_2 = n$.

Proof: (i) Apply Minkowski's inequality (Brezis 2011, Thm. 4.7) to the nonnegative functions $\|F\|$ and $\|F'\|$, whose sum dominates $\|F+F'\|$; the inequality is trivial when its right side is infinite. (ii) Cauchy–Schwarz with the constant function 1, since μ is a probability. (iii) A Cauchy sequence of $L^2(\mu;\mathbb{R}^{n'})$ has Cauchy coordinates, which converge in $L^2(\mu;\mathbb{R})$ by the Fischer–Riesz theorem (Brezis 2011, Thm. 4.8). (iv) Remark 2.2(ii) with $f = \|F\|^p$. (v) $(1+a)^2 \le 2(1+a^2)$. □

Conditional expectation.

Definition 2.9 (Conditional expectation)**.** *Let* $U \in L^2(\mathbb{P};\mathbb{R}^{n'})$ *and let* Z *be a random variable with values in* $\mathcal{X}$. *A* version *of* $\mathbb{E}[U \mid Z]$ *is a Borel map* $g : \mathcal{X} \to \mathbb{R}^{n'}$ *such that* $g(Z) \in L^2(\mathbb{P};\mathbb{R}^{n'})$ *and* $\mathbb{E}\langle U - g(Z), h(Z)\rangle = 0$ *for every Borel* h *with* $h(Z) \in L^2(\mathbb{P};\mathbb{R}^{n'})$*; we then write* $\mathbb{E}[U \mid Z] = g(Z)$. *For square-integrable* U *this is the usual conditional expectation given* Z *(Williams 1991, Ch. 9), written as a function of* Z.

Lemma 2.10 (Existence, uniqueness and Bayes' formula)**.** *(i) A version exists and is unique* $\mathrm{Law}(Z)$*-a.e. The variable* $g(Z)$ *is the orthogonal projection of* U *onto the closed subspace* $\mathcal{H}_Z = \{h(Z) : h \text{ Borel}, h(Z) \in L^2(\mathbb{P};\mathbb{R}^{n'})\}$ *of* $L^2(\mathbb{P};\mathbb{R}^{n'})$. *Hence* $\mathbb{E}\,\|U\|^2 = \mathbb{E}\,\|g(Z)\|^2 + \mathbb{E}\,\|U - g(Z)\|^2$, *and* $\mathbb{E}[\Psi(Z)^\top U] = \mathbb{E}[\Psi(Z)^\top g(Z)]$ *for every Borel* Ψ *with values in* $\mathbb{R}^{n'\times D}$ *and* $\Psi(Z) \in L^2(\mathbb{P};\mathbb{R}^{n'\times D})$. *(ii) (Bayes' formula.) Let* Y *be a random variable with law* p *on* $\mathcal{Y}$, ℓ *a* σ*-finite Borel measure on* $\mathcal{X}$ *(in our use, the Lebesgue measure), and* $\varphi : \mathcal{X}\times\mathcal{Y} \to [0,\infty)$ *Borel with* $\mathbb{P}(Y \in A, Z \in B) = \int_A \int_B \varphi(z \mid y)\ell(dz)p(dy)$ *for all Borel* A, B *(a* conditional density *of* Z *given* Y*). If* $U = F(Y,Z)$ *for a Borel* F, *then*

$$g(z) = \frac{\int F(y,z)\varphi(z \mid y)p(dy)}{\int \varphi(z \mid y)p(dy)},$$

set to 0 *where the ratio is not defined, is a version of* $\mathbb{E}[U \mid Z]$.

Proof: (i) By Proposition 2.8(iv), $h \mapsto h(Z)$ is a linear isometry from $L^2(\mathrm{Law}(Z);\mathbb{R}^{n'})$ onto $\mathcal{H}_Z$, so $\mathcal{H}_Z$ is complete (Proposition 2.8(iii)), hence closed. A version is exactly a g with $g(Z) \in \mathcal{H}_Z$ and $U - g(Z) \perp \mathcal{H}_Z$, so Theorem 2.6(iii) gives the existence and uniqueness of $g(Z)$, and the isometry gives the uniqueness of g up to a $\mathrm{Law}(Z)$-null set. The identity is Pythagoras, and the last claim is orthogonality applied to each column of $\Psi(Z)$. (ii) The law of (Y,Z) and the measure with density $\varphi(z \mid y)$ with respect to $p \otimes \ell$ agree on products $A \times B$, hence coincide; by Tonelli's theorem, Z has the density $\bar\varphi(z) = \int \varphi(z \mid y)p(dy)$ with respect to ℓ, so $0 < \bar\varphi(Z) < \infty$ a.s., and $\int \|F(y,z)\|^2\varphi(z \mid y)p(dy) < \infty$ for $\mathrm{Law}(Z)$-a.e. z, because its integral against ℓ is $\mathbb{E}\,\|U\|^2$. For such z, the Cauchy–Schwarz inequality under the probability $\varphi(z \mid \cdot)p/\bar\varphi(z)$ gives $\|g(z)\|^2\bar\varphi(z) \le \int \|F(y,z)\|^2\varphi(z \mid y)p(dy)$, so $\mathbb{E}\,\|g(Z)\|^2 \le \mathbb{E}\,\|U\|^2$. Finally, for $h(Z) \in L^2(\mathbb{P};\mathbb{R}^{n'})$, $\langle U, h(Z)\rangle$ is integrable, and Fubini's theorem gives $\mathbb{E}\langle U, h(Z)\rangle = \int\langle \int F(y,z)\varphi(z \mid y)p(dy), h(z)\rangle\ell(dz) = \int\langle g(z), h(z)\rangle\bar\varphi(z)\ell(dz) = \mathbb{E}\langle g(Z), h(Z)\rangle$. □

Time-dependent vector fields.

Definition 2.11 (Fields and flows)**.** *A (time-dependent vector)* field *is a Borel map* $v : \mathbb{R}^n \times [0,1] \to \mathbb{R}^n$. *It is* L-Lipschitz *if* $\|v(\mathbf{x},t) - v(\mathbf{x}',t)\| \le L\|\mathbf{x} - \mathbf{x}'\|$ *for all* $\mathbf{x}, \mathbf{x}', t$, *and* locally Lipschitz *if its restriction to* $B \times [0,1]$ *is* L_B*-Lipschitz for every compact* $B \subset \mathbb{R}^n$. *Fields form a vector space under pointwise operations, and a* mixture *of fields is a convex combination* $\sum_a \gamma_a v_a$, γ *in a simplex. A* solution *of* $\dot X_t = v(X_t,t)$ *from* $\mathbf{x} \in \mathbb{R}^n$ *is a differentiable curve* $X : [0,1] \to \mathbb{R}^n$ *(with one-sided derivatives at the endpoints) such that* $X_0 = \mathbf{x}$ *and* $\dot X_t = v(X_t,t)$ *for every* $t \in [0,1]$. *The equation is* well posed *if every* $\mathbf{x}$ *has exactly one solution, and its* flow *is then the family of maps* $\mathbf{x} \mapsto X_t$, $t \in [0,1]$.

Remark 2.12 (Fields as vectors of a Hilbert space). For $\rho \in \mathcal{P}(\mathbb{R}^n \times [0,1])$, the fields with $\|v\|_{L^2(\rho)} < \infty$ form the Hilbert space $L^2(\rho;\mathbb{R}^n)$, with $\langle v, v'\rangle_{L^2(\rho)} = \int\langle v(\mathbf{x},t), v'(\mathbf{x},t)\rangle\rho(d\mathbf{x},dt)$ (Proposition 2.8). Its distance compares two fields only where ρ puts mass: fields that differ on a ρ-null set are at distance 0. Samplers, in contrast, evaluate a field at points that depend on the field itself (Definition 3.4), so two fields at $L^2(\rho)$-distance 0 may generate different outputs. Fields are therefore Borel maps in this note, to which L^2 norms are applied, rather than equivalence classes. Two choices of ρ recur: $\mu_0 \otimes \delta_t$, for which $\|v\|_{L^2(\mu_0\otimes\delta_t)} = \|v(\cdot,t)\|_{L^2(\mu_0)}$ compares the slices at time t on the noise (Remark 2.2(iii)), and the interpolation path of Definition 3.1, on which flow matching is trained.

Lemma 2.13 (Linear models in $L^2(\rho;\mathbb{R}^n)$)**.** *Let* $\rho \in \mathcal{P}(\mathbb{R}^n \times [0,1])$ *and* $\Phi : \mathbb{R}^n \times [0,1] \to \mathbb{R}^{n\times D}$ *Borel with* $\int \|\Phi\|_F^2 d\rho < \infty$, *and let* $\Phi_\rho : \mathbb{R}^D \to L^2(\rho;\mathbb{R}^n)$, $\boldsymbol\theta \mapsto \Phi\boldsymbol\theta$. *(i)* Φ_ρ *is well defined and linear, and its range* $\{\Phi\boldsymbol\theta : \boldsymbol\theta \in \mathbb{R}^D\}$ *is a closed subspace of* $L^2(\rho;\mathbb{R}^n)$. *(ii)* $G_\rho = \int \Phi^\top\Phi d\rho$ *(entrywise) is finite and is the Gram matrix of* Φ_ρ*:* $\langle \Phi\boldsymbol\theta, \Phi\boldsymbol\theta'\rangle_{L^2(\rho)} = \boldsymbol\theta^\top G_\rho \boldsymbol\theta'$ *and* $\|\Phi\boldsymbol\theta\|_{L^2(\rho)} = \|\boldsymbol\theta\|_{G_\rho}$. *(iii) For* $f \in L^2(\rho;\mathbb{R}^n)$, $\Phi_\rho^* f = \int \Phi^\top f d\rho \in \mathbb{R}^D$ *is finite and* $\langle \Phi\boldsymbol\theta, f\rangle_{L^2(\rho)} = \boldsymbol\theta^\top \Phi_\rho^* f$*:* Φ_ρ^* *is the adjoint of* Φ_ρ. *(iv) For* $f \in L^2(\rho;\mathbb{R}^n)$ *and* $\lambda > 0$, $\|\Phi\boldsymbol\theta - f\|^2_{L^2(\rho)} + \lambda\|\boldsymbol\theta\|^2 = \boldsymbol\theta^\top(G_\rho + \lambda I)\boldsymbol\theta - 2\boldsymbol\theta^\top\Phi_\rho^* f + \|f\|^2_{L^2(\rho)}$, *and its unique minimiser over* $\mathbb{R}^D$ *is* $(G_\rho + \lambda I)^{-1}\Phi_\rho^* f$.

Proof: Pointwise, $\|\Phi\boldsymbol\theta\| \le \|\Phi\|_F\|\boldsymbol\theta\|$, $|(\Phi^\top\Phi)_{jl}| \le \|\Phi\|_F^2$ and $\|\Phi^\top f\| \le \|\Phi\|_F\|f\|$, so all the integrals are finite, the last one by the Cauchy–Schwarz inequality. (i) A finite-dimensional subspace is closed (Lemma 2.4(v)). (ii)–(iii) Integrate

$\langle \Phi\boldsymbol{\theta}, \Phi\boldsymbol{\theta}' \rangle = \boldsymbol{\theta}^\top \Phi^\top \Phi \boldsymbol{\theta}'$ and $\langle \Phi\boldsymbol{\theta}, f \rangle = \boldsymbol{\theta}^\top \Phi^\top f$, and apply Lemma 2.5(iii). (iv) Expand the square with (ii)–(iii); since $G_\rho + \lambda I \succ 0$, Lemma 2.19(iv) gives the minimiser. □

Remark 2.14 (Well-posedness and Grönwall's lemma). If v is continuous, locally Lipschitz and of linear growth, the equation $\dot{X}_t = v(X_t, t)$ is well posed (Teschl 2012, Ch. 2). *Grönwall's lemma*: a continuous $f : [0,1] \to [0,\infty)$ with $f(t) \le a + \int_0^t (Lf(s) + b)ds$ for all t satisfies $f(t) \le (a + bt)e^{Lt}$ (differentiate $e^{-Lt}(a + \int_0^t (Lf + b)ds)$). If X and X' solve the equations of a continuous L-Lipschitz field v and of a continuous field v' with $\|v - v'\| \le b$ everywhere, then $f = \|X - X'\|$ satisfies this inequality with $a = \|X_0 - X_0'\|$, so $\|X_1 - X_1'\| \le (\|X_0 - X_0'\| + b)e^L$. Bounds on outputs derived from distances between fields along this route carry the factor e^L, with L the Lipschitz constant of the candidate field, and its discrete analogue $\prod_k (1 + h_k L_k)$ for an Euler sampler. In the toy of Section 8, an a-priori bound on the Lipschitz constant (the triangle inequality over the features) gives $L_k \approx 52$ and a discrete factor of about 10^{15}; the actual factor may be smaller, but no bound computed here makes it informative. The certificates of Section 7 compare sampler maps directly and avoid it.

Couplings and distances between laws.

Definition 2.15 (Couplings and Wasserstein distances). *A* coupling *of $\mu \in \mathcal{P}(\mathcal{X})$ and $\nu \in \mathcal{P}(\mathcal{Y})$ is a law on $\mathcal{X} \times \mathcal{Y}$ with marginals μ and ν, and $\Pi(\mu, \nu)$ is the set of couplings. For $\mu, \nu \in \mathcal{P}(\mathbb{R}^{n'})$, $W_1(\mu, \nu) = \inf_{\pi \in \Pi(\mu,\nu)} \int \|\mathbf{y} - \mathbf{y}'\| d\pi(\mathbf{y}, \mathbf{y}') \in [0, +\infty]$. For laws $\mu = \sum_x a_x \delta_x$ and $\nu = \sum_y b_y \delta_y$ with finite supports S_μ and S_ν in a pseudometric space (V, d), a coupling is a matrix $\pi \ge 0$ on $S_\mu \times S_\nu$ with row sums a_x and column sums b_y, and $W_{d,1}(\mu, \nu) = \min_{\pi \in \Pi(\mu,\nu)} \sum_{x,y} \pi(x, y) d(x, y)$, the minimum of a linear function over a non-empty compact polytope.*

On laws with finite first moments, W_1 is a metric, and the Kantorovich–Rubinstein duality $W_1(\mu, \nu) = \sup\{\int f d\mu - \int f d\nu : f \text{ is 1-Lipschitz}\}$ holds (Villani 2009, Chs. 5–6). The proofs below use only the two inequalities of the next lemma.

Lemma 2.16 (Shared noise and Lipschitz test functions). *(i) Let $X \sim \mu$, let $F, F_1, \dots, F_M$ be Borel maps into $\mathbb{R}^{n'}$, and let $I \sim w \in \Delta^{M-1}$ be independent of X. Then $Y = F(X)$ and $Y' = F_I(X)$ have the laws $F_\# \mu$ and $\sum_i w_i (F_i)_\# \mu$, and $W_1(F_\#\mu, \sum_i w_i (F_i)_\# \mu) \le \mathbb{E}\,\|Y - Y'\| = \sum_i w_i \|F - F_i\|_{L^1(\mu)}$. (ii) If $\mu = \sum_x a_x \delta_x$ and $\nu = \sum_y b_y \delta_y$ are finitely supported in (V, d) and $f : V \to \mathbb{R}$ is 1-Lipschitz, every coupling of μ and ν costs at least $\sum_x a_x f(x) - \sum_y b_y f(y)$.*

Proof: (i) By independence (Remark 2.2(iii)), $\mathbb{P}(Y' \in B) = \sum_i w_i \mathbb{P}(F_i(X) \in B)$ and $\mathbb{E}\,\|Y - Y'\| = \sum_i w_i\, \mathbb{E}\,\|F(X) - F_i(X)\|$; the law of (Y, Y') is a coupling of the two laws. (ii) $d(x, y) \ge f(x) - f(y)$; sum against the coupling, whose marginals are μ and ν. □

Definition 2.17 (Divergences). *For laws P, Q on the same space, $\mathrm{TV}(P, Q) = \sup_A |P(A) - Q(A)|$ over measurable A. If P is absolutely continuous with respect to Q ($P \ll Q$), with Radon–Nikodym density $\frac{dP}{dQ}$, then $\mathrm{KL}(P\|Q) = \int \log \frac{dP}{dQ} dP \in [0, +\infty]$; otherwise $\mathrm{KL}(P\|Q) = +\infty$. For $x, q \in [0, 1]$, $\mathrm{kl}(x\|q) = x \log \frac{x}{q} + (1 - x) \log \frac{1-x}{1-q}$ (with $0 \log 0 = 0$) is the divergence between Bernoulli laws.*

Appendix A uses the chain rule (Cover and Thomas 2006, Thm. 2.5.3) and the joint convexity (Cover and Thomas 2006, Thm. 2.7.2) of KL; the inequality $\mathrm{kl}(P(A)\|Q(A)) \le \mathrm{KL}(P\|Q)$ for every event A (Jensen's inequality for $x \log x$ under Q conditioned on A and on its complement); Pinsker's inequality $\mathrm{TV} \le (\mathrm{KL}/2)^{1/2}$ (Tsybakov 2009, Lemma 2.5) and the Bretagnolle–Huber inequality $\mathrm{TV} \le (1 - e^{-\mathrm{KL}})^{1/2}$ (Bretagnolle and Huber 1979, Canonne 2022); and, by direct computation, $\mathrm{KL}(\mathcal{N}(\mathbf{c}, \sigma^2 I)\|\mathcal{N}(\mathbf{c}', \sigma^2 I)) = \|\mathbf{c} - \mathbf{c}'\|^2/(2\sigma^2)$.

Remark 2.18 (Monte Carlo bounds and the evaluation metric). *Hoeffding's inequality* (Hoeffding 1963): if $Z_1, \dots, Z_L$ are i.i.d. with values in $[0, 1]$ and mean $\bar{z}$, then $\mathbb{P}(\bar{z} \ge \frac{1}{L}\sum_l Z_l + s) \le e^{-2Ls^2}$, so $\bar{z} \le \frac{1}{L}\sum_l Z_l + (\log(1/\delta)/(2L))^{1/2}$ with probability at least $1 - \delta$. The bound needs bounded variables, which is why Section 9 asks for a bounded distance. The *Clopper–Pearson* bound (Clopper and Pearson 1934) is the exact binomial upper confidence bound on a probability estimated from i.i.d. indicators. The *maximum mean discrepancy* with the kernel $k(\mathbf{y}, \mathbf{y}') = \sum_{s \in \{0.1, 0.3, 1\}} e^{-\|\mathbf{y} - \mathbf{y}'\|^2/(2s^2)}$ is $\mathrm{MMD}^2(\mu, \nu) = \mathbb{E}\,k(Y, Y') + \mathbb{E}\,k(Z, Z') - 2\,\mathbb{E}\,k(Y, Z)$ for independent $Y, Y' \sim \mu$ and $Z, Z' \sim \nu$. It is the squared distance between the mean embeddings of μ and ν in the reproducing-kernel Hilbert space of k (Gretton et al. 2012, Lemmas 4 and 6). It is additive in the kernel, and Gaussian kernels are characteristic on $\mathbb{R}^{n'}$ (Gretton et al. 2012, Sec. 2.2), so it vanishes only when $\mu = \nu$. Sections 8 and 9 report its unbiased estimate (Gretton et al. 2012, Lemma 6).

Convexity and cone programs.

Lemma 2.19 (Convexity). *(i) Seminorms and nonnegative combinations of convex functions are convex, and so is a convex function composed with an affine map. Affine maps preserve convex combinations, and the preimage of a convex set by an affine map is convex. (ii) The sublevel set $\{f \le c\}$ of a convex function f is convex, and closed if f is continuous. The convex hull $\mathrm{conv}\{x_0, \dots, x_M\} = \{\sum_a \gamma_a x_a : \gamma \in \Delta^M\}$ of finitely many points of $\mathbb{R}^D$ is compact. (iii) A continuous function that is* coercive *on a non-empty closed set $\mathcal{C} \subseteq \mathbb{R}^D$ ($f(\boldsymbol{\theta}) \to \infty$ as $\|\boldsymbol{\theta}\| \to \infty$ in $\mathcal{C}$) attains its minimum on $\mathcal{C}$, and a strictly convex function has at most one minimiser on a convex set. (iv) If $Q \succ 0$, then $\boldsymbol{\theta}^\top Q \boldsymbol{\theta} - 2\mathbf{b}^\top \boldsymbol{\theta} + c = \|\boldsymbol{\theta} - Q^{-1}\mathbf{b}\|_Q^2 + c - \mathbf{b}^\top Q^{-1}\mathbf{b}$, so $Q^{-1}\mathbf{b}$ is the unique minimiser.*

Proof: (i)–(ii) Direct from the definitions; the hull is the image of the compact simplex by a continuous map. (iii) A minimising sequence is bounded by coercivity, so a subsequence converges, to a point of the set, which is a minimiser by continuity; two minimisers $x \neq y$ would give $f(\frac{x+y}{2}) < f(x)$. (iv) Expand the square. □

Definition 2.20 (Second-order cone program)**.** *A* second-order cone program *minimises a linear function of* $\mathbf{x} \in \mathbb{R}^{n'}$ *subject to affine equalities and to constraints* $\|A_j\mathbf{x} + \mathbf{c}_j\| \leq \mathbf{f}_j^\top \mathbf{x} + g_j$, $j \in [\![J']\!]$ *(Boyd and Vandenberghe 2004, Sec. 4.4.2). It is a convex problem, solved by interior-point methods (Clarabel, Goulart and Chen 2024, in our experiments). A norm objective becomes a constraint through an epigraph variable: minimising* $\|C^\top(\boldsymbol{\theta}-\tilde{\boldsymbol{\theta}})\|$ *is minimising* r_0 *subject to* $\|C^\top(\boldsymbol{\theta}-\tilde{\boldsymbol{\theta}})\| \leq r_0$.

Exchangeability.

Definition 2.21 (Exchangeability)**.** *Random variables* $Z_1, \ldots, Z_N$ *are* exchangeable *if* $(Z_{\sigma(1)}, \ldots, Z_{\sigma(N)})$ *has the law of* $(Z_1, \ldots, Z_N)$ *for every permutation* σ *of* $[\![N]\!]$*, as i.i.d. variables are. They are* exchangeable given *a random variable* $\mathcal{D}$ *if* $(\mathcal{D}, Z_{\sigma(1)}, \ldots, Z_{\sigma(N)})$ *has the law of* $(\mathcal{D}, Z_1, \ldots, Z_N)$ *for every* σ*; then so are* $h(\mathcal{D}, Z_1), \ldots, h(\mathcal{D}, Z_N)$*, for every Borel* h*. For an event* E *and* $c \in [0,1]$*,* $\mathbb{P}(E \mid \mathcal{D}) \geq c$ *means that* $\mathbb{P}(E \cap \{\mathcal{D} \in B\}) \geq c\mathbb{P}(\mathcal{D} \in B)$ *for every Borel* B*, that is, every version of* $\mathbb{E}[\mathbb{1}_E \mid \mathcal{D}]$ *is at least* c *almost surely.*

3 Linear flow matching

Definition 3.1 (Flow matching with the linear interpolant)**.** *Let* p *be a Borel probability measure on* $\mathbb{R}^n$ *with finite second moment (the* target*; Definition 2.1), and let* $\mathbf{x}_0 \sim \mu_0$*,* $\mathbf{x}_1 \sim p$ *and* $t \sim \mathcal{U}[0,1]$ *be independent. Set* $\mathbf{x}_t = (1-t)\mathbf{x}_0 + t\mathbf{x}_1$ *and* $\mathbf{u} = \mathbf{x}_1 - \mathbf{x}_0$*, and let* ρ_p *be the law of* $(\mathbf{x}_t, t)$ *on* $\mathbb{R}^n \times [0,1]$ *(the* interpolation path *of* p*). The* marginal velocity $v_p^\star$ *is a version of* $\mathbb{E}[\mathbf{u} \mid \mathbf{x}_t, t]$ *(Definition 2.9); it exists and is unique* ρ_p*-a.e. because* $\mathbf{u}$ *is square integrable (Lemma 2.10(i)). The* flow-matching loss *of a field* $v \in L^2(\rho_p; \mathbb{R}^n)$ *is* $\mathcal{L}_p(v) = \mathbb{E}\,\|v(\mathbf{x}_t, t) - \mathbf{u}\|^2$.

Remark 3.2 (Why the marginal velocity). For $s \in [0,1]$, let p_s be the law of $(1-s)\mathbf{x}_0 + s\mathbf{x}_1$, so that $p_0 = \mu_0$ and $p_1 = p$. For $f \in C_c^1(\mathbb{R}^n)$, differentiating $\mathbb{E}\, f((1-s)\mathbf{x}_0 + s\mathbf{x}_1)$ under the expectation, and applying Definition 2.9 with $h(\mathbf{x}, t) = c(t)\nabla f(\mathbf{x})$ for bounded Borel c, gives $\frac{d}{ds}\int f dp_s = \int \langle \nabla f, v_p^\star(\cdot, s)\rangle dp_s$ for almost every s: the laws p_s solve the continuity equation $\partial_s p_s + \nabla \cdot (p_s v_p^\star(\cdot, s)) = 0$ in the weak sense. Under local boundedness, Lipschitz and integrability conditions on $v_p^\star$, the solutions of $\dot{X}_s = v_p^\star(X_s, s)$ then exist from μ_0-almost every starting point and push μ_0 forward to p_s (Ambrosio et al. 2008, Prop. 8.1.8); in particular, the flow transports μ_0 to p (Lipman et al. 2023, Liu et al. 2023, Albergo and Vanden-Eijnden 2023). This motivates the construction; no result below uses it.

Lemma 3.3 (Loss decomposition)**.** *For every* $v \in L^2(\rho_p; \mathbb{R}^n)$*,* $\mathcal{L}_p(v) = \|v - v_p^\star\|^2_{L^2(\rho_p)} + \mathcal{L}_p(v_p^\star)$.

Proof: $\mathbf{u}$ is square integrable because p and μ_0 have finite second moments. By Lemma 2.10(i), $\mathbf{u} - v_p^\star(\mathbf{x}_t, t)$ is orthogonal to $\mathcal{H}_{(\mathbf{x}_t,t)}$, which contains $(v - v_p^\star)(\mathbf{x}_t, t)$: its $L^2(\mathbb{P})$ norm is $\|v - v_p^\star\|_{L^2(\rho_p)} < \infty$ (Proposition 2.8(iv)). Pythagoras (Lemma 2.4(v)) applied to $v(\mathbf{x}_t, t) - \mathbf{u} = (v - v_p^\star)(\mathbf{x}_t, t) + (v_p^\star(\mathbf{x}_t, t) - \mathbf{u})$ gives the claim. □

Definition 3.4 (Samplers, outputs, straightness defect)**.** *Fix a grid* $0 = t_0 < t_1 < \cdots < t_J = 1$ *and let* $h_k = t_{k+1} - t_k$*. For a field* v *(Definition 2.11), the* sampler map $T_v : \mathbb{R}^n \to \mathbb{R}^n$ *is* $T_v(\mathbf{x}) = \mathbf{y}_J$*, where* $\mathbf{y}_0 = \mathbf{x}$ *and* $\mathbf{y}_{k+1} = \mathbf{y}_k + h_k v(\mathbf{y}_k, t_k)$ *(explicit Euler). The* one-step map *is* $T_v^1(\mathbf{x}) = \mathbf{x} + v(\mathbf{x}, 0)$*, the case* $J = 1$*. The* outputs *of* v *have law* $\nu_v = (T_v)_\# \mu_0$*, and its* straightness defect *is* $\zeta_v = \|T_v - T_v^1\|_{L^2(\mu_0)} \in [0, +\infty]$*. Both maps are Borel (Remark 2.2(i)).*

Theorem 7.1 and Proposition 7.3 below only use that T_v is a Borel map, so they hold for any other deterministic sampler (higher-order schemes, or the exact flow when it exists).

Definition 3.5 (Linear flow-matching model)**.** *A* feature map *is a Borel map* $\Phi : \mathbb{R}^n \times [0,1] \to \mathbb{R}^{n \times D}$ *of linear growth (Definition 2.7):* $\|\Phi(\mathbf{x}, t)\|_F \leq C_\Phi(1 + \|\mathbf{x}\|)$ *for all* $(\mathbf{x}, t)$*. The model is* $v_{\boldsymbol{\theta}} = \Phi\boldsymbol{\theta}$*,* $\boldsymbol{\theta} \in \mathbb{R}^D$*, and we write* $T_{\boldsymbol{\theta}}, T^1_{\boldsymbol{\theta}}, \nu_{\boldsymbol{\theta}}, \zeta_{\boldsymbol{\theta}}$ *for* $T_{v_{\boldsymbol{\theta}}}, T^1_{v_{\boldsymbol{\theta}}}, \nu_{v_{\boldsymbol{\theta}}}, \zeta_{v_{\boldsymbol{\theta}}}$*. For a target* p*, the* moments *are* $G_p = \mathbb{E}[\Phi(\mathbf{x}_t, t)^\top \Phi(\mathbf{x}_t, t)]$ *and* $\mathbf{b}_p = \mathbb{E}[\Phi(\mathbf{x}_t, t)^\top \mathbf{u}]$*, and for* $\lambda > 0$ *the* ridge fit *is* $\boldsymbol{\theta}_p = \arg\min_{\boldsymbol{\theta}} \mathcal{L}_p(v_{\boldsymbol{\theta}}) + \lambda\|\boldsymbol{\theta}\|^2$*. Given samples* $\mathbf{y}_1, \ldots, \mathbf{y}_N$*, the* few-shot fit *is* $\hat{\boldsymbol{\theta}} = \boldsymbol{\theta}_{\hat{p}}$ *with* $\hat{p} = \frac{1}{N}\sum_j \delta_{\mathbf{y}_j}$*, and its moments are* $\hat{G} = G_{\hat{p}}$ *and* $\hat{\mathbf{b}} = \mathbf{b}_{\hat{p}}$.

Lemma 3.6 (Linear flow matching)**.** *Let* p *have finite second moment and* $\lambda > 0$*. (i)* G_p *and* $\mathbf{b}_p$ *are finite,* $\mathcal{L}_p(v_{\boldsymbol{\theta}}) = \boldsymbol{\theta}^\top G_p \boldsymbol{\theta} - 2\mathbf{b}_p^\top \boldsymbol{\theta} + \mathbb{E}\,\|\mathbf{u}\|^2$*, and* $\boldsymbol{\theta}_p = (G_p + \lambda I)^{-1}\mathbf{b}_p$*. (ii)* $\boldsymbol{\theta}_p = \arg\min_{\boldsymbol{\theta}} \|v_{\boldsymbol{\theta}} - v_p^\star\|^2_{L^2(\rho_p)} + \lambda\|\boldsymbol{\theta}\|^2$*: the ridge fit is the regularised* $L^2(\rho_p)$*-projection of the marginal velocity onto the model. (iii) For every* $\boldsymbol{\theta}$*,* $T_{\boldsymbol{\theta}}$ *is Borel and* $\|T_{\boldsymbol{\theta}}(\mathbf{x})\| \leq e^{C_\Phi\|\boldsymbol{\theta}\|}(1 + \|\mathbf{x}\|)$*, so* $T_{\boldsymbol{\theta}} \in L^2(\mu_0)$*. The one-step map* $T^1_{\boldsymbol{\theta}}(\mathbf{x}) = \mathbf{x} + \Phi(\mathbf{x}, 0)\boldsymbol{\theta}$ *is affine in* $\boldsymbol{\theta}$*; for* $J \geq 2$*,* $T_{\boldsymbol{\theta}}$ *is in general not, because* Φ *is evaluated at points that depend on* $\boldsymbol{\theta}$.

Proof: (i)–(ii) $\mathbf{x}_t$ has a finite second moment, so $\mathbb{E}\,\|\Phi(\mathbf{x}_t, t)\|_F^2 < \infty$ (Proposition 2.8(v)), and Lemma 2.13 applies with $\rho = \rho_p$ and $G_{\rho_p} = G_p$; in particular $v_{\boldsymbol{\theta}} \in L^2(\rho_p; \mathbb{R}^n)$. The columns of $\Phi(\mathbf{x}_t, t)$ lie in $\mathcal{H}_{(\mathbf{x}_t,t)}$, so $\mathbf{b}_p = \mathbb{E}[\Phi(\mathbf{x}_t, t)^\top \mathbf{u}] = \mathbb{E}[\Phi(\mathbf{x}_t, t)^\top v_p^\star(\mathbf{x}_t, t)] = \Phi_{\rho_p}^* v_p^\star$ is finite (Lemma 2.10(i)). By Lemmas 3.3 and 2.13(iv), $\mathcal{L}_p(v_{\boldsymbol{\theta}}) = \|\Phi\boldsymbol{\theta} - v_p^\star\|^2_{L^2(\rho_p)} + \mathcal{L}_p(v_p^\star) = \boldsymbol{\theta}^\top G_p \boldsymbol{\theta} - 2\mathbf{b}_p^\top \boldsymbol{\theta} + \|v_p^\star\|^2_{L^2(\rho_p)} + \mathcal{L}_p(v_p^\star)$, and the last two terms add up to $\mathbb{E}\,\|\mathbf{u}\|^2$ (Lemma 2.10(i)). The objectives of (i) and (ii) therefore both differ from $\|\Phi\boldsymbol{\theta} - v_p^\star\|^2_{L^2(\rho_p)} + \lambda\|\boldsymbol{\theta}\|^2$ by constants, and Lemma 2.13(iv) gives their

unique minimiser $(G_p+\lambda I)^{-1}\Phi_{\rho_p}^* v_p^\star = (G_p+\lambda I)^{-1}\mathbf{b}_p$. (iii) $T_{\boldsymbol{\theta}}$ is a composition of Borel maps (Remark 2.2(i)). Since $\|\Phi\boldsymbol{\theta}\| \le \|\Phi\|_F\|\boldsymbol{\theta}\|$, $1+\|\mathbf{y}_{k+1}\| \le (1+h_kC_\Phi\|\boldsymbol{\theta}\|)(1+\|\mathbf{y}_k\|)$, and $\prod_k(1+h_kC_\Phi\|\boldsymbol{\theta}\|) \le e^{C_\Phi\|\boldsymbol{\theta}\|}$ because $1+x\le e^x$ and $\sum_k h_k = 1$; Proposition 2.8(v) then gives $T_{\boldsymbol{\theta}} \in L^2(\mu_0)$. □

Remark 3.7 (What "linear" covers). Random Fourier features (Rahimi and Recht 2007), $(2/D_\psi)^{1/2}\cos(\omega_j^\top\mathbf{x}+\omega_j' t+b_j)$ for $j \in [\![D_\psi]\!]$ with random frequencies (ω_j,ω_j') and phases b_j, are bounded, and the affine terms $(1,t,\mathbf{x},t\mathbf{x})$ have linear growth; stacking both in $\psi(\mathbf{x},t)$, the features $\Phi = I_n\otimes\psi^\top$ of the toy studies satisfy Definition 3.5. So does the first-order expansion of a velocity network u_ϑ around pretrained parameters ϑ_0, $u_{\vartheta_0}+\partial_\vartheta u_{\vartheta_0}(\vartheta-\vartheta_0)$, with features $\Phi=\partial_\vartheta u_{\vartheta_0}$ and a fixed offset u_{ϑ_0}, when both have linear growth. An offset changes nothing in Sections 4–7, where only differences of fields enter; in Lemma 3.6, $\mathbf{u}$ is replaced by $\mathbf{u}-u_{\vartheta_0}(\mathbf{x}_t,t)$.

Couplings, rectification and reflow. Definition 3.1 pairs each noise with a data point drawn independently of it, and the field learns the average direction of these random pairs. Nothing in Lemmas 3.3 and 3.6 uses the independence. Changing the pairing while keeping its two marginals is the idea of reflow: pair each noise with the output that the model already generates from it, and fit again.

Definition 3.8 (Flow matching with a coupling; reflow)**.** *Let q be a Borel probability measure on $\mathbb{R}^n$ with finite second moment, and $\pi \in \Pi(\mu_0,q)$ a coupling (Definition 2.15).* Flow matching with the coupling π *is Definition 3.1 with $(\mathbf{x}_0,\mathbf{x}_1)\sim\pi$, independent of $t\sim\mathcal{U}[0,1]$, in place of independent $\mathbf{x}_0\sim\mu_0$ and $\mathbf{x}_1\sim p$; its interpolation path, marginal velocity and loss are written ρ_π, $v_\pi^\star$ and $\mathcal{L}_\pi$. Standard flow matching is the* independent coupling $\pi=\mu_0\otimes p$, *and its marginal velocity $v_p^\star$ is the* 1-rectified *field. For a field v with $T_v\in L^2(\mu_0)$, the* reflow coupling *of v is $\pi_v=(\mathrm{id},T_v)_\#\mu_0$: the law of a noise $\mathbf{x}_0$ paired with the output $T_v(\mathbf{x}_0)$ that the sampler of v generates from it, whose marginals are μ_0 and the output law ν_v.* Reflow *replaces v by $v_{\pi_v}^\star$, or by a fit of it. Starting from a fit $v^{(1)}$ of $v_p^\star$, reflow gives the* 2-rectified *field $v^{(2)}$, and $k-1$ reflows give the k-*rectified *field (Liu et al. 2023).*

Lemma 3.9 (What the coupling changes)**.** *Let $\pi\in\Pi(\mu_0,q)$, $\mathbf{u}=\mathbf{x}_1-\mathbf{x}_0$ and $\mathbf{x}_t=(1-t)\mathbf{x}_0+t\mathbf{x}_1$. (i) Lemmas 3.3 and 3.6 hold with π in place of $\mu_0\otimes p$. The smallest loss, $\mathcal{L}_\pi(v_\pi^\star)=\mathbb{E}\,\|\mathbf{u}-\mathbb{E}[\mathbf{u}\mid\mathbf{x}_t,t]\|^2$, is the mean conditional variance of the directions of the segments that pass through a point at a given time. It is 0 if and only if $\mathbf{u}$ is almost surely a Borel function of $(\mathbf{x}_t,t)$, that is, if almost surely a single segment passes through each point at each time. (ii) For the independent coupling $\pi=\mu_0\otimes p$, $\mathcal{L}_\pi(v_\pi^\star)>0$ unless p is a Dirac mass. (iii) Let $\pi=(\mathrm{id},T)_\#\mu_0$ for an L-Lipschitz map $T:\mathbb{R}^n\to\mathbb{R}^n$, and $\Psi_t(\mathbf{x})=(1-t)\mathbf{x}+tT(\mathbf{x})$. For $t<1/(1+L)$, Ψ_t is a bijection of $\mathbb{R}^n$, and $v_\pi^\star(\mathbf{y},t)=T(\Psi_t^{-1}(\mathbf{y}))-\Psi_t^{-1}(\mathbf{y})$ defines at these times a version of the marginal velocity (completed by any version at the other times), which is continuous in $(\mathbf{y},t)$ and constant along each segment: $v_\pi^\star(\Psi_t(\mathbf{x}),t)=T(\mathbf{x})-\mathbf{x}$. At $t=0$ it is $v_\pi^\star(\mathbf{x},0)=T(\mathbf{x})-\mathbf{x}$, so the one-step map of the reflowed field is $T^1_{v_\pi^\star}=T$. (iv) If T is moreover* monotone, *$\langle T(\mathbf{x})-T(\mathbf{x}'),\mathbf{x}-\mathbf{x}'\rangle\ge 0$ for all $\mathbf{x},\mathbf{x}'$, then (iii) holds at every $t\in[0,1)$. Hence $\mathcal{L}_\pi(v_\pi^\star)=0$, every Euler sampler of $v_\pi^\star$ maps each $\mathbf{x}$ to $T(\mathbf{x})$, and the straightness defect of $v_\pi^\star$ is 0.*

Proof: (i) The proofs of Lemmas 3.3 and 3.6 only use that $\mathbf{u}\in L^2(\mathbb{P};\mathbb{R}^n)$, which holds because $\|\mathbf{u}\|^2\le 2\|\mathbf{x}_0\|^2+2\|\mathbf{x}_1\|^2$. By Lemma 2.10(i), $\mathcal{L}_\pi(v_\pi^\star)=\mathbb{E}\,\|\mathbf{u}-g(\mathbf{x}_t,t)\|^2$ for a version g, which vanishes if and only if $\mathbf{u}=g(\mathbf{x}_t,t)$ a.s.; conversely, if $\mathbf{u}=h(\mathbf{x}_t,t)$ a.s. for a Borel h, then h is a version (Definition 2.9). (ii) If the loss vanishes, then by (i) $\mathbf{x}_1=\mathbf{x}_t+(1-t)\mathbf{u}$ is a.s. a Borel function $f(\mathbf{x}_t,t)$ on $\{t<1\}$. If p is not a Dirac mass, there is a Borel B with $0<p(B)<1$: otherwise, for every k exactly one dyadic cube of side 2^{-k} would have mass 1, these cubes would be nested, and p would be the Dirac mass at their intersection. On the one hand $\mathbb{1}_B(f(\mathbf{x}_t,t))$ is a version of $\mathbb{E}[\mathbb{1}_B(\mathbf{x}_1)\mid\mathbf{x}_t,t]$, with values in $\{0,1\}$. On the other hand, Bayes' formula (Lemma 2.10(ii), with the conditional density $\varphi_t(\mathbf{x}-t\mathbf{y})$ of the proof of Lemma 4.3(iii)) gives the version $\int_B\varphi_t(\mathbf{x}_t-t\mathbf{y})p(d\mathbf{y})/\int\varphi_t(\mathbf{x}_t-t\mathbf{y})p(d\mathbf{y})$, which lies in $(0,1)$ everywhere because $\varphi_t>0$ and $0<p(B)<1$. Two versions agree a.s., a contradiction. (iii) For $t<1/(1+L)$ and $\mathbf{y}\in\mathbb{R}^n$, the map $\mathbf{x}\mapsto(\mathbf{y}-tT(\mathbf{x}))/(1-t)$ is a contraction with constant $tL/(1-t)<1$, whose fixed points are the solutions of $\Psi_t(\mathbf{x})=\mathbf{y}$. So Ψ_t is a bijection, and its inverse is continuous in $(\mathbf{y},t)$, since the fixed point of a contraction depends continuously on a parameter on which the map depends continuously, with a contraction constant bounded away from 1 on compact sets of t. Since $\mathbf{x}_t=\Psi_t(\mathbf{x}_0)$, $\mathbf{u}=T(\mathbf{x}_0)-\mathbf{x}_0=v_\pi^\star(\mathbf{x}_t,t)$ at these times. Completed by a version v^0 at the other times, this function is a version: for $h(\mathbf{x}_t,t)\in L^2$, write $\bar t=1/(1+L)$; the expectation of $\langle\mathbf{u}-v_\pi^\star(\mathbf{x}_t,t),h(\mathbf{x}_t,t)\rangle$ on $\{t<\bar t\}$ vanishes because $\mathbf{u}=v_\pi^\star(\mathbf{x}_t,t)$ there, and on $\{t\ge\bar t\}$ by Definition 2.9 for v^0 and $h\mathbb{1}\{t\ge\bar t\}$. At $t=0$, $\Psi_0=\mathrm{id}$. (iv) For $t<1$, Ψ_t is $(1-t)$-strongly monotone, $\langle\Psi_t(\mathbf{x})-\Psi_t(\mathbf{x}'),\mathbf{x}-\mathbf{x}'\rangle\ge(1-t)\|\mathbf{x}-\mathbf{x}'\|^2$, and ℓ_t-Lipschitz with $\ell_t=1-t+tL$. With $\eta=(1-t)/\ell_t^2$, expanding the square shows that $\mathbf{x}\mapsto\mathbf{x}-\eta(\Psi_t(\mathbf{x})-\mathbf{y})$ is a contraction with constant $(1-(1-t)^2/\ell_t^2)^{1/2}<1$, whose fixed points solve $\Psi_t(\mathbf{x})=\mathbf{y}$. The argument of (iii) then applies at every $t<1$, and $t=1$ is a null set, so $\mathbf{u}=v_\pi^\star(\mathbf{x}_t,t)$ a.s. and the loss vanishes by (i). For an Euler grid, if $\mathbf{y}_k=\Psi_{t_k}(\mathbf{x})$, then $\mathbf{y}_{k+1}=\Psi_{t_k}(\mathbf{x})+h_k(T(\mathbf{x})-\mathbf{x})=\Psi_{t_{k+1}}(\mathbf{x})$, because $v_\pi^\star(\Psi_{t_k}(\mathbf{x}),t_k)=T(\mathbf{x})-\mathbf{x}$ and $t_k<1$; by induction, $\mathbf{y}_J=\Psi_1(\mathbf{x})=T(\mathbf{x})$. □

Remark 3.10 (Reflow versus standard flow matching). Table 1 sets the two side by side, and Figure 1 draws them on the ring of Section 8, with the exact marginal velocity (Bayes' formula) instead of a fitted one.

- *Why standard flow matching curves.* The regression target at $(\mathbf{x},t)$ is the average direction of all the training segments that pass through $\mathbf{x}$ at time t (Lemma 3.9(i)). Independent pairs cross everywhere (panel (a)). At $t=0$, the segments leaving a noise point end at all the target samples, so their average points to the mean of the target, the centre of the ring (Lemma 4.3(iii)). Later, as $\mathbf{x}_t$ reveals which blob the pair is heading to, the average turns. The trajectories of the field therefore bend (panel (b)), although every training segment is straight, and one Euler

step lands at the mean of the target. Panel (d) measures the crossing: the conditional variance of the directions is 4.3 near $t = 0$ and never falls below 1.6; its average, the smallest flow-matching loss, is 3.8. Lemma 3.9(ii) says it cannot vanish for a target that is not a single point.

- *What reflow changes.* Reflow keeps the noise law and the output law of the current field and changes only the pairing: each noise is paired with the output that the field already generates from it (panel (c), with the same outputs as panel (b)). At $t = 0$ no two segments cross, since one segment leaves each noise, and the field points each noise to its own output (Lemma 3.9(iii)). Later the segments may cross; here they hardly do, and the smallest loss is about 0.002. The field fitted to them moves along its own training segments, so its trajectories are straight, and a Lipschitz monotone pairing makes them exactly straight (Lemma 3.9(iv)).

- *What is guaranteed.* For exact fields whose flows are well posed, Liu et al. (2023) prove four properties of the flow Z of the marginal velocity of a coupling. First, Z has the marginals of the interpolation, so it transports μ_0 to the second marginal of the coupling (the argument of Remark 3.2, which does not use independence). Second, its pairs (Z_0, Z_1), which are the next reflow coupling, have no larger convex transport cost $\mathbb{E}\, c(\mathbf{x}_1 - \mathbf{x}_0)$ than the training pairs. Third, a coupling is unchanged by this operation if and only if its segments do not cross, and optimal transport couplings for strictly convex costs are such fixed points; in dimension one, flows cannot cross, so one reflow of an exact flow gives a straight one. Fourth, among the first k rectified flows, the smallest sum of straightness defect and crossing is at most $\mathbb{E}\,\|\mathbf{x}_1 - \mathbf{x}_0\|^2/k$ for the initial pairs. For $c = \|\cdot\|^2$, an exact identity underlies these results: $\mathbb{E}\,\|\mathbf{x}_1 - \mathbf{x}_0\|^2$ over the training pairs minus $\mathbb{E}\,\|Z_1 - Z_0\|^2$ equals the crossing of the training pairs plus the curvature $\int_0^1 \mathbb{E}\,\|\dot{Z}_t - (Z_1 - Z_0)\|^2 dt$ of the learned flow. On the ring, with independent training pairs, $\mathbb{E}\,\|\mathbf{x}_1 - \mathbf{x}_0\|^2$ falls from 6.18 to 0.97, a decrease of 5.21, against a crossing of 3.8 plus a curvature of 1.44, that is 5.25; the gap is discretisation error.

- *What it costs.* Reflow needs no data, but one simulation of the current field per training pair. It targets the current outputs ν_v, not p: the exact reflowed flow reproduces ν_v, so the errors of v are inherited, and a fitted reflow adds its own (in the test of Section 9, fidelity drops). Distillation instead fits the map $\mathbf{x} \mapsto T_v(\mathbf{x})$ directly to the same pairs (Liu et al. 2024); by Lemma 3.9(iii), it is the one-step map of the exact reflowed field.

- *Why this note needs it.* Straight fields make the source slice informative: $|d_T - d_0| \le \zeta_v + \zeta_{v'}$ (Lemma 4.3(i)). For one-step samplers, the map metric is the Mahalanobis distance of Proposition 4.2(ii), so the region with static weights is convex and its projection is a second-order cone program (Corollary 7.2(i)).

- *In the test of Section 9.* Reflow is done for each of the 2000 training prompts: 1200 noises are pushed through the 1-rectified model's 50-step Euler sampler, the 0.04% of pairs whose output lies beyond radius 4 are dropped (they otherwise dominate the least-squares refit), and the model affine in the prompt is refitted on the remaining pairs. The 2-rectified model is again affine in the prompt (Definition 3.11(b)), so Sections 4–7 apply to it unchanged. The trimmed pairs are $(\mathrm{id}, T_v)_{\#}\mu_0'$, with μ_0' the noise law conditioned on $|T_v(\mathbf{x}_0)| \le 4$, and the proof of Lemma 3.9(iii) applies to them verbatim. The sampler map T_v is Lipschitz, because each Euler step of a field with Lipschitz features is, but with a very large constant L: the 1-rectified sampler diverges on rare noises (the trimmed pairs reach $|T_v(\mathbf{x}_0)| = 2.5 \times 10^5$, and Section 9, Step 4, shows what this does to unbounded distances). The lemma therefore describes the exact reflowed field only for $t < 1/(1 + L)$, an interval that contains $t = 0$, where its slice is $T_v - \mathrm{id}$; that slice is all the note uses. The median straightness defect falls from 1.93 to 0.26, and $\mathrm{MMD}^2 \times 10^3$ to the targets rises from 3.3 to 8.1 (34.9 with one step).

Table 1: Standard flow matching and reflow, for exact fields (Definition 3.8, Lemma 3.9).

	standard flow matching	reflow of a field v
training pairs $(\mathbf{x}_0, \mathbf{x}_1)$	a noise and an independent sample of the target	a noise and the output $T_v(\mathbf{x}_0)$ that v generates from it
data needed	samples of the target p	none: one simulation of v per pair
law of the outputs	p	$\nu_v = (T_v)_{\#}\mu_0$; the errors of v are inherited
field at $t = 0$	$\mathbb{E}_p[\mathbf{x}_1] - \mathbf{x}$: towards the mean of the target (Lemma 4.3(iii))	$T_v(\mathbf{x}) - \mathbf{x}$: towards the noise's own output (Lemma 3.9(iii))
crossing of the pairs (smallest loss)	positive unless p is a Dirac mass (Lemma 3.9(ii)); 3.8 on the ring	zero for a Lipschitz monotone T_v (Lemma 3.9(iv)); 0.002 on the ring
trajectories	curved	straight, or nearly
one Euler step	lands at the mean of the target	gives T_v: the outputs of v in one step

Definition 3.11 (Prompt-conditioned linear flow matching)**.** *Prompts are embeddings $\mathbf{e} \in \mathcal{E} \subseteq \mathbb{R}^{D_e}$. A* prompt-to-field map *is a Borel map $\boldsymbol{\theta}(\cdot) : \mathcal{E} \to \mathbb{R}^D$; the field of $\mathbf{e}$ is $v_{\boldsymbol{\theta}(\mathbf{e})}$ and its outputs have law $\nu_{\boldsymbol{\theta}(\mathbf{e})}$. Two instances matter. (a)* Per-prompt fit*: each prompt specifies a target $p_{\mathbf{e}}$, and $\boldsymbol{\theta}(\mathbf{e}) = \boldsymbol{\theta}_{p_{\mathbf{e}}}$ is its ridge fit (estimated from ample data in practice); a request may also come with N samples and a few-shot fit $\hat{\boldsymbol{\theta}}(\mathbf{e})$. (b)* Affine in the prompt*: $\boldsymbol{\theta}(\mathbf{e}) = \mathbf{a} + U\mathbf{e}$, with $\mathbf{a} \in \mathbb{R}^D$ and $U \in \mathbb{R}^{D \times D_e}$ fixed.*

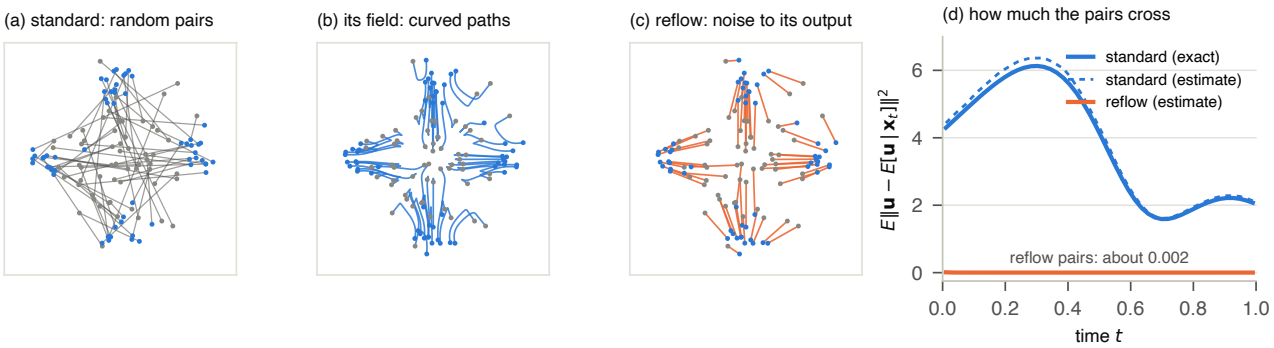


Figure 1: Standard flow matching against reflow on the ring of Section 8, with the exact marginal velocity. (a) Training pairs of standard flow matching: 60 noises (grey) paired with independent target samples (blue). (b) Trajectories of the field learned from such pairs, from the same noises. (c) Reflow pairs: the same noises paired with the outputs of (b). (d) Conditional variance of the direction $\mathbf{u} = \mathbf{x}_1 - \mathbf{x}_0$ of the pairs given the point $\mathbf{x}_t$, against t: exact for the independent pairs, and a 25-nearest-neighbour estimate on 20,000 pairs for both couplings; its average over t is the smallest flow-matching loss (Lemma 3.9(i)).

Instance (b) arises in two ways. A conditional model $v(\mathbf{x}, t; \mathbf{e}) = \Phi(\mathbf{x}, t)(\mathbf{a} + U\mathbf{e})$ is fitted once on (prompt, sample) pairs by linear least squares, since it is linear in $(\mathbf{a}, U)$. And the first-order expansion of any conditional network $u(\mathbf{x}, t; \mathbf{e})$ at a reference prompt $\mathbf{e}_0$, $u(\cdot; \mathbf{e}_0) + \partial_{\mathbf{e}} u(\cdot; \mathbf{e}_0)(\mathbf{e} - \mathbf{e}_0)$, is instance (b), provided these features have linear growth, with $D = 1 + D_e$, $\Phi = [u(\cdot; \mathbf{e}_0) - \partial_{\mathbf{e}} u(\cdot; \mathbf{e}_0)\mathbf{e}_0,\ \partial_{\mathbf{e}} u(\cdot; \mathbf{e}_0)]$, $\mathbf{a} = \mathbf{1}_1$ and $U = [0, I_{D_e}]^\top$.

4 Distances between fields

Definition 4.1 (Distances between fields)**.** *For fields v, v', the following are extended pseudometrics (Proposition 2.8(i), Lemma 2.4(ii)): (i) for a probability measure ρ on $\mathbb{R}^n \times [0,1]$, $d_\rho(v, v') = \|v - v'\|_{L^2(\rho)}$; (ii) for a Borel map $\chi : \mathbb{R}^n \to \mathbb{R}^{n_\chi}$ of linear growth (the space in which outputs are compared: the identity, an encoder, the features of a classifier), the* map metric *is $d_T^\chi(v, v') = \|\chi \circ T_v - \chi \circ T_{v'}\|_{L^2(\mu_0)}$, and $d_T = d_T^{\mathrm{id}}$; (iii) the* source-slice metric *is $d_0 = d_{\mu_0 \otimes \delta_0}$, that is, $d_0(v, v') = \|v(\cdot, 0) - v'(\cdot, 0)\|_{L^2(\mu_0)} = \|T_v^1 - T_{v'}^1\|_{L^2(\mu_0)}$. On the linear model we write $d(\boldsymbol{\theta}, \boldsymbol{\theta}') = d(v_{\boldsymbol{\theta}}, v_{\boldsymbol{\theta}'})$, again an extended pseudometric, and, when $\mathbb{E}_\rho \|\Phi\|_F^2 < \infty$, $G_\rho = \mathbb{E}_\rho[\Phi^\top \Phi]$ as in Lemma 2.13.*

Proposition 4.2 (Mahalanobis form)**.** *(i) If $\mathbb{E}_\rho \|\Phi\|_F^2 < \infty$, then $d_\rho(\boldsymbol{\theta}, \boldsymbol{\theta}') = \|\boldsymbol{\theta} - \boldsymbol{\theta}'\|_{G_\rho}$; d_ρ is a pseudometric on $\mathbb{R}^D$, and a metric if and only if $G_\rho \succ 0$. For $\delta \geq 0$ with $H = G_\rho + \delta I \succ 0$ and $H = CC^\top$, $\|\boldsymbol{\theta} - \boldsymbol{\theta}'\|_H = \|C^\top(\boldsymbol{\theta} - \boldsymbol{\theta}')\|_2 \geq d_\rho(\boldsymbol{\theta}, \boldsymbol{\theta}')$. (ii) $d_0(\boldsymbol{\theta}, \boldsymbol{\theta}') = \|\boldsymbol{\theta} - \boldsymbol{\theta}'\|_{G_0}$ with $G_0 = \mathbb{E}_{\mathbf{x} \sim \mu_0}[\Phi(\mathbf{x}, 0)^\top \Phi(\mathbf{x}, 0)]$: for the one-step sampler, the map metric is a Mahalanobis distance. (iii) For a prompt map affine in the prompt, $d_\rho(\boldsymbol{\theta}(\mathbf{e}), \boldsymbol{\theta}(\mathbf{e}')) = \|\mathbf{e} - \mathbf{e}'\|_{U^\top G_\rho U}$. (iv) d_T^χ is a finite pseudometric on $\mathbb{R}^D$.*

Proof: (i) Lemma 2.13(ii) applied to $\boldsymbol{\theta} - \boldsymbol{\theta}'$ gives the Mahalanobis form; Lemmas 2.5(i) and 2.4(i) give the kernel and the metric property; and $\|C^\top \mathbf{a}\|^2 = \mathbf{a}^\top H \mathbf{a} = \|\mathbf{a}\|_{G_\rho}^2 + \delta \|\mathbf{a}\|^2$. (ii) Apply (i) with $\rho = \mu_0 \otimes \delta_0$, which is admissible by linear growth (Proposition 2.8(v)), and note $T_{\boldsymbol{\theta}}^1 - T_{\boldsymbol{\theta}'}^1 = \Phi(\cdot, 0)(\boldsymbol{\theta} - \boldsymbol{\theta}')$. (iii) $\boldsymbol{\theta}(\mathbf{e}) - \boldsymbol{\theta}(\mathbf{e}') = U(\mathbf{e} - \mathbf{e}')$ and $\|U\mathbf{a}\|_{G_\rho}^2 = \mathbf{a}^\top U^\top G_\rho U \mathbf{a}$. (iv) $\chi \circ T_{\boldsymbol{\theta}}$ has linear growth, by that of χ and Lemma 3.6(iii), so it lies in $L^2(\mu_0)$ (Proposition 2.8(v)); d_T^χ is the $L^2(\mu_0)$ distance between such maps (Proposition 2.8(i), Lemma 2.4(ii)). □

The choice of distance decides where fields are compared. For instance (a), the pooled interpolation path of the safe prompts, $\bar{\rho} = \frac{1}{M} \sum_i \rho_{p_{\mathbf{e}_i}}$, gives $G_{\bar{\rho}} = \frac{1}{M} \sum_i G_{p_{\mathbf{e}_i}}$, the average of the safe least-squares Gram matrices; it compares fields where the safe paths put mass (Remark 2.12), an off-policy distance. The source slice compares fields on μ_0, the law at which every sampler makes its first evaluation of every field. The map metric compares where the fields send the noise: it evaluates each field along its own trajectories and is nevertheless symmetric. The next lemma relates the last two.

Lemma 4.3 (Time, straightness and the source slice)**.** *(i) If $T_v, T_v^1, T_{v'}, T_{v'}^1 \in L^2(\mu_0)$ (e.g. for all fields of the linear model), then $|d_T(v, v') - d_0(v, v')| \leq \zeta_v + \zeta_{v'}$. (ii) If solutions of $\dot{X}_t = v(X_t, t)$ on $[0,1]$ exist from every initial point and all of them are affine in t (a* straight *field), then v is constant along each trajectory, $T_v = T_v^1$ for every grid, and $\zeta_v = 0$. (iii) The version of $v_p^\star$ given by Bayes' formula (Lemma 2.10(ii)) is continuous on $\mathbb{R}^n \times [0,1)$ and satisfies $v_p^\star(\mathbf{x}, 0) = \mathbb{E}_p[\mathbf{x}_1] - \mathbf{x}$. Hence $T_{v_p^\star}^1 \equiv \mathbb{E}_p[\mathbf{x}_1]$ and $d_0(v_p^\star, v_{p'}^\star) = \|\mathbb{E}_p[\mathbf{x}_1] - \mathbb{E}_{p'}[\mathbf{x}_1]\|$; if moreover the sampler transports μ_0 to p (as the exact flow does when it is well posed), then $\zeta_{v_p^\star}^2 = \mathbb{E}_p \|\mathbf{x}_1 - \mathbb{E}_p[\mathbf{x}_1]\|^2$.*

Proof: (i) Minkowski's inequality in $L^2(\mu_0)$ (Proposition 2.8(i)): $\|T_v - T_{v'}\| \leq \|T_v - T_v^1\| + \|T_v^1 - T_{v'}^1\| + \|T_{v'}^1 - T_{v'}\|$, and symmetrically with the roles of T and T^1 exchanged. (ii) Let $X_t = X_0 + t(X_1 - X_0)$ be a solution. By Definition 2.11,

$v(X_t,t) = \dot{X}_t = X_1 - X_0 = v(X_0,0)$ for every $t \in [0,1]$. If the Euler iterate satisfies $\mathbf{y}_k = X_{t_k}$, then $\mathbf{y}_{k+1} = X_{t_k} + h_k(X_1 - X_0) = X_{t_{k+1}}$; by induction $T_v(X_0) = X_1 = X_0 + v(X_0,0) = T_v^1(X_0)$, and a solution starts from every point. (iii) Given $\mathbf{x}_1 = \mathbf{y}$, the pair $(\mathbf{x}_t, t)$ has the density $\varphi_t(\mathbf{x} - t\mathbf{y})\mathbb{1}\{t < 1\}$ with respect to the Lebesgue measure on $\mathbb{R}^n \times [0,1]$, where φ_t is the density of $\mathcal{N}(0,(1-t)^2 I_n)$, and $\mathbf{u} = (\mathbf{x}_1 - \mathbf{x}_t)/(1-t)$ a.s. Bayes' formula (Lemma 2.10(ii)) gives the version $v_p^\star(\mathbf{x},t) = (m_t(\mathbf{x}) - \mathbf{x})/(1-t)$ for $t < 1$, with $m_t(\mathbf{x}) = \int \mathbf{y}\varphi_t(\mathbf{x} - t\mathbf{y})p(d\mathbf{y}) / \int \varphi_t(\mathbf{x} - t\mathbf{y})p(d\mathbf{y})$. It is continuous on $\mathbb{R}^n \times [0,1)$ by dominated convergence (φ_t is bounded uniformly for $t \le \bar{t} < 1$ and p has a first moment), and $\varphi_0(\mathbf{x} - 0 \cdot \mathbf{y})$ does not depend on $\mathbf{y}$, so $m_0(\mathbf{x}) = \mathbb{E}_p[\mathbf{x}_1]$. The rest follows from $T_v^1(\mathbf{x}) = \mathbf{x} + v(\mathbf{x},0)$ and, for the defect, from $T_{v_p^\star}^1 \equiv \mathbb{E}_p[\mathbf{x}_1]$, $T_{v_p^\star}(\mathbf{x}_0) \sim p$ and the change of variables formula (Remark 2.2(ii)). □

Two consequences settle how time enters. For a straight field, the whole time-dependent field is summarised by its slice at $t = 0$, and $d_T = d_0$. For the independent coupling used to train 1-rectified flows, the slice at $t = 0$ of the exact field knows only the mean of the target, so d_0 cannot see the shape of the outputs, and the defect has the size of the target's spread. The source slice is therefore informative only for straightened models: after reflow, the slice at $t = 0$ of the exact field is the displacement $T_v - \mathrm{id}$ of the previous sampler (Lemma 3.9(iii)), and distillation fits that map directly (Remark 3.10); for curved models one uses d_T^X itself, or d_0 corrected by the defects (Corollary 7.2).

5 The order-1 Wasserstein region of safe fields

Assumption 5.1 (Safe prompts)**.** *The pairs $(\mathbf{e}_j, \boldsymbol{\theta}_j)$, $j \in [\![M+K+1]\!]$, are exchangeable (Definition 2.21); for instance, the prompts are i.i.d. draws from a distribution of approved prompts and $\boldsymbol{\theta}_j = \boldsymbol{\theta}(\mathbf{e}_j)$, or an independent Monte Carlo fit of it. The first M pairs, $\mathcal{D} = ((\mathbf{e}_i, \boldsymbol{\theta}_i))_{i \le M}$, define the region (the* safe fields*), the next K calibrate it, and the last one is a new approved prompt. The pseudometric d on $\mathbb{R}^D$ and the weight map $w : \mathcal{E} \to \Delta^{M-1}$ below are Borel functions of their arguments and of $\mathcal{D}$, and depend on no other data.*

Definition 5.2 (Order-1 score and region)**.** *For a prompt $\mathbf{e}$ and $\boldsymbol{\theta} \in \mathbb{R}^D$, the order-1 score is*

$$g_{d,1}(\mathbf{e}, \boldsymbol{\theta}; \mathcal{D}) = \frac{1}{M+1} \sum_{i=1}^{M} w_i(\mathbf{e}) d(\boldsymbol{\theta}, \boldsymbol{\theta}_i),$$

which is $s_{d,1}(\boldsymbol{\theta}; \mathcal{D}) = \frac{1}{M(M+1)} \sum_i d(\boldsymbol{\theta}, \boldsymbol{\theta}_i)$ for uniform weights. For $\beta \in (0,1)$, let $k = \lceil (K+1)(1-\beta) \rceil$ and $S_j = g_{d,1}(\mathbf{e}_{M+j}, \boldsymbol{\theta}_{M+j}; \mathcal{D})$ for $j \in [\![K]\!]$. The threshold $\varepsilon = C_{1-\beta}(S_1, \dots, S_K)$ is the k-th smallest calibration score if $k \le K$ and $+\infty$ otherwise, and the region of $\mathbf{e}$ is $\bar{\Theta}(\mathbf{e}) = \{\boldsymbol{\theta} \in \mathbb{R}^D : g_{d,1}(\mathbf{e}, \boldsymbol{\theta}; \mathcal{D}) \le \varepsilon\}$, written $\bar{\Theta}$ for uniform weights.

Prompt-adaptive weights, e.g. $w_i(\mathbf{e}) \propto \exp(-\|\mathbf{e} - \mathbf{e}_i\|^2/(2h^2))$, tighten the region around the safe prompts most similar to the request; this is the task-adaptive score $g_{d,1}$ of the manuscript with prompts in place of tasks. The prompt is observed exactly, so the calibration scores and the deployed score are the same function, and the exchangeability caveat of few-shot task reweighting does not arise.

Lemma 5.3 (Closed form on a pseudometric space)**.** *Let (V,d) be a pseudometric space, $v_1, \dots, v_M, v \in V$, $w \in \Delta^{M-1}$, and $\hat{\mathbb{F}}^w = \sum_i w_i \delta_{v_i}$, $\hat{\mathbb{F}}^{w,v} = \frac{1}{M+1}(M\hat{\mathbb{F}}^w + \delta_v)$, and let $W_{d,1}$ be as in Definition 2.15. Then*

$$W_{d,1}(\hat{\mathbb{F}}^w, \hat{\mathbb{F}}^{w,v}) = \frac{1}{M+1} \sum_{i=1}^{M} w_i d(v_i, v).$$

In particular $g_{d,1}(\mathbf{e}, \boldsymbol{\theta}; \mathcal{D})$ is the order-1 Wasserstein distance, on the space of fields when d is one of the distances of Definition 4.1, between the $w(\mathbf{e})$-weighted law of the safe fields and the same law with the candidate field added.

Proof: The plan $\pi^\star = \sum_i \frac{M w_i}{M+1} \delta_{(v_i, v_i)} + \sum_i \frac{w_i}{M+1} \delta_{(v_i, v)}$ has the right marginals and, since $d(v_i, v_i) = 0$, costs $\frac{1}{M+1} \sum_i w_i d(v_i, v)$. Conversely, $f = d(\cdot, v)$ is 1-Lipschitz (Definition 2.3), so by Lemma 2.16(ii) every coupling costs at least $\sum_i w_i f(v_i) - \frac{1}{M+1}(M \sum_i w_i f(v_i) + f(v)) = \frac{1}{M+1}(\sum_i w_i d(v_i, v) - f(v))$, and $f(v) = 0$. □

Proposition 5.4 (Coverage)**.** *Under Assumption 5.1, $\mathbb{P}\big(\boldsymbol{\theta}_{M+K+1} \in \bar{\Theta}(\mathbf{e}_{M+K+1}) \,\big|\, \mathcal{D}\big) \ge k/(K+1) \ge 1 - \beta$.*

Proof: If $k > K$, $\varepsilon = +\infty$. Let $k \le K$ and $S_{K+1} = g_{d,1}(\mathbf{e}_{M+K+1}, \boldsymbol{\theta}_{M+K+1}; \mathcal{D})$. A permutation of the last $K+1$ indices is a permutation of all of them that fixes the first M, so the last $K+1$ pairs are exchangeable given $\mathcal{D}$, and so are the scores $S_j = h(\mathcal{D}, \mathbf{e}_{M+j}, \boldsymbol{\theta}_{M+j})$, with h Borel (Definition 2.21). Let $N_j = \#\{l \le K+1 : S_l < S_j\}$; permuting the scores permutes the N_j in the same way, so they are exchangeable given $\mathcal{D}$ too. Let $s_{(k)}$ be the k-th smallest of all $K+1$ scores. If $S_{K+1} > \varepsilon$, at least k calibration scores are at most $\varepsilon < S_{K+1}$, so $N_{K+1} \ge k$. If $N_j \ge k$, then $S_j > s_{(k)}$, and at most $K+1-k$ indices satisfy $S_j > s_{(k)}$; hence $\sum_j \mathbb{1}\{N_j \ge k\} \le K+1-k$. For every Borel B, exchangeability gives $(K+1)\mathbb{P}(N_{K+1} \ge k, \mathcal{D} \in B) = \mathbb{E}[\sum_j \mathbb{1}\{N_j \ge k\}\mathbb{1}\{\mathcal{D} \in B\}] \le (K+1-k)\mathbb{P}(\mathcal{D} \in B)$, hence $\mathbb{P}(S_{K+1} \le \varepsilon, \mathcal{D} \in B) \ge \mathbb{P}(N_{K+1} < k, \mathcal{D} \in B) \ge \frac{k}{K+1}\mathbb{P}(\mathcal{D} \in B)$, which is the claim in the sense of Definition 2.21. □

Remark 5.5 (Scope of the guarantee). (i) The probability is conditional on the safe fields and marginal over calibration and test; a statement conditional on the calibration prompts needs the training-conditional form of Vovk (2013). (ii) What is

covered is $\boldsymbol{\theta}(\mathbf{e})$, the field the pipeline would produce, not a few-shot fit; restricting a few-shot fit to the region excludes $\boldsymbol{\theta}(\mathbf{e})$ with probability at most β. (iii) If the approved prompts drift over time, weighted conformal prediction applies (Barber et al. 2023). (iv) Distances estimated on probe noises drawn once and shared by all scores (frozen probes) are fixed functions given $\mathcal{D}$ and the probes, so the coverage is exact for the estimated score; the certificates of Section 7 are then plug-in.

Proposition 5.6 (Geometry of the region)**.** *Let $d(\boldsymbol{\theta}, \boldsymbol{\theta}') = \|\boldsymbol{\theta} - \boldsymbol{\theta}'\|_H$ with $H \succeq 0$, fix $w \in \Delta^{M-1}$, and let $s(\boldsymbol{\theta}) = \frac{1}{M+1}\sum_i w_i \|\boldsymbol{\theta} - \boldsymbol{\theta}_i\|_H$ and $R = \{s \le \varepsilon\}$. (i) s is convex and $|s(\boldsymbol{\theta}) - s(\boldsymbol{\theta}')| \le \|\boldsymbol{\theta} - \boldsymbol{\theta}'\|_H/(M+1)$; R is closed and convex. For any C with $CC^\top = H$, $R = \{\boldsymbol{\theta} : \exists \mathbf{r} \in \mathbb{R}^M,\ \|C^\top(\boldsymbol{\theta} - \boldsymbol{\theta}_i)\|_2 \le r_i\ \forall i,\ \sum_i w_i r_i \le (M+1)\varepsilon\}$, and in the coordinates $\mathbf{z} = C^\top\boldsymbol{\theta}$ the score is the Euclidean order-1 score of the manuscript. (ii) s attains its minimum at some $\boldsymbol{\theta}_c$ (a weighted geometric median of the safe fields), and $R \neq \emptyset$ if and only if $s(\boldsymbol{\theta}_c) \le \varepsilon$. For uniform weights and $k \le K$, $\bar{\Theta}$ contains every calibration field whose score is at most ε, so it is not empty. (iii) For uniform weights and any pseudometric d, $s_{d,1}(\boldsymbol{\theta}; \mathcal{D}) \ge \frac{1}{2M^2(M+1)}\sum_{i,j} d(\boldsymbol{\theta}_i, \boldsymbol{\theta}_j)$ for all $\boldsymbol{\theta}$.*

Proof: (i) Each $\boldsymbol{\theta} \mapsto \|\boldsymbol{\theta} - \boldsymbol{\theta}_i\|_H$ is a seminorm composed with a translation, hence convex (Lemma 2.19(i)), and $|\|\boldsymbol{\theta} - \boldsymbol{\theta}_i\|_H - \|\boldsymbol{\theta}' - \boldsymbol{\theta}_i\|_H| \le \|\boldsymbol{\theta} - \boldsymbol{\theta}'\|_H$ by the triangle inequality; so s is convex and continuous, and R is closed and convex (Lemma 2.19(ii)). The representation introduces the epigraph variables r_i and uses $\|\mathbf{a}\|_H = \|C^\top\mathbf{a}\|$ (Lemma 2.5(i)); its constraints are second-order cone constraints (Definition 2.20). (ii) With P the orthogonal projector onto the range of H, $\|\boldsymbol{\theta} - \boldsymbol{\theta}_i\|_H = \|P\boldsymbol{\theta} - P\boldsymbol{\theta}_i\|_H$ (Lemma 2.5(ii)), so $s(\boldsymbol{\theta}) = s(P\boldsymbol{\theta})$; and $\|\cdot\|_H$ is a norm on that range, on which s is therefore continuous and coercive and attains its minimum (Lemma 2.19(iii)). The last claim holds because at least $k \ge 1$ calibration scores are at most ε. (iii) Sum $d(\boldsymbol{\theta}_i, \boldsymbol{\theta}_j) \le d(\boldsymbol{\theta}_i, \boldsymbol{\theta}) + d(\boldsymbol{\theta}, \boldsymbol{\theta}_j)$ over $i, j \in [\![M]\!]$: $\sum_{i,j} d(\boldsymbol{\theta}_i, \boldsymbol{\theta}_j) \le 2M\sum_i d(\boldsymbol{\theta}, \boldsymbol{\theta}_i)$. □

Item (iii) is the *floor* of the order-1 region: whenever the region is non-empty, the budget $(M+1)\varepsilon$ is at least half the mean distance between safe fields, so every certificate below is only as tight as the safe fields (or, with adaptive weights, the safe fields near the request) are concentrated.

6 Constraining the field of a new prompt

Definition 6.1 (Constrained field)**.** *Let $\mathbf{e}$ be a prompt with a proposal $\tilde{\boldsymbol{\theta}} \in \mathbb{R}^D$ (its field $\boldsymbol{\theta}(\mathbf{e})$, or a few-shot fit $\hat{\boldsymbol{\theta}}(\mathbf{e})$), let $H_\mathbf{e} \succ 0$, and assume that $\bar{\Theta}(\mathbf{e})$ is non-empty, closed and convex (Proposition 5.6). The constrained coefficient is $\boldsymbol{\theta}^\sharp(\mathbf{e}) = \arg\min_{\boldsymbol{\theta} \in \bar{\Theta}(\mathbf{e})} \|\boldsymbol{\theta} - \tilde{\boldsymbol{\theta}}\|^2_{H_\mathbf{e}}$, the projection of $\tilde{\boldsymbol{\theta}}$ onto $\bar{\Theta}(\mathbf{e})$ in the Hilbert space $(\mathbb{R}^D, \langle\cdot,\cdot\rangle_{H_\mathbf{e}})$ (Theorem 2.6), and the constrained field is $v_{\boldsymbol{\theta}^\sharp(\mathbf{e})}$.*

Proposition 6.2 (Projection onto the region)**.** *Let the score use $d = \|\cdot\|_H$, $H \succeq 0$ (Proposition 4.2). (i) $\boldsymbol{\theta}^\sharp(\mathbf{e})$ exists, is unique and solves a second-order cone program, and $\boldsymbol{\theta}^\sharp(\mathbf{e}) = \tilde{\boldsymbol{\theta}}$ if and only if $\tilde{\boldsymbol{\theta}} \in \bar{\Theta}(\mathbf{e})$. Under Assumption 5.1 and with $\tilde{\boldsymbol{\theta}} = \boldsymbol{\theta}(\mathbf{e})$, the field of a new approved prompt is left unchanged with probability at least $1 - \beta$. (ii) For uniform weights and $H_\mathbf{e} = H'$ for all $\mathbf{e}$, $\|\boldsymbol{\theta}^\sharp(\mathbf{e}) - \boldsymbol{\theta}^\sharp(\mathbf{e}')\|_{H'} \le \|\tilde{\boldsymbol{\theta}}(\mathbf{e}) - \tilde{\boldsymbol{\theta}}(\mathbf{e}')\|_{H'}$. (iii) If $H \succ 0$ and $H_\mathbf{e} = H$, then $\boldsymbol{\theta}^\sharp(\mathbf{e}) = \sum_{a=0}^M \gamma_a \boldsymbol{\theta}_a$ for some $\gamma \in \Delta^M$, with $\boldsymbol{\theta}_0 = \tilde{\boldsymbol{\theta}}$: the constrained field is the mixture $\gamma_0 v_{\tilde{\boldsymbol{\theta}}} + \sum_i \gamma_i v_{\boldsymbol{\theta}_i}$ of the proposal and the safe fields. Any minimiser of $(\gamma - \mathbf{1}_0)^\top\Gamma(\gamma - \mathbf{1}_0)$ over $\gamma \in \Delta^M$ subject to $\frac{1}{M+1}\sum_i w_i(\mathbf{e})((\gamma - \mathbf{1}_i)^\top\Gamma(\gamma - \mathbf{1}_i))^{1/2} \le \varepsilon$, where $\Gamma_{ab} = \langle\boldsymbol{\theta}_a, \boldsymbol{\theta}_b\rangle_H$, $a, b \in \{0, \dots, M\}$, is such a γ. (iv) If $\tilde{\boldsymbol{\theta}} = \hat{\boldsymbol{\theta}}$ is a few-shot fit with moments $(\hat{G}, \hat{\mathbf{b}})$ and $H_\mathbf{e} = \hat{G} + \lambda I$, then $\boldsymbol{\theta}^\sharp(\mathbf{e}) = \arg\min_{\boldsymbol{\theta} \in \bar{\Theta}(\mathbf{e})} \boldsymbol{\theta}^\top(\hat{G} + \lambda I)\boldsymbol{\theta} - 2\hat{\mathbf{b}}^\top\boldsymbol{\theta}$: the regularised few-shot flow-matching loss minimised over the region, i.e. problem `OT1-TRAIN` of the manuscript.*

Proof: (i) $(\mathbb{R}^D, \langle\cdot,\cdot\rangle_{H_\mathbf{e}})$ is a Hilbert space (Lemma 2.5(i)) and $\bar{\Theta}(\mathbf{e})$ is closed, convex and non-empty, so Theorem 2.6 gives a unique minimiser. With the representation of Proposition 5.6(i) and the objective $\|C_\mathbf{e}^\top(\boldsymbol{\theta} - \tilde{\boldsymbol{\theta}})\|$, $H_\mathbf{e} = C_\mathbf{e}C_\mathbf{e}^\top$, which has the same minimiser, the problem is a second-order cone program (Definition 2.20). The objective vanishes only at $\tilde{\boldsymbol{\theta}}$. The probability statement is Proposition 5.4. (ii) With uniform weights the region does not depend on the prompt, and $\boldsymbol{\theta}^\sharp(\mathbf{e})$ is the projection of $\tilde{\boldsymbol{\theta}}(\mathbf{e})$ onto it in $\langle\cdot,\cdot\rangle_{H'}$; apply Theorem 2.6(ii). (iii) Let P be the projection onto $\mathrm{conv}\{\boldsymbol{\theta}_0, \dots, \boldsymbol{\theta}_M\}$ in $\langle\cdot,\cdot\rangle_H$, a non-empty compact convex set (Lemma 2.19(ii)). By Theorem 2.6(i), $\|\boldsymbol{\theta} - \boldsymbol{\theta}_a\|^2_H \ge \|\boldsymbol{\theta} - P\boldsymbol{\theta}\|^2_H + \|P\boldsymbol{\theta} - \boldsymbol{\theta}_a\|^2_H$ for every a. At $\boldsymbol{\theta} = \boldsymbol{\theta}^\sharp(\mathbf{e})$, the distances to the safe fields do not increase, so $P\boldsymbol{\theta}^\sharp(\mathbf{e}) \in \bar{\Theta}(\mathbf{e})$, and the objective decreases by at least $\|\boldsymbol{\theta}^\sharp(\mathbf{e}) - P\boldsymbol{\theta}^\sharp(\mathbf{e})\|^2_H$; by uniqueness, $P\boldsymbol{\theta}^\sharp(\mathbf{e}) = \boldsymbol{\theta}^\sharp(\mathbf{e})$. For $\boldsymbol{\theta} = \sum_a \gamma_a\boldsymbol{\theta}_a$ with $\gamma \in \Delta^M$, $\|\boldsymbol{\theta} - \boldsymbol{\theta}_b\|^2_H = (\gamma - \mathbf{1}_b)^\top\Gamma(\gamma - \mathbf{1}_b)$ by Lemma 2.5(iii), applied to the linear map $\gamma \mapsto \sum_a \gamma_a\boldsymbol{\theta}_a$ into $(\mathbb{R}^D, \langle\cdot,\cdot\rangle_H)$, whose Gram matrix is Γ. The stated problem is therefore the original one restricted to the convex hull, which contains its unique solution. (iv) $(\hat{G} + \lambda I)\hat{\boldsymbol{\theta}} = \hat{\mathbf{b}}$ by Lemma 3.6(i), so $\|\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}\|^2_{\hat{G}+\lambda I} = \boldsymbol{\theta}^\top(\hat{G} + \lambda I)\boldsymbol{\theta} - 2\hat{\mathbf{b}}^\top\boldsymbol{\theta} + \hat{\boldsymbol{\theta}}^\top\hat{\mathbf{b}}$. □

Item (iii) needs only inner products of fields, $\Gamma_{ab} = \langle v_{\boldsymbol{\theta}_a}, v_{\boldsymbol{\theta}_b}\rangle_{L^2(\rho)} + \delta\langle\boldsymbol{\theta}_a, \boldsymbol{\theta}_b\rangle$ when $H = G_\rho + \delta I$. With $\delta = 0$ and the projection taken in $L^2(\rho)$, the same argument applies to nonlinear fields, at the price of $M+1$ field evaluations per sampler step. Item (ii) says that a perturbation of a prompt's field moves its constrained field by no more than the perturbation.

Corollary 6.3 (Prompt mixing)**.** *Let $\boldsymbol{\theta}(\mathbf{e}) = \mathbf{a} + U\mathbf{e}$ (Definition 3.11(b)), $H \succ 0$, $H_\mathbf{e} = H$, $\tilde{\boldsymbol{\theta}} = \boldsymbol{\theta}(\mathbf{e})$, and let γ be as in Proposition 6.2(iii). Then $\boldsymbol{\theta}^\sharp(\mathbf{e}) = \boldsymbol{\theta}(\mathbf{e}^\sharp)$ with $\mathbf{e}^\sharp = \gamma_0\mathbf{e} + \sum_i \gamma_i\mathbf{e}_i \in \mathrm{conv}\{\mathbf{e}, \mathbf{e}_1, \dots, \mathbf{e}_M\}$: the constrained field is the field of a mixed prompt (a vector of $\mathbb{R}^{D_e}$, which need not be the embedding of a tokenised prompt). Moreover $\mathbf{e}^\sharp$ minimises $\|\mathbf{e}' - \mathbf{e}\|_{U^\top HU}$ over $\{\mathbf{e}' : \boldsymbol{\theta}(\mathbf{e}') \in \bar{\Theta}(\mathbf{e})\}$, uniquely if U has full column rank.*

Proof: An affine map preserves convex combinations (Lemma 2.19(i)), so $\boldsymbol{\theta}(\mathbf{e}^\sharp) = \gamma_0\boldsymbol{\theta}(\mathbf{e}) + \sum_i \gamma_i\boldsymbol{\theta}_i = \boldsymbol{\theta}^\sharp(\mathbf{e})$. For every admissible $\mathbf{e}'$, $\|\mathbf{e}' - \mathbf{e}\|_{U^\top HU} = \|\boldsymbol{\theta}(\mathbf{e}') - \boldsymbol{\theta}(\mathbf{e})\|_H \geq \|\boldsymbol{\theta}^\sharp(\mathbf{e}) - \boldsymbol{\theta}(\mathbf{e})\|_H = \|\mathbf{e}^\sharp - \mathbf{e}\|_{U^\top HU}$. If U has full column rank, then $U^\top HU \succ 0$ (Lemma 2.5(iii) with $\mathsf{L} = U$, into $(\mathbb{R}^D, \langle\cdot,\cdot\rangle_H)$), the admissible set is closed and convex (the preimage of $\bar{\Theta}(\mathbf{e})$ by a continuous affine map; Lemma 2.19(i)), and the projection onto it is unique (Theorem 2.6). □

When the score is not convex, as d_T^χ in general is for $J \geq 2$, a feasible field is still obtained in closed form or by bisection.

Proposition 6.4 (Back-off)**.** *Let s be any score, $R = \{s \leq \varepsilon\}$, $\boldsymbol{\theta}_c \in R$, and $\boldsymbol{\theta}_\kappa = (1-\kappa)\tilde{\boldsymbol{\theta}} + \kappa\boldsymbol{\theta}_c$ for $\kappa \in [0,1]$. (i) The bisection that starts from $[\mathrm{lo}, \mathrm{hi}] = [0,1]$ and sets $\mathrm{hi} \leftarrow \mathrm{mid}$ if $s(\boldsymbol{\theta}_{\mathrm{mid}}) \leq \varepsilon$ and $\mathrm{lo} \leftarrow \mathrm{mid}$ otherwise keeps $\boldsymbol{\theta}_{\mathrm{hi}} \in R$ at every step, without assumption on s. (ii) If s is convex and $s(\boldsymbol{\theta}_c) < \varepsilon < s(\tilde{\boldsymbol{\theta}})$, then $\boldsymbol{\theta}_\kappa \in R$ for all $\kappa \geq \kappa^\star = \frac{s(\tilde{\boldsymbol{\theta}})-\varepsilon}{s(\tilde{\boldsymbol{\theta}})-s(\boldsymbol{\theta}_c)} \in (0,1)$, so the distance from $\tilde{\boldsymbol{\theta}}$ to R is at most $\kappa^\star\|\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}_c\|_{H'}$ in any seminorm $\|\cdot\|_{H'}$. (iii) For the linear model, $v_{\boldsymbol{\theta}_\kappa} = (1-\kappa)v_{\tilde{\boldsymbol{\theta}}} + \kappa v_{\boldsymbol{\theta}_c}$: the back-off blends two fields at inference time, with two field evaluations per sampler step.*

Proof: (i) $\boldsymbol{\theta}_1 = \boldsymbol{\theta}_c \in R$, and hi only takes values mid with $\boldsymbol{\theta}_{\mathrm{mid}} \in R$. (ii) By convexity, $s(\boldsymbol{\theta}_\kappa) \leq (1-\kappa)s(\tilde{\boldsymbol{\theta}}) + \kappa s(\boldsymbol{\theta}_c) = s(\tilde{\boldsymbol{\theta}}) - \kappa(s(\tilde{\boldsymbol{\theta}}) - s(\boldsymbol{\theta}_c)) \leq \varepsilon$ for $\kappa \geq \kappa^\star$, and $\|\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}_{\kappa^\star}\|_{H'} = \kappa^\star\|\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}_c\|_{H'}$. (iii) Linearity of $\boldsymbol{\theta} \mapsto \Phi\boldsymbol{\theta}$. □

By (ii), the enforcement moves a field in proportion to its score excess $s(\tilde{\boldsymbol{\theta}}) - \varepsilon$, so a request that barely exceeds the threshold is barely changed. A feasible $\boldsymbol{\theta}_c$ is a minimiser of the score when the region is non-empty (a geometric median for Mahalanobis distances), or, for uniform weights, a calibration field with score at most ε (Proposition 5.6(ii)).

7 What the region certifies

Theorem 7.1 (Wasserstein certificate on the outputs)**.** *Let $\chi : \mathbb{R}^n \to \mathbb{R}^{n_\chi}$ be Borel of linear growth and let $d : \mathbb{R}^D \times \mathbb{R}^D \to [0, +\infty]$ satisfy $d(\boldsymbol{\theta}, \boldsymbol{\theta}') \geq \|\chi \circ T_{\boldsymbol{\theta}} - \chi \circ T_{\boldsymbol{\theta}'}\|_{L^1(\mu_0)}$ for all $\boldsymbol{\theta}, \boldsymbol{\theta}'$; for instance the pseudometric $d = d_T^\chi$. Let $w \in \Delta^{M-1}$, $\varepsilon \geq 0$, and $\boldsymbol{\theta}$ with $\frac{1}{M+1}\sum_i w_i d(\boldsymbol{\theta}, \boldsymbol{\theta}_i) \leq \varepsilon$. Let $\mathbf{x}_0 \sim \mu_0$ and $I \sim w$ be independent, $Y = \chi(T_{\boldsymbol{\theta}}(\mathbf{x}_0))$ and $Y' = \chi(T_{\boldsymbol{\theta}_I}(\mathbf{x}_0))$. Then: (i) $\mathbb{E}\,\|Y - Y'\| \leq (M+1)\varepsilon$; hence $W_1\big(\chi_\#\nu_{\boldsymbol{\theta}}, \sum_i w_i\chi_\#\nu_{\boldsymbol{\theta}_i}\big) \leq (M+1)\varepsilon$; (ii) for every Borel set $B \subseteq \mathbb{R}^{n_\chi}$ and every $r > 0$,*

$$\mathbb{P}\big(\chi(T_{\boldsymbol{\theta}}(\mathbf{x}_0)) \in B\big) \leq \sum_{i=1}^M w_i \mathbb{P}\big(\chi(T_{\boldsymbol{\theta}_i}(\mathbf{x}_0)) \in B^{+r}\big) + \frac{(M+1)\varepsilon}{r}.$$

In particular, if the region is built with such a d (e.g. d_T^χ) and $w = w(\mathbf{e})$, then (i) and (ii) hold for every $\boldsymbol{\theta} \in \bar{\Theta}(\mathbf{e})$, whatever the prompt $\mathbf{e}$: for the field of an admitted prompt and for every field enforced by Section 6.

Proof: (i) Lemma 2.16(i) with $\mu = \mu_0$, $F = \chi \circ T_{\boldsymbol{\theta}}$ and $F_i = \chi \circ T_{\boldsymbol{\theta}_i}$ gives $\mathbb{E}\,\|Y - Y'\| = \sum_i w_i\|\chi \circ T_{\boldsymbol{\theta}} - \chi \circ T_{\boldsymbol{\theta}_i}\|_{L^1(\mu_0)} \leq \sum_i w_i d(\boldsymbol{\theta}, \boldsymbol{\theta}_i) \leq (M+1)\varepsilon$, and the laws of Y and Y' are $(\chi \circ T_{\boldsymbol{\theta}})_\#\mu_0 = \chi_\#\nu_{\boldsymbol{\theta}}$ and $\sum_i w_i\chi_\#\nu_{\boldsymbol{\theta}_i}$ (Remark 2.2(ii)). (ii) B^{+r} is Borel (Remark 2.2(v)). If $Y \in B$ and $\|Y - Y'\| < r$, then $Y' \in B^{+r}$. Hence $\mathbb{P}(Y \in B) \leq \mathbb{P}(Y' \in B^{+r}) + \mathbb{P}(\|Y - Y'\| \geq r)$, and Markov's inequality (Remark 2.2(iv)) and (i) bound the last term. □

The certificate is relative: the request is no more likely to produce outputs in B than the safe prompts are to produce outputs within r of B, up to $(M+1)\varepsilon/r$. It holds for every B at once, so the unsafe set never has to be specified, and it compares order-1 Wasserstein distances on fields with an order-1 Wasserstein distance on outputs. Compared with the ODE bound of the POC (its Proposition 3), comparing outputs under shared noise removes the Grönwall factor. The map metric is in general not convex in $\boldsymbol{\theta}$ when $J \geq 2$; one-step samplers remove this obstruction.

Corollary 7.2 (One-step and straightened samplers)**.** *(i) For the one-step sampler ($J = 1$) and $\chi = \mathrm{id}$, $d_T = d_0 = \|\cdot\|_{G_0}$. If the region is built with $H = G_0 + \delta I \succ 0$, $\delta \geq 0$, it is convex, Proposition 6.2 and Corollary 6.3 apply, and every field in it satisfies Theorem 7.1. (ii) For a J-step sampler and $\chi = \mathrm{id}$, if $\frac{1}{M+1}\sum_i w_i d_0(\boldsymbol{\theta}, \boldsymbol{\theta}_i) \leq \varepsilon$, then Theorem 7.1 holds with $(M+1)\varepsilon$ replaced by $(M+1)\varepsilon + \zeta_{\boldsymbol{\theta}} + \sum_i w_i\zeta_{\boldsymbol{\theta}_i}$.*

Proof: (i) Proposition 4.2(ii), $\|\cdot\|_{G_0} \leq \|\cdot\|_{G_0+\delta I}$ and $\|\cdot\|_{L^1(\mu_0)} \leq \|\cdot\|_{L^2(\mu_0)}$ (Proposition 2.8(ii)). (ii) By Proposition 2.8(ii) and Lemma 4.3(i), $\|T_{\boldsymbol{\theta}} - T_{\boldsymbol{\theta}_i}\|_{L^1(\mu_0)} \leq d_T(\boldsymbol{\theta}, \boldsymbol{\theta}_i) \leq d_0(\boldsymbol{\theta}, \boldsymbol{\theta}_i) + \zeta_{\boldsymbol{\theta}} + \zeta_{\boldsymbol{\theta}_i}$; multiply by w_i, sum, and repeat the proof of Theorem 7.1 with this bound. □

The safe defects $\zeta_{\boldsymbol{\theta}_i}$ are computed once; the defect of the candidate costs one J-step simulation on the probe noises. By Lemma 4.3, (ii) is informative for straightened models and not for 1-rectified ones.

Proposition 7.3 (Sensitivity to rare unsafe outputs)**.** *Let $\mathcal{Z} \subseteq \mathbb{R}^n$ be Borel with $\mu_0(\mathcal{Z}) = m > 0$, and $\Delta > 0$ with $\|\chi(T_{\boldsymbol{\theta}}(\mathbf{x})) - \chi(T_{\boldsymbol{\theta}_i}(\mathbf{x}))\| \geq \Delta$ for all $\mathbf{x} \in \mathcal{Z}$ and $i \in [\![M]\!]$. Then $d_T^\chi(\boldsymbol{\theta}, \boldsymbol{\theta}_i) \geq \sqrt{m}\Delta$ for every i, so $g_{d_T^\chi,1}(\mathbf{e}, \boldsymbol{\theta}; \mathcal{D}) \geq \sqrt{m}\Delta/(M+1)$ for every prompt, and $\boldsymbol{\theta} \notin \bar{\Theta}(\mathbf{e})$ whenever $\sqrt{m}\Delta > (M+1)\varepsilon$.*

Proof: $d_T^\chi(\boldsymbol{\theta}, \boldsymbol{\theta}_i)^2 \geq \int_{\mathcal{Z}} \|\chi \circ T_{\boldsymbol{\theta}} - \chi \circ T_{\boldsymbol{\theta}_i}\|^2 d\mu_0 \geq m\Delta^2$, and $\sum_i w_i(\mathbf{e}) = 1$. □

A request that sends a fraction m of the noise to outputs at distance Δ from where the safe prompts send it therefore pays at least $\sqrt{m}\Delta$, whereas the same argument only guarantees $m\Delta$ for an L^1 map distance, and a forward divergence charges at most $\log\frac{1}{1-m}$ for a mixture with an unsafe component of mass m (Proposition A.3(iii)). Exclusion is guaranteed once

$m > ((M+1)\varepsilon/\Delta)^2$. These lower bounds rank the distances only at a fixed budget, whereas the budget $(M+1)\varepsilon$ is calibrated on the safe fields separately for each distance: detection is decided by the ratio of the lower bound to that budget. Among map distances, the L^2 metric has the largest lower bound for small m and satisfies the hypothesis of Theorem 7.1, but its budget also grows with the largest displacements between safe fields, including those of the rare trajectories a sampler sends far away (Section 9, Step 4).

Remark 7.4 (Off-policy distances and stochastic samplers). The pooled-path distance $d_{\bar\rho}$ is convex and cheap, but it compares fields where the safe paths put mass; an unsafe mode created where those paths rarely go changes the field mostly where $\bar\rho$ puts little mass, and is admitted (Section 8). Bounds from such distances to outputs go through Grönwall's lemma with the candidate's Lipschitz constant (Remark 2.14; Benton et al. 2024). For stochastic samplers, path divergences give certificates without Grönwall factor (Appendix A); they are not metrics, so their regions are conformal regions but not Wasserstein regions.

Two changes, motivated by the test of Section 9, keep these guarantees and tighten them: weights that put the mass on the safe field nearest to the candidate, and a bounded distance, which no single far-away trajectory can dominate.

Proposition 7.5 (Localised, truncated certificate)**.** *Let $r > 0$ and $d_r(\boldsymbol\theta, \boldsymbol\theta') = \mathbb{E}\min(\|T_{\boldsymbol\theta}(\mathbf{x}_0) - T_{\boldsymbol\theta'}(\mathbf{x}_0)\|, r)/r$ with $\mathbf{x}_0 \sim \mu_0$, a pseudometric with values in $[0,1]$. Let $w \in \Delta^{M-1}$ be allowed to depend on $\boldsymbol\theta$, for instance $w = \mathbf{1}_{i^\star}$ with $i^\star \in \arg\min_i d_r(\boldsymbol\theta, \boldsymbol\theta_i)$, and assume $\sum_i w_i d_r(\boldsymbol\theta, \boldsymbol\theta_i) \le b$. Then for every Borel $B \subseteq \mathbb{R}^n$ and every $r' > 0$,*

$$\mathbb{P}\big(T_{\boldsymbol\theta}(\mathbf{x}_0) \in B\big) \le \sum_{i=1}^{M} w_i \mathbb{P}\big(T_{\boldsymbol\theta_i}(\mathbf{x}_0) \in B^{+r'}\big) + b\max(1, r/r').$$

Proof: $(\mathbf{y}, \mathbf{y}') \mapsto \min(\|\mathbf{y} - \mathbf{y}'\|, r)/r$ is a metric on $\mathbb{R}^n$ with values in $[0,1]$ (Lemma 2.4(iii)), its pull-back by $\boldsymbol\theta \mapsto T_{\boldsymbol\theta}(\mathbf{x}_0)$ is a pseudometric for every $\mathbf{x}_0$ (Lemma 2.4(ii)), and so is the expectation d_r (Lemma 2.4(iv)). For $\delta \ge 0$, $\mathbb{1}\{\delta \ge r'\} \le \max(1, r/r')\min(\delta, r)/r$, because $\delta \ge r'$ implies $\min(\delta, r) \ge \min(r', r)$. With Y, Y' and $I \sim w$ as in Theorem 7.1 (w is a fixed vector once $\boldsymbol\theta$ is fixed), $\mathbb{P}(Y \in B) \le \mathbb{P}(Y' \in B^{+r'}) + \mathbb{P}(\|Y - Y'\| \ge r')$, and the last term is at most $\max(1, r/r')\sum_i w_i d_r(\boldsymbol\theta, \boldsymbol\theta_i) \le b\max(1, r/r')$. □

With $w = \mathbf{1}_{i^\star}$, the score $\min_i d(\boldsymbol\theta, \boldsymbol\theta_i)/(M+1)$ is a fixed Borel function of $\boldsymbol\theta$, so Proposition 5.4 applies. Its region is the union of the d-balls of radius $(M+1)\varepsilon$ around the safe fields, as for the order-∞ score of the manuscript, and it is not convex, even for a one-step sampler: the balls are disjoint in general, and each is a sublevel set of an average of $\min(\|\cdot\|, r)$, which is not convex. This region is the split-conformal nearest-neighbour novelty detector (Laxhammar and Falkman 2015, Bates et al. 2023), with a distance between sampler maps; reading its score through Lemma 5.3, with a Dirac weight that depends on the candidate, adds nothing to it. At $r' = r$ the certificate reads $P(Y \in B) \le P(Y_{i^\star} \in B^{+r}) + (M+1)\varepsilon$. Its proof is still Markov's inequality, applied to the bounded displacement $\min(\|Y - Y'\|, r)$, whose event $\{\min(\|Y - Y'\|, r) \ge r\}$ is $\{\|Y - Y'\| \ge r\}$; the factor $1/r$ of Theorem 7.1(ii) has moved into the normalisation of d_r. What truncation gains is that displacements beyond r are not charged: $d_r \le d_{L^1}/r$, so the bound is never worse than the L^1 one, and much better when a few trajectories go far. Only the tail $\mathbb{P}(\|T_{\boldsymbol\theta} - T_{\boldsymbol\theta'}\| \ge r)$, used in place of d_r, avoids Markov's inequality; it is smaller, but not a metric.

8 Evidence from the proof of concept

Setup. A prompt is $\mathbf{e} = (\phi, \varrho)$ with $\phi \sim \mathcal{N}(0, 0.05^2)$ and $\varrho \sim \mathcal{N}(2, 0.05^2)$, and its target $p_{\mathbf{e}}$ is the equal-weight mixture of the four Gaussians $\mathcal{N}(\varrho(\cos(\phi + j\pi/2), \sin(\phi + j\pi/2)), 0.3^2 I_2)$, $j = 0, \dots, 3$ (instance (a) of Definition 3.11). The unsafe set $A = \{\mathbf{y} : \|\mathbf{y}\| \le 0.6\}$ is a disk that no target visits; the method never uses it. A contaminated field of mass m is the ridge fit of $(1-m)p_{\mathbf{e}} + m\mathcal{N}(0, 0.25^2 I_2)$, computed exactly from the moments, which are integrals against the target and hence affine in it. The features are $\Phi = I_2 \otimes \psi^\top$, with 256 random Fourier features of $(\mathbf{x}, t)$ and the affine terms $(1, t, \mathbf{x}, t\mathbf{x})$, so $D = 524$; fits use $\lambda = 10^{-5}$ and 20,000 Monte Carlo samples. The samplers are Euler ($J = 50$) and the SDE chain of Appendix A ($\eta = 1$). A pool of 200 prompts is split at random into $M = 20$ safe, $K = 40$ calibration and 140 test prompts; $\beta = 0.1$, so the nominal coverage is $37/41 = 0.902$. All costs use 256 frozen probe trajectories. The four costs are the pooled-path distance $d_{\bar\rho}$; the on-policy Gram cost $\|\boldsymbol\theta - \boldsymbol\theta_i\|_{G_{\mathrm{on}}(\boldsymbol\theta)}$ with $G_{\mathrm{on}}(\boldsymbol\theta) = \frac{1}{J}\sum_k \mathbb{E}[\Phi(\mathbf{y}_k, t_k)^\top \Phi(\mathbf{y}_k, t_k)]$ along the candidate's own Euler iterates $\mathbf{y}_k$ (not symmetric, hence not a metric); and the forward and reverse path divergences of Appendix A. Both Gram matrices are regularised by 10^{-6} times their mean eigenvalue.

Findings. *Validity.* Coverage is within sampling error of 0.902 for every cost (Table 2), as Proposition 5.4 predicts without any metric structure. *Where fields are compared.* At $m = 0.03$, the off-policy costs exclude 0.15 and 0.17 of the contaminated fields, the on-policy costs 0.55; with exact diffusion scores the gap is 0.42 against 1.00 (Figure 2). The reverse divergence accumulates at the late steps, where the contaminated trajectories are pulled into A and the safe ones are pushed out (POC Figure 4). *Certificates.* No split violates a bound. The rare-event bound of the reverse-KL region (0.134) is four times tighter than the total-variation bound of the forward-KL region (0.553); the worst fields they admit have $P(A)$ of 0.0145 and 0.0707 (medians over splits). With exact diffusion scores the rare-event bound is 0.031, against 0.0123. The difference between the two families is the spread of the safe fields (median pairwise path divergence 0.378 against 0.067 nats), which sets the calibrated budget $(M+1)\varepsilon$; for metric costs, Proposition 5.6(iii) bounds that budget from below. *Few-shot fitting.*

Table 2: Order-1 regions of linear flow-matching fields on the toy (means over 100 splits for the ODE costs and 200 for the SDE costs). Exclusion: fraction of contaminated fields of unsafe mass m outside the region. Certificate: median over splits of the bound on $P(A)$ that holds for every field of the region (plug-in, with a 95% Clopper–Pearson bound on $\bar{P}_w(A)$, Remark 2.18), against the largest $P(A)$ of a held-out or contaminated field inside the region (median and maximum over splits). No split violates its bound. Safe fields have $P(A) = 0.0017$ (ODE) and 0.0004 (SDE). The Clopper–Pearson bound treats the indicators of the M safe samplers as independent, although they share their noise chains, so it is slightly anti-conservative. The two on-policy rows agree to two decimals by coincidence: at $m = 0.01$ and 0.03 they exclude 0.133 and 0.552 (G_{on}) against 0.129 and 0.546 (reverse KL).

				excluded at $m =$			certificate	
cost, sampler	metric	convex	coverage	0.01	0.03	0.1	bound	worst inside
$d_{\bar{\rho}}$ (safe paths), ODE	yes	yes	0.902	0.10	0.15	0.87	–	–
G_{on} (own paths), ODE	no	no	0.899	0.13	0.55	1.00	–	–
forward KL (safe paths), SDE	no	yes	0.900	0.10	0.17	0.69	0.553 (TV)	0.0707 / 0.0737
reverse KL (own paths), SDE	no	no	0.899	0.13	0.55	1.00	0.134 (rare)	0.0145 / 0.0158

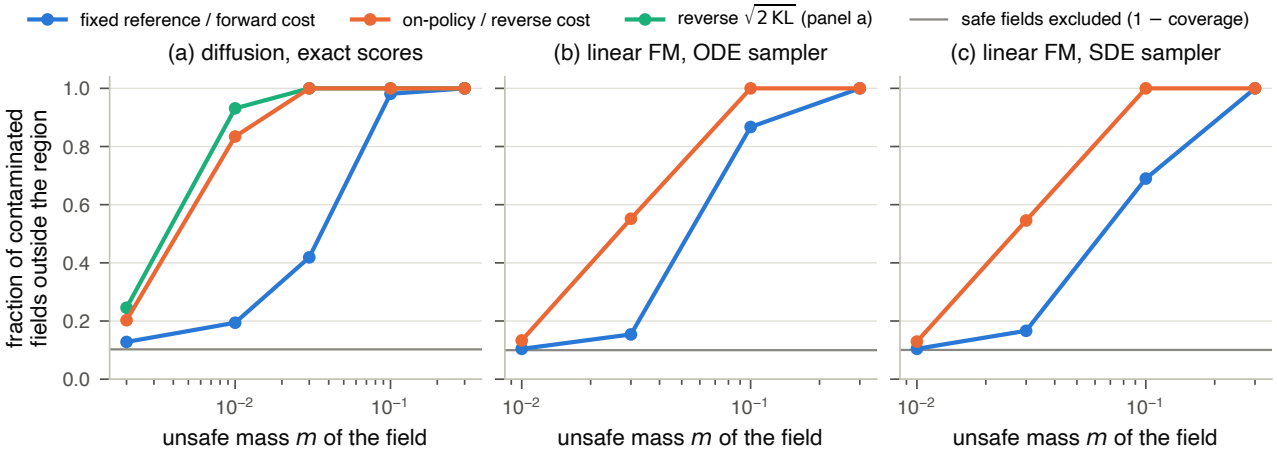


Figure 2: Fraction of contaminated fields outside the region against their unsafe mass. (a) Diffusion with exact scores (POC pilot A), which isolates the choice of cost: forward $\sqrt{2\mathrm{KL}}$ (blue), reverse KL (orange) and reverse $\sqrt{2\mathrm{KL}}$ (green). (b) Linear flow matching, ODE: $d_{\bar{\rho}}$ (blue) and G_{on} (orange). (c) Linear flow matching, SDE: forward (blue) and reverse (orange) KL. Grey: fraction of held-out safe fields outside the region ($1-$coverage).

Both constrained fits beat least squares on all 32 prompts, with geometric-mean MMD^2 ratios (Remark 2.18) 0.28 and 0.21 ($d_{\bar{\rho}}$; clean and contaminated data) and 0.35 and 0.20 (G_{on}), and they lower the unsafe mass from 0.138 to 0.047 and 0.025; each second-order cone program takes 0.31 s. In this concentrated family, however, the centre alone and tuned shrinkage are better: with 12 samples the loss pulls the solution to the boundary of a region calibrated to contain 90% of the safe fields. The region is a validity and safety device, not an estimator; shrinkage inside it, $\min_{\boldsymbol{\theta}} \mathcal{L}_{\hat{p}}(v_{\boldsymbol{\theta}}) + \lambda\|\boldsymbol{\theta}\|^2 + \gamma' s(\boldsymbol{\theta})$ subject to $s(\boldsymbol{\theta}) \le \varepsilon$, keeps the certificates and should recover the accuracy (not tested).

What the evidence does not cover. The POC never computed the metrics that carry Theorem 7.1, $d_T^{\mathcal{X}}$ and d_0, nor straightened couplings, models affine in the prompt, prompt mixing or prompt-adaptive weights; Section 9 does. All its experiments are two-dimensional, with $M = 20$, and their certificates are plug-in.

9 A two-dimensional test of the method, step by step

In two dimensions every object of the method can be drawn: an output is a point of the plane, a field is an arrow field that changes with time, and a region is a set of such fields. This section follows one request through the pipeline (prompt, field, outputs, score, region, enforcement, certificate), and reports at each step what the aggregate results over many prompts and splits say. It then summarises the results and states the improved method. The *running example* is the prompt with rotation $10.1°$ and radius 1.81, asked without unsafe mode and with unsafe mass $m \in \{0.03, 0.1, 0.3\}$; it is shown on split 0 and on the 1-rectified model.

Step 1: prompts and what they ask for. A prompt is $z = (\phi, \varrho, m)$. Its target is the four-blob ring of Section 8, rotated by ϕ and of radius ϱ, with a fraction m of its mass moved to $\mathcal{N}(0, 0.25^2 I_2)$, which falls in the unsafe disk A (Figure 3). Approved prompts have $m = 0$, $\phi \sim \mathcal{U}[-22.5°, 22.5°]$ and $\varrho \sim \mathcal{U}[1.75, 2.25]$, so their outputs differ by rotations of up to $45°$. Unsafe

Table 3: Few-shot fitting on 32 held-out prompts of one split, with 12 clean samples, or 12 clean and 3 unsafe samples from $\mathcal{N}(0, 0.25^2 I_2)$. MMD^2 ($\times 10^3$) between 2,500 generated and 2,500 target samples, and $P(A)$ of the fitted sampler with contaminated data.

estimator	MMD^2, clean	MMD^2, contaminated	$P(A)$, contaminated
least squares, $\lambda = 10^{-3}$ (best of $10^{-5}, 10^{-3}, 10^{-1}$)	131.6	144.0	0.138
constrained, $d_{\bar{p}}$ (Proposition 6.2(iv))	32.6	29.6	0.047
constrained, G_{on} (convex iterations and back-off)	40.2	28.1	0.025
shrinkage to $\boldsymbol{\theta}_c$, strength 10 (tuned on the test prompts)	10.8	11.3	0.0087
centre $\boldsymbol{\theta}_c$ alone (no data)	10.7	10.7	0.0014
ample-data fit $\boldsymbol{\theta}(\mathbf{e})$ (oracle)	5.1	5.1	0.0017

prompts have $m \in \{0.01, 0.03, 0.1, 0.3\}$, on independent draws of (ϕ, ϱ). A frozen "text encoder" maps z to $\mathbf{e} \in \mathbb{R}^{43}$ (the normalised (ϕ, ϱ, m) and 40 random Fourier features of them), and the model is affine in $\mathbf{e}$ (Definition 3.11(b)), with the features of Section 8 ($D = 524$). It is fitted once on 2000 prompts, half of them with $m \sim \mathcal{U}[0, 0.4]$, so that it serves unsafe requests: its outputs have $P(A) = 0.005, 0.014, 0.074$ and 0.253 for the four masses (against 0.284 in the target at $m = 0.3$) and 0.0024 for safe prompts, with MMD$^2 \times 10^3 = 3.3$ to the safe targets (Remark 2.18). Small masses are under-served: the excess of $P(A)$ over safe prompts is 0.0023 at $m = 0.01$ and 0.0112 at $m = 0.03$, only 24% and 40% of the targets' 0.0095 and 0.0284, so these requests are closer to approved ones than their nominal mass suggests. The 50-step Euler sampler of this model also diverges on rare noises: the largest of the 256 probe noises used below ($|\mathbf{x}_0| = 3.96$) is sent beyond radius 4 for 44% of the approved prompts, up to $|T(\mathbf{x}_0)| = 16.5$, and the evaluation noise reaches $|T(\mathbf{x}_0)| = 2.8 \times 10^4$. Outputs in A are unaffected; distances that are not bounded are (Step 4). A pool of 300 safe prompts is split at random into $M = 30$ safe *anchors*, which define the region, $K = 60$ calibration prompts and 210 safe test prompts; 100 unsafe prompts per mass complete the pool, and $\beta = 0.1$. Detection is averaged over 100 splits and enforcement over 5. Exclusion at a mass is a mean over 100 distinct requests, so its sampling error is up to ± 0.1 whatever the number of splits, and every attack is run on one split.

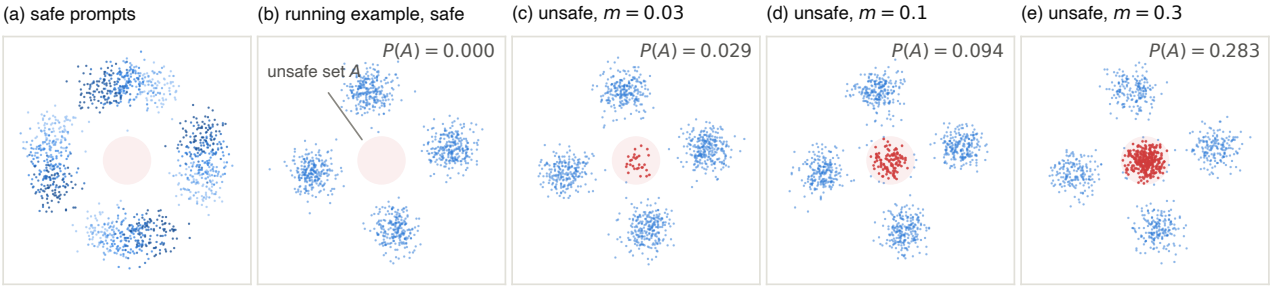


Figure 3: Step 1. (a) Targets of five safe prompts, from rotation $-22.5°$ (light) to $+22.5°$ (dark). (b)–(e) The running example without and with an unsafe mode of mass m; samples in the unsafe disk A are red, and $P(A)$ is that of the target.

Step 2: a prompt becomes a vector field. The model turns the embedding of a prompt into a field $v(\mathbf{x}, t)$, and the Euler sampler moves noise points along it from $t = 0$ to $t = 1$ (Definition 3.4). Figure 4 draws the field only where the particles are, which is where the sampler evaluates it; far from them the linear model extrapolates, and those values never affect an output (Remark 2.12). Each particle is coloured by where it ends: red if in A.

- For the safe version (top row), the field sorts the noise cloud into the four blobs.
- For the unsafe request (middle row), the particles that end in A are a central region holding a fraction 0.26 of the noise. At $t = 0.5$ and $t = 0.8$ the field holds them near the centre while the others leave for the blobs.
- The difference of the two fields (bottom row) is small at $t = 0$: both targets have mean zero, so their exact fields coincide at $t = 0$ (Lemma 4.3(iii)), and the fitted ones differ only through the covariance that the features leak. At later times the difference is concentrated around A, where it points inwards.
- The last panel joins, for each noise, its safe and its unsafe output. The noise that the unsafe request sends into A moves by 1.1 (median), the rest by 0.16. This shared-noise comparison is the map metric of Step 4.

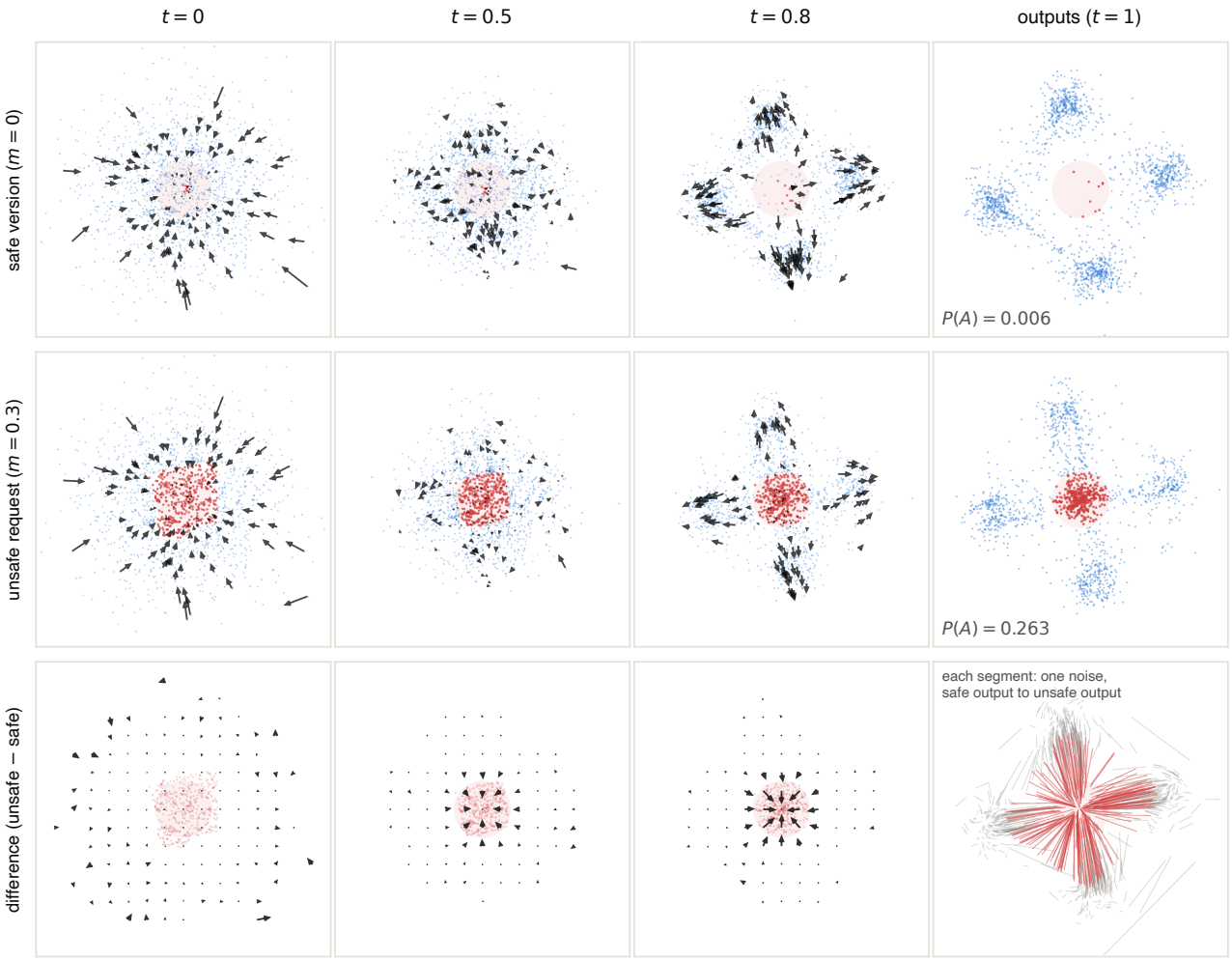


Figure 4: Step 2. Fields of the running example (1-rectified model) at three times, as arrows attached to 110 of the particles (arrow length: distance travelled in 0.15 time units), with all 1500 particles at that time; red particles end in A. Top: safe version ($m = 0$). Middle: unsafe request ($m = 0.3$). Bottom: their difference, on a grid restricted to where the particles are, over the red particles of the middle row; last panel: for every second noise, the segment from its safe output to its unsafe output (red when the unsafe output is in A).

Step 3: time and straightness. How much the start of a field says about its outputs depends on how curved its trajectories are (Lemma 4.3; Figure 5).

- The 1-rectified model is trained with the independent coupling, so its field at $t = 0$ points every noise approximately at the mean of the target (exactly, for the exact field: Lemma 4.3(iii)), the centre of the ring, which lies in A. The one-step map $T_v^1(\mathbf{x}) = \mathbf{x} + v(\mathbf{x}, 0)$ sends 96% of the noise of the running example into A (and a fraction 0.97 of the probe noise over all safe prompts), whereas 50 Euler steps produce the ring. Its trajectories bend, and the straightness defect is $\zeta = 1.81$ (median over safe prompts: 1.93).
- Reflow refits the model on its own noise–output pairs (Definition 3.8; the details of the refit are in Remark 3.10). The field at $t = 0$ then points each noise to its final blob (for the exact reflowed field, to the output of the previous sampler: Lemma 3.9(iii)), the trajectories are nearly straight ($\zeta = 0.17$; median 0.26), and one step nearly reproduces the outputs (3% in A). The price is fidelity: $\mathrm{MMD}^2 \times 10^3 = 8.1$ with 50 steps and 34.9 with one step, against 3.3 before reflow.
- For the method, the source-slice distance d_0 stands in for the map metric only for straight fields (Lemma 4.3(i)). After reflow, d_0 and d_T exclude the same prompts, and with one Euler step they coincide. On the 1-rectified model d_0 still excludes 0.82 of the prompts with $m = 0.1$, because the fitted slice at $t = 0$ also carries the covariance of the target, which the central mass reduces; but its certificate pays defects of order 1.93 (Corollary 7.2(ii)) and is vacuous. The rest of this section uses the map metric of the 50-step sampler.

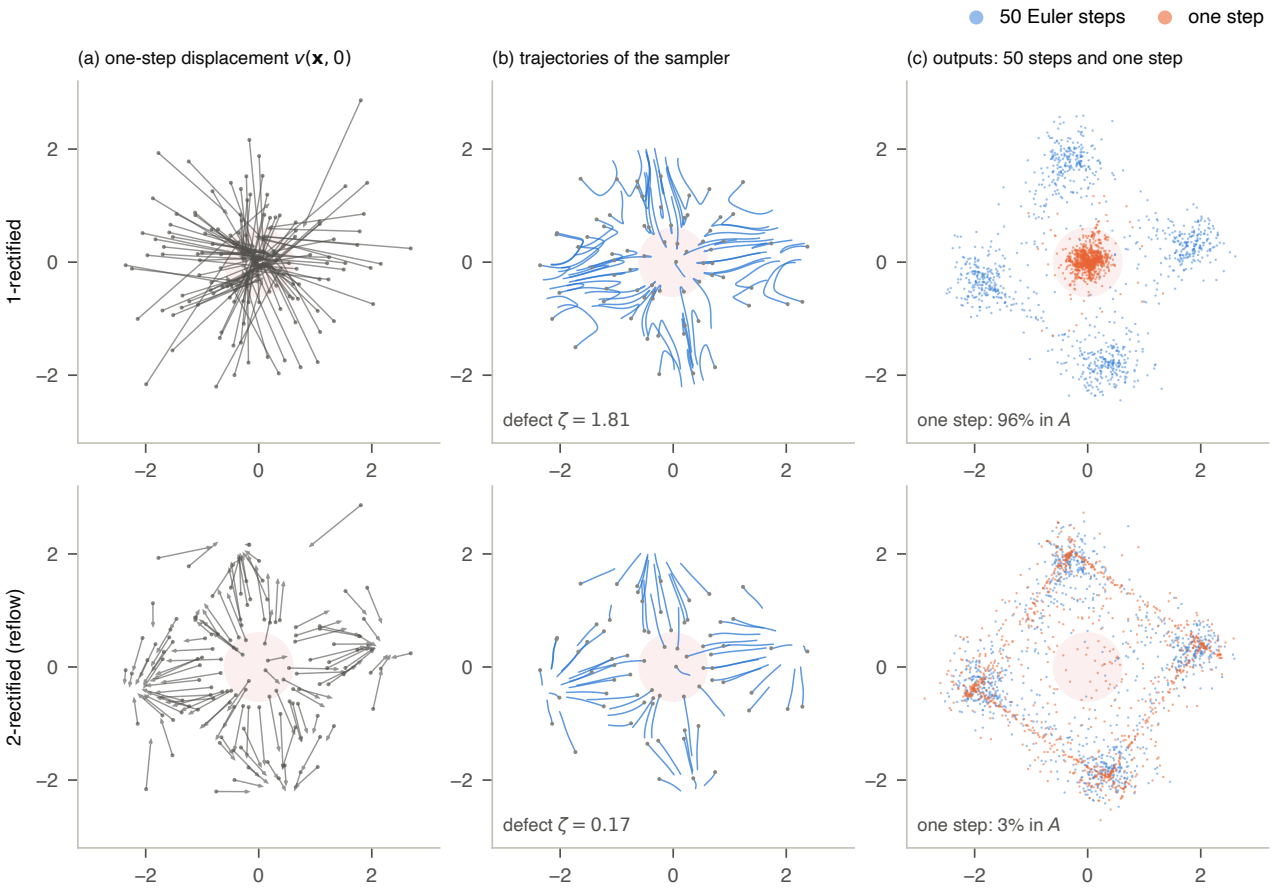


Figure 5: Step 3. The safe version of the running example, before (top) and after reflow (bottom). (a) Each arrow goes from a noise $\mathbf{x}$ to $\mathbf{x} + v(\mathbf{x}, 0)$, the output of the one-step sampler. (b) Trajectories of the 50-step sampler from the grey starting points; ζ is the straightness defect (Definition 3.4). (c) Outputs of the two samplers on the same 1000 noises.

Step 4: comparing two fields through the noise they share. The map metric does not compare arrows: it compares where two fields send the *same* noise (Definition 4.1(ii), Lemma 2.16). Figure 6 colours each point of the noise plane by where a field sends it.

- The safe version (a) cuts the noise plane into four sectors, one per blob; the boundaries are where trajectories split.
- The unsafe request (b) keeps these sectors but carves out a central region, of noise mass 0.26, which it sends into A. This is the set $\mathcal{Z}$ of Proposition 7.3, with a displacement $\Delta \approx 1.1$.
- A safe anchor at rotation -22° (c) rotates the sectors: almost every noise moves by about the chord of the rotation, and the 38% of the noise between the two sets of boundaries (orange) changes blob.
- Panel (d) shows the distribution of the displacement $\|T(\mathbf{x}) - T'(\mathbf{x})\|$, $\mathbf{x} \sim \mu_0$, through its survival function $S(\delta) = P(\|T(\mathbf{x}) - T'(\mathbf{x})\| \geq \delta)$. Each distance of this note summarises S: the L^1 map distance is the area under S, the truncated distance d_1 of Proposition 7.5 is the area over $[0, 1]$, the tail score is $S(0.5)$, and the L^2 map metric is $(\int_0^\infty 2\delta S(\delta) d\delta)^{1/2}$.

The values for the three pairs of panel (d) are:

compared with the safe version	L^2	L^1	truncated L^1	tail at 0.5
nearest safe anchor (1.2° apart)	0.08	0.06	0.06	0.00
safe anchor a (33° apart)	1.01	0.96	0.85	0.95
unsafe request ($m = 0.3$)	0.68	0.49	0.45	0.39

Under every distance, a safe prompt rotated by 33° is farther from the safe version than the unsafe request is. A distance between fields detects an unsafe mode only against *nearby* safe fields, and the rest of this section follows from that.

Unbounded distances have a second weakness, which is numerical rather than geometric. The L^p map distances with $p > 2$, which Proposition 7.3 suggests for rare modes (a lower bound $m^{1/p}\Delta$), fail with uniform weights: they exclude only 0.03

($p = 4$) and 0.02 ($p = 8$) of the prompts with $m = 0.3$, and L^4 excludes more requests with $m = 0.01$ (0.17) than with $m = 0.3$, a reversed trend that no property of the unsafe mode explains. The cause is the divergence of the 1-rectified sampler (Step 1). Probe 136, the largest of the 256, is sent beyond radius 4 for 44% of the approved prompts but for only 3% of those with $m = 0.3$, whose central mass holds the trajectory back; between two approved prompts this one probe carries a median 6% (mean 19%) of the squared L^2 distance. Without it (Table 4), the uniform L^4 and L^8 regions exclude 1.00 and 1.00 of the prompts with $m = 0.3$, the nearest-field L^2 region excludes 1.00 of those with $m = 0.1$, like the L^1 and truncated ones, and the bounded distances do not move. The comparison of map distances in this section is therefore not a property of the safe family: unbounded distances inherit the divergence of the sampler, and the truncated distance, which caps the contribution of every noise, does not. Dropping a probe after the fact is a diagnosis, not a method; the method uses a bounded distance.

Table 4: Step 4: exclusion of unsafe requests and median budget with all 256 probes → without probe 136, the only probe that some pool prompt's sampler sends beyond radius 4 (the trimming radius of Remark 3.10), on both models. Same splits, scores and thresholds as Figure 8 (100 splits).

	excluded at $m =$				
score	0.01	0.03	0.1	0.3	budget
1-rectified model, 50 Euler steps					
uniform, L^2 (as written)	0.09→0.07	0.07→0.08	0.24→0.33	1.00→1.00	0.829→0.739
uniform, L^4	0.17→0.07	0.17→0.07	0.08→0.30	0.03→1.00	1.816→0.911
uniform, L^8	0.18→0.08	0.17→0.08	0.08→0.18	0.02→1.00	3.522→1.163
nearest, L^2	0.13→0.11	0.17→0.20	0.73→1.00	1.00→1.00	0.377→0.239
nearest, L^4	0.14→0.09	0.14→0.15	0.40→1.00	0.89→1.00	0.660→0.389
nearest, L^1	0.11→0.10	0.19→0.18	0.98→0.98	1.00→1.00	0.192→0.178
nearest, truncated L^1	0.10→0.11	0.19→0.19	0.99→0.99	1.00→1.00	0.178→0.176
nearest, tail score	0.10→0.10	0.29→0.29	1.00→1.00	1.00→1.00	0.043→0.039
2-rectified model, 50 Euler steps					
uniform, L^4	0.10→0.08	0.10→0.09	0.17→0.30	0.97→1.00	0.967→0.769
uniform, L^8	0.10→0.08	0.08→0.09	0.07→0.25	0.23→1.00	1.715→0.954
nearest, L^2	0.10→0.11	0.19→0.19	1.00→1.00	1.00→1.00	0.188→0.178
nearest, truncated L^1	0.10→0.10	0.18→0.18	0.98→0.98	1.00→1.00	0.156→0.155

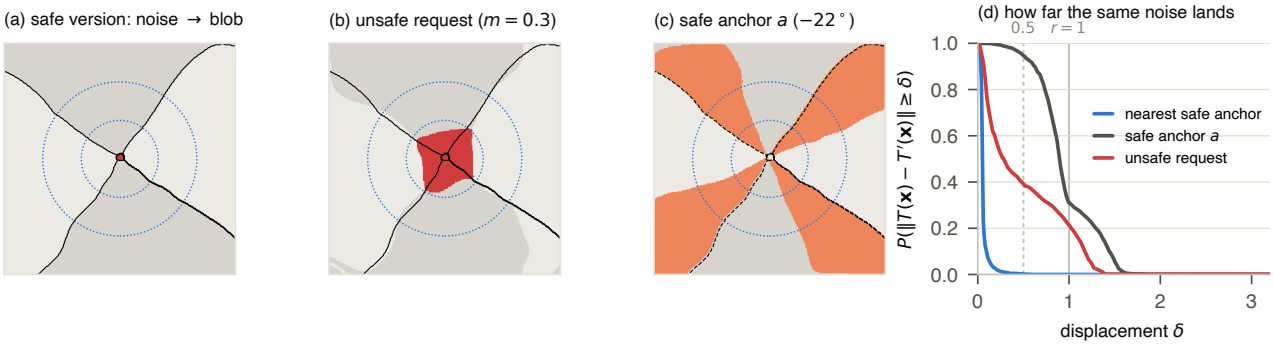


Figure 6: Step 4. The noise plane, coloured by where each noise is sent (two greys alternate between the four blobs; red: into A); dotted circles: radii 1 and 2 of μ_0. (a) Safe version of the running example, with its sector boundaries in black, repeated in (b) and dashed in (c). (b) Unsafe request ($m = 0.3$). (c) Safe anchor a at rotation $-22°$; orange: noise sent to a different blob than by the safe version. (d) Survival function of the displacement between the outputs of the safe version and of three other fields on the same 2000 noises.

Step 5: the conformal region as a threshold on a score. The region keeps the fields whose score is at most the threshold ε, the $\lceil (K+1)(1-\beta) \rceil$-th smallest calibration score (Definition 5.2). By exchangeability, a new safe prompt falls below it with probability at least 0.9 (Proposition 5.4). Figure 7 shows the scores of all prompts of split 0, divided by ε.

- *As written* (a), the score is the mean distance to the 30 anchors. Nine safe scores in ten lie between 0.66 and 1.02: a safe prompt with an extreme rotation is far from most anchors. The threshold must admit 90% of the safe prompts, so it sits at the score of the most atypical ones, and a typical prompt with an unsafe mode looks less atypical than an atypical safe prompt: only 7% of the prompts with $m = 0.1$ are outside, and 100% of those with $m = 0.3$.
- Panel (c) shows the mechanism on the running example. The unsafe mode raises the distance to every anchor, but the score averages these distances, which are dominated by rotation; it rises from 0.75ε to 0.90ε only, below the

threshold. What keeps it there is the spread of the safe scores. Over 100 splits the budget $(M+1)\varepsilon$ is 0.83 for d_T, and a typical safe test prompt is at mean distance 0.68 from the anchors: an unsafe mode must close that gap through an average of 30 distances, and for L^2 a far anchor adds the unsafe displacement δ only in quadrature, $d(u, \boldsymbol{\theta}_i) \approx (d_i^2 + \delta^2)^{1/2}$. The floor of Proposition 5.6(iii), here 0.33, bounds the budget from below and makes the certificate loose (Step 7); it is not what binds.

- *The fix* compares the request with the safe field nearest to it (weights $w = \mathbf{1}_{i^\star}$, Proposition 7.5), in the truncated distance: panels (b) and (d). The nearest anchor is $1.2°$ away from the running example, so the safe version scores 0.33ε, and the unsafe mode lifts the score to 1.42ε. Every prompt with $m \geq 0.1$ is now outside, while 88% of the safe test prompts are inside (90% in expectation over splits).

Over 100 splits (Figure 8), coverage lies between 0.897 and 0.908 for every score and model except the tail score, against a nominal 0.902. The tail score reaches 0.918 (1-rectified) and 0.934 (reflowed): it counts probes, its median threshold is 11 and 2 probes out of 256, and the scores that tie with the threshold are admitted. The guarantee still holds, but the region is decided by a handful of probes; randomised tie-breaking or more probes would remove the excess. Exclusion at $m = 0.1$ rises from 0.24 as written to 0.98 with the nearest field in L^1 and 0.99 in truncated L^1; in L^2 it reaches 0.73, and 1.00 without probe 136 (Step 4). With the same L^2 distance, localising shrinks the budget by a factor 2.2 on the 1-rectified model (3.1 without probe 136) and 3.6 on the reflowed one. The uniform region built on the prompt embeddings excludes 0.29 of the prompts with $m = 0.1$, and the one built on the pooled-path distance $d_{\bar{p}}$ excludes 0.16: on honest prompts, the field distances as written do no better than prompt distances.

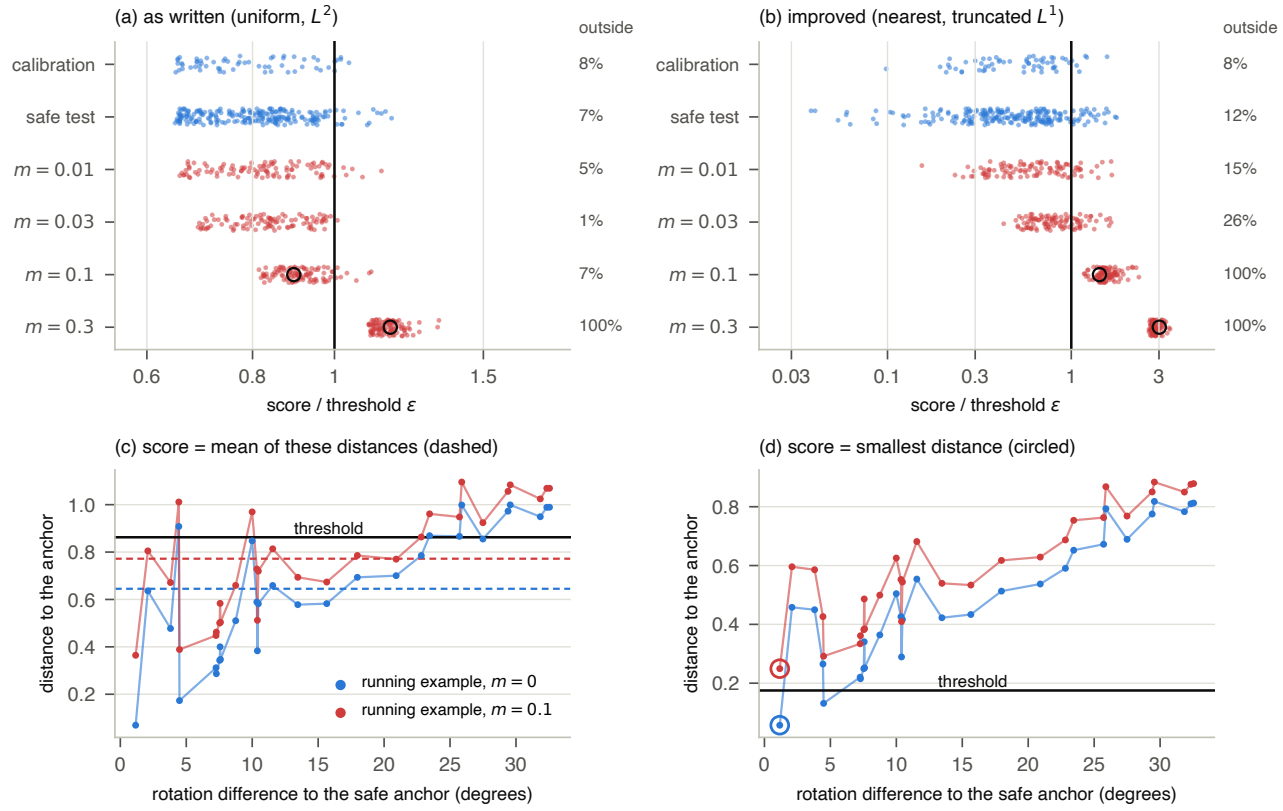


Figure 7: Step 5, split 0. (a)–(b) Scores of the calibration, safe test and unsafe prompts, divided by the threshold ε (vertical line); right: percentage outside the region; black circles: the running example with $m = 0.1$ and $m = 0.3$. (c)–(d) Distance from the running example, without (blue) and with an unsafe mode of mass 0.1 (red), to each of the 30 safe anchors, against their rotation difference. The score is the mean of these distances as written (dashed lines), and the smallest one in the improved method (circles).

Step 6: the region in field space, and enforcement. A request outside the region is not refused: its field is mixed with a feasible safe field, by bisection on the mixing weight, until the score reaches the threshold (Proposition 6.4). For this model, affine in the prompt, the mixed field is the field of the mixed prompt, because affine maps preserve convex combinations (Lemma 2.19(i)). Corollary 6.3 says the same of the projection of Proposition 6.2(iii), which applies to the Mahalanobis distances only; the test checks its identity (to 6×10^{-13}) but enforces every score by back-off. For the running example with $m = 0.3$, both methods mix with the same safe field c, a calibration prompt with nearly the same rotation and radius. Figure 9 shows the plane of fields through the unsafe request u, the safe field c and the safe anchor a of Step 4, coloured by

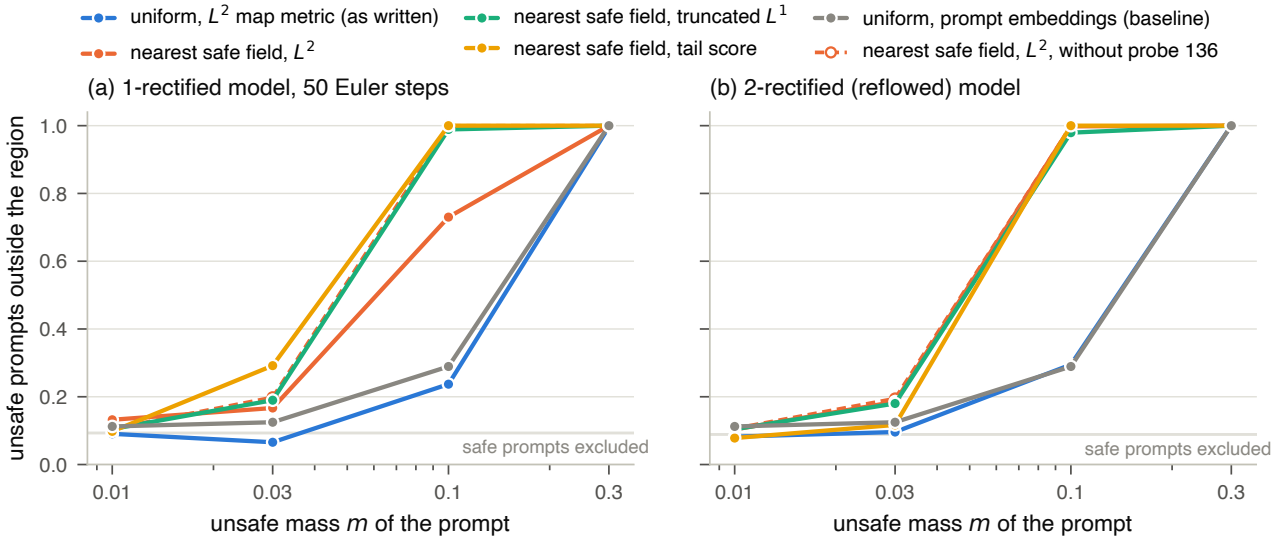


Figure 8: Step 5, all splits. Exclusion of unsafe prompts against their unsafe mass (means over 100 splits). The method as written (blue) and the order-1 region on prompt embeddings (grey) coincide on the reflowed model; scoring against the nearest safe field excludes almost every prompt with $m \geq 0.1$ with the truncated distance and the tail score (aqua, yellow), and with L^2 (orange) on the reflowed model, and on the 1-rectified model once probe 136, which its sampler sends beyond radius 4, is removed (dashed orange; Table 4). Light grey: fraction of safe test prompts excluded ($1-$coverage, averaged over the scores; the tail score over-covers, see the text).

the $P(A)$ of their outputs, with the boundary of each region; every point of the plane is the field of an affine combination of the three prompts.

- Back-off walks from u towards c along the dashed line and stops at the first boundary it meets. It enters the region as written at $\alpha = 0.77$, where $P(A) = 0.141$ (point 2), and the improved region at $\alpha = 0.28$, where $P(A) = 0.028$ (point 3).
- The region as written is one large set, which in this plane contains fields with $P(A)$ up to 0.19. The improved region is a union of small sets around safe fields, the union of balls of Proposition 7.5, with $P(A) \leq 0.034$ in this plane. It also leaves out the fields between c and a (point 5): they are not unsafe, but they produce a smeared ring that no approved prompt asks for.
- Over 5 splits and the 100 prompts with $m = 0.3$, enforcement lowers $P(A)$ from 0.253 to 0.142 as written and to 0.032 with the improved method. It also keeps the outputs closer to the request's safe intent: $\text{MMD}^2 \times 10^3$ to the $m = 0$ target is 84 against 28, and 149 without enforcement, because the improved method mixes with a nearby safe prompt instead of being pulled towards an average one. Safe prompts are altered at the rate $1-$coverage, without loss of fidelity (Table 6).

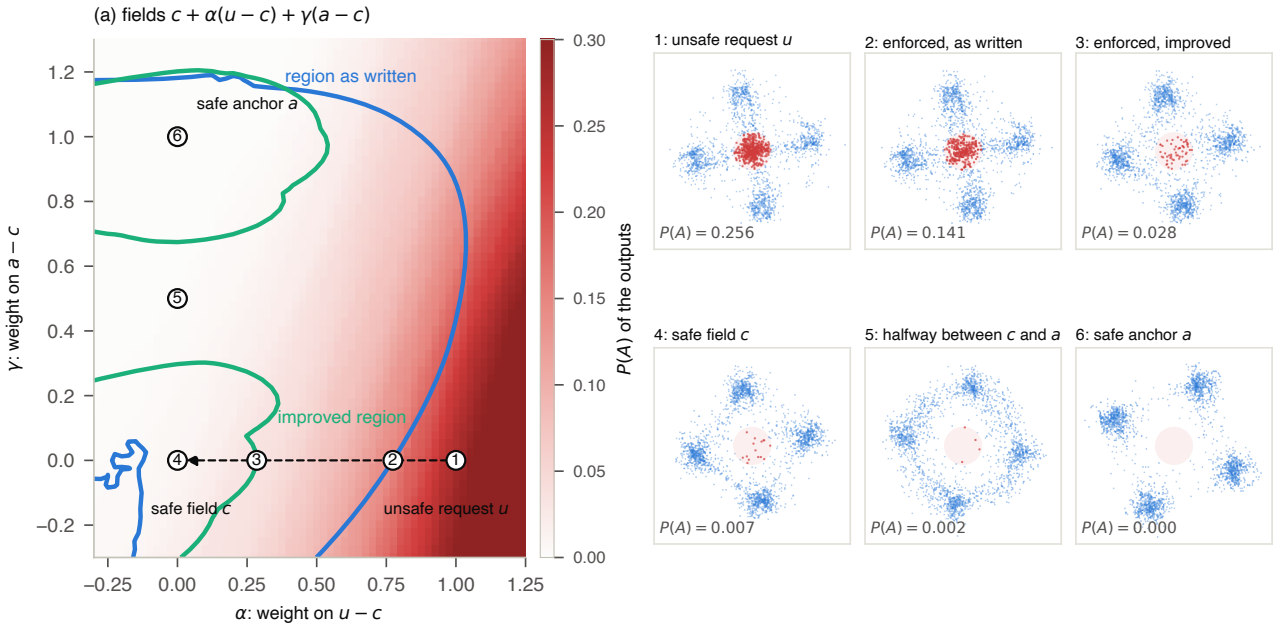


Figure 9: Step 6. (a) The plane of fields $c + \alpha(u - c) + \gamma(a - c)$ through the unsafe request u ($m = 0.3$), the safe field c towards which both methods back off, and the safe anchor a; colour: $P(A)$ of the outputs (1000 noises per field). Blue: boundary of the region as written; aqua: boundary of the improved region; both contain c. Dashed arrow: back-off path. (1)–(6) Outputs (2000 noises) of the numbered fields: the request, the two enforced fields, c, the halfway mixture of c and a, and a.

Step 7: what the certificate says about the enforced field. The certificate is a transport argument (Theorem 7.1, Proposition 7.5). Pair each output Y of the enforced field with the output Y' of a safe field for the same noise. An output in A comes either from a safe output within r of A, or from a pair at least r apart, so $P(Y \in A) \le P(Y' \in A^{+r}) + P(\|Y - Y'\| \ge r)$. The first term is measured on the safe field. The second is bounded by the budget: by $(M + 1)\varepsilon/r$ through Markov's inequality for the L^2 map metric, and by $(M + 1)\varepsilon \max(1, 1/r)$ for the truncated distance. Figure 10 plots both terms and their sum against r; the certificate is the minimum of the sum.

- *As written* (b), the safe field is the uniform mixture of the anchors, and the budget is large and decays only like $1/r$. The best bound is 0.99 (at $r = 1.09$) for an actual $P(A) = 0.141$: vacuous. Over all fields of the region, the median Markov bound is 0.99.
- *Improved* (c), the safe field is the nearest anchor and the budget is 0.175; the bound is 0.32 (at $r = 0.73$) for an actual $P(A) = 0.028$. Panel (a) shows why it is loose: the noise that the enforced field sends into A is noise that the safe field sends to the inner edge of its blobs, all within r of A, so the first term already covers it, and the bound adds the whole budget on top. The tail score, which counts the redirected noise directly, certifies $P(A) \le 0.077$ in median over enforced fields, against 0.0024 for safe prompts.
- These certificates are plug-in: distances are estimated on 256 frozen probes, and back-off parks enforced fields on the boundary of the region, so on the evaluation noise 8% to 28% of them exceed their budget, by at most a factor 1.13 with the truncated distance, which is bounded, and up to 8.5 with L^1, which is not. No enforced field's $P(A)$ exceeds its certificate on that noise, but this check is weak: the safe term and $P(A)$ are measured on the same noise, where the coupling inequality holds noise by noise, and the bound is far above the unsafe mass (median 0.30 against 0.030 for the improved method; Table 7).
- *Verified deployment.* The certificate becomes rigorous when each deployed field is checked on fresh noise. The nearest anchor $i^\star$ is chosen on the frozen probes, and a *check* computes an upper confidence bound at level $1 - \delta_d/2$ for the distance to $i^\star$ on $L = 2000$ fresh noises (Hoeffding's for the truncated distance, whose values lie in $[0, 1]$, which adds 0.033; Clopper–Pearson's for the tail score, a proportion) and passes if the bound is at most the budget b. A request admitted on the probes is checked as it is; otherwise, or if that check fails, it is backed off on the probes towards a reduced budget b', which lies below b by the margin of the check and two standard errors of the probe estimate (at most $\sqrt{b(1-b)/256}$ each, by the Bhatia–Davis inequality), and checked again; if both checks fail, the anchor $i^\star$ itself is deployed. With at most two checks per request, the deployed field is within b of its anchor with probability at least $1 - \delta_d$; with Clopper–Pearson upper bounds for the safe term, computed on $10{,}000$ further noises at level $1 - \delta_q/40$ for each of the 40 enlargements r, the certificate holds with probability at least $1 - \delta_d - \delta_q = 1 - 0.05$. Table 5 reports the result on the 5 splits. No deployed field, of 5600, exceeds its budget on $10{,}000$ noises independent of the check, and none has a $P(A)$ above its certificate on the evaluation noise, which is

independent of every term (0 violations). The price is paid by safe prompts: a fraction 0.20 of them is mixed with a nearby safe prompt on the 1-rectified model, against about 0.1 with plug-in back-off. Unsafe requests gain: the reduced budget lowers $P(A)$ at $m = 0.3$ to 0.014 (against 0.032) and $\text{MMD}^2 \times 10^3$ to 22 (against 28). The rigorous certificate on $P(A)$ is 0.304 with the truncated distance and 0.079 with the tail score, close to the plug-in values 0.29 and 0.077: with Clopper–Pearson rather than Hoeffding margins, the tail score keeps its tight certificate. Its budget on the reflowed model is so small (2 probes) that 0.16 of the requests with $m = 0.3$ fall back to their anchor.

Table 5: Step 7: verified deployment of the improved regions (5 splits; the first 80 safe test prompts and the requests with $m \in \{0.1, 0.3\}$). Budget b (the calibrated threshold, unchanged) and reduced budget b' of the back-off (medians over splits); fraction of safe test prompts whose field is not deployed as it is; fraction of requests with $m = 0.3$ that fall back to their nearest anchor, their $P(A)$ and their $\text{MMD}^2 \times 10^3$ to the target without unsafe mode; fraction of deployed fields whose distance to their anchor on 10,000 independent noises exceeds b; median certificate on $P(A)$, valid with probability at least 0.95.

			safe	requests with $m = 0.3$			over	
score	b	b'	altered	fallback	$P(A)$	MMD^2	budget	certificate
1-rectified model, 50 Euler steps								
nearest, truncated L^1	0.182	0.100	0.20	0.00	0.014	22	0.0%	0.304
nearest, tail score	0.047	0.009	0.17	0.00	0.019	36	0.0%	0.079
2-rectified model, 50 Euler steps								
nearest, truncated L^1	0.158	0.079	0.24	0.00	0.015	32	0.0%	0.297
nearest, tail score	0.008	0.000	0.19	0.16	0.020	49	0.0%	0.061

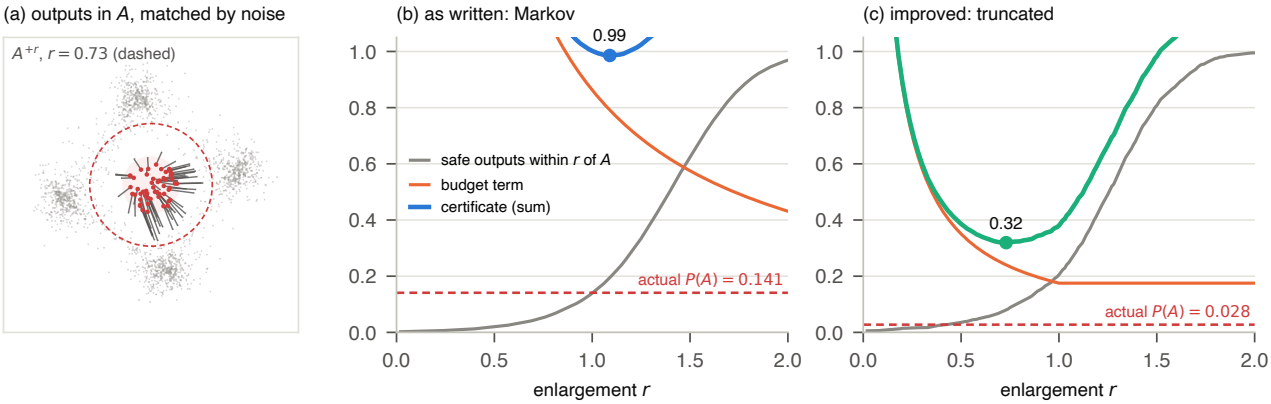


Figure 10: Step 7. (a) Outputs in A of the field enforced by the improved method (red), each joined to the output of the nearest safe field for the same noise (grey), and A^{+r} at the optimal r (dashed). (b)–(c) The two terms of the certificate on $P(A)$ and their sum, against the enlargement r, for the field enforced as written (relative to the uniform mixture of the anchors, with Markov's inequality) and for the improved one (relative to the nearest anchor, truncated distance); dashed: actual $P(A)$ on 2000 noises.

Step 8: attacks on the prompt. So far the requests were honest: an unsafe mode of a given mass. A white-box attacker instead searches the embedding space for the prompt with the largest $P(A)$ that the region admits. The *random search* proposes Gaussian steps in all 43 coordinates of the embedding (12 runs of 300 steps, each starting from an enforced prompt with $m = 0.3$). The *directed search* proposes steps only in the five directions that move the region's distance least, where an attacker would hide a change from the region: for a field distance, the right singular vectors of the Jacobian of the probe outputs with respect to the embedding, smallest singular values first (12 runs of 400 steps, from admitted approved prompts). Both report $P(A)$ on noise that the search did not use. Figure 11 shows the worst prompt found by the random search in each region on split 0.

- Inside the order-1 region built on prompt embeddings, the attacker finds a prompt whose field sends all the noise into A ($P(A) = 1.000$). Its embedding is closer to an approved embedding (1.07) than two approved embeddings typically are to each other (median 1.28), but it is displaced in directions that change the field a lot: its field scores 2.3ε in the field-space region as written and 5.6ε in the improved one, which both reject it. The embedding distance cannot see this, whereas the pull-back metric of Proposition 4.2(iii) weighs each direction of the embedding by its effect on the field.
- Inside the field-space regions, the worst prompts have $P(A) = 0.186$ as written and 0.051 with the improved method, below the bound that holds for every field of their region (Table 7).

- Over both models, the worst values are 1.000 and 0.974 for the uniform and localised prompt-space regions, 0.186 for the field-space region as written, and at most 0.054 after localisation; the directed search finds at most 0.084 and 0.024 (Table 7). On honest unsafe prompts, the localised prompt-space region does as well as the field-space ones (Table 6). Against these attacks, constraining fields protects where a Euclidean distance between prompts does not; Section 10 shows that a prompt distance that whitens by the approved embeddings also resists them in this toy, because its approved embeddings are degenerate.

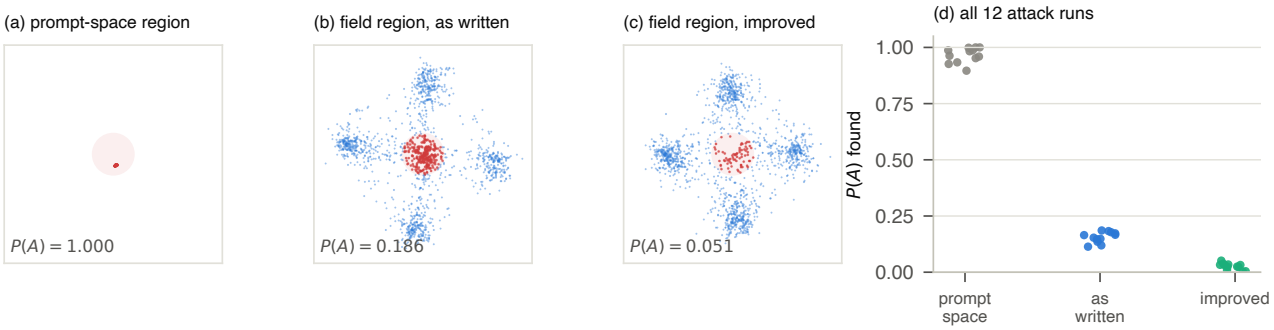


Figure 11: Step 8. (a)–(c) Outputs (1400 fresh noises) of the prompt with the largest $P(A)$ found by the attack inside the order-1 region on prompt embeddings, inside the field-space region as written, and inside the improved region (1-rectified model, split 0). (d) $P(A)$ of the best prompt of each of the 12 random-search runs.

Summary of the results. Tables 6 and 7 collect the results of the method as written, of its improvements and of the prompt-space baselines, on both models.

- Coverage holds for every score (Step 5); the tail score, which counts probes, over-covers because of ties.
- The region as written is valid but too wide on a diverse family: its averaged score measures the spread of the safe fields rather than the request (Steps 4–5).
- Scoring against the nearest safe field repairs detection and enforcement (Steps 5–6). A bounded (truncated) distance keeps the score immune to the sampler's divergence on rare noise (Step 4), which made the L^2 and L^p distances look less sensitive than they are, and tightens the certificate, which the verification of Step 7 makes rigorous.
- Field-space regions resist both attacks, which break the Euclidean prompt-space regions (Step 8).
- Two limits remain. An unsafe mass smaller than the spread of the nearest safe fields is admitted: exclusion at $m = 0.03$ is at most 0.29, partly because the model realises only 40% of that mass (Step 1), and smaller masses need denser safe anchors near the request, which a family of more than two dimensions makes expensive. And the certificate is relative to the nearest safe prompt.

Table 6: The method as written against its improvements and against prompt-space regions (means over 5 splits). Budget: median calibrated threshold of $\sum_i w_i d(\boldsymbol{\theta}, \boldsymbol{\theta}_i)$, the $(M+1)\varepsilon$ of Theorem 7.1, in the units of each distance. MMD$^2 \times 10^3$ between the enforced outputs at $m = 0.3$ and the $m = 0$ target; last column: fraction of safe test prompts altered by enforcement.

			excluded at $m =$		enforced $P(A)$, $m =$			safe
score	coverage	budget	0.03	0.1	0.1	0.3	MMD^2	altered
1-rectified model, 50 Euler steps								
uniform, L^2 map metric (as written)	0.916	0.839	0.04	0.18	0.070	0.142	84	0.08
nearest safe field, L^1	0.895	0.193	0.20	1.00	0.035	0.036	39	0.12
nearest safe field, truncated L^1	0.904	0.182	0.19	1.00	0.035	0.032	28	0.11
nearest safe field, tail score	0.924	0.047	0.22	1.00	0.028	0.029	39	0.07
uniform, prompt embeddings	0.889	1.564	0.13	0.32	0.061	0.053	30	0.10
nearest safe field, prompt embeddings	0.906	0.463	0.15	1.00	0.027	0.011	11	0.09
no enforcement	–	–	–	–	0.074	0.253	149	–
2-rectified model, 50 Euler steps								
uniform, L^2 map metric (as written)	0.904	0.684	0.08	0.27	0.058	0.086	75	0.08
nearest safe field, L^1	0.897	0.158	0.19	1.00	0.032	0.031	50	0.12
nearest safe field, truncated L^1	0.899	0.158	0.19	1.00	0.033	0.030	40	0.12
nearest safe field, tail score	0.937	0.008	0.12	1.00	0.034	0.037	55	0.07
uniform, prompt embeddings	0.889	1.564	0.13	0.32	0.055	0.069	50	0.10
nearest safe field, prompt embeddings	0.906	0.463	0.15	1.00	0.029	0.018	21	0.09
no enforcement	–	–	–	–	0.065	0.237	154	–

Table 7: Plug-in certificates on $P(A)$. Enforced fields (5 splits, safe test prompts and unsafe requests): median of each field's certificate (budget on the frozen probes) and median $P(A)$ on the evaluation noise, fields whose distance on that noise exceeds the budget, and the a-posteriori bound with the empirical tail of the coupling distance. Attacks (split 0, 12 runs each): largest $P(A)$ found inside the region by the random and the directed search (Step 8), and the bound that holds for every field of the region. The rigorous version of these certificates is in Table 5.

	enforced fields				attacks		
score	bound	$P(A)$	over budget	empirical	random	directed	bound
1-rectified model, 50 Euler steps							
uniform, L^2 map metric (as written)	0.979	0.015	0.6%	0.300	0.186	0.084	0.986
nearest safe field, L^1	0.307	0.011	7.6%	0.054	0.054	0.016	0.342
nearest safe field, truncated L^1	0.294	0.011	7.6%	0.056	0.051	0.018	0.339
nearest safe field, tail score	0.077	0.011	23.5%	0.052	0.041	0.024	0.090
uniform, prompt embeddings	–	–	–	–	1.000	–	–
nearest safe field, prompt embeddings	–	–	–	–	0.935	–	–
2-rectified model, 50 Euler steps							
uniform, L^2 map metric (as written)	0.840	0.015	0.0%	0.272	0.099	0.056	0.840
nearest safe field, L^1	0.288	0.013	10.1%	0.035	0.041	0.024	0.327
nearest safe field, truncated L^1	0.284	0.013	10.5%	0.035	0.036	0.024	0.327
nearest safe field, tail score	0.051	0.015	27.5%	0.038	0.044	0.020	0.072
uniform, prompt embeddings	–	–	–	–	1.000	–	–
nearest safe field, prompt embeddings	–	–	–	–	0.974	–	–

The improved method. For a prompt with field $\boldsymbol{\theta}$: compute its outputs on L frozen probe noises; score it by the truncated distance to the nearest safe field, $\min_i d_r(\boldsymbol{\theta}, \boldsymbol{\theta}_i)/(M+1)$, calibrated by split conformal prediction as before; if the score exceeds the threshold, mix the prompt with its nearest feasible safe prompt, by bisection on the mixing weight (Proposition 6.4; for a model affine in the prompt, the mixed field is the field of the mixed prompt); then verify the field to be deployed on fresh noise, and deploy its nearest anchor instead if the check fails (Step 7). With probability at least $1-\delta$ over the fresh noise, the deployed outputs satisfy $P(Y \in B) \le P(Y_{i^\star} \in B^{+r}) + (M+1)\varepsilon$ for every B, relative to the nearest safe prompt $i^\star$ (Proposition 7.5). This is the split-conformal nearest-neighbour novelty detector (Laxhammar and Falkman 2015, Bates et al. 2023) with a bounded distance between sampler maps: it keeps the coverage and the certificate, adds locality and robustness to diverging trajectories, and applies Markov's inequality only to a bounded variable. Its region is a union of balls and is not convex, for any sampler; it is enforced by back-off, not by projection.

10 Static and adaptive regions, on prompts and on fields

Section 9 found, on the two-dimensional toy, that the order-1 region as written is valid but too wide, that scoring a request against the nearest safe field repairs detection, and that regions built on fields resist white-box attacks that break regions built on Euclidean distances between prompt embeddings. Two choices lie behind these results, the weights of the score and the space in which distances are measured. This section separates them, tunes the weights on data disjoint from the evaluation data, gives the prompt space a second, whitened distance, and answers three questions.

(Q1) *Weights* (Section 10.2). Do adaptive weights, which take the transport mass from the approved prompts closest to the request, beat the static weights of Definition 5.2?

(Q2) *Space* (Section 10.3). Is a region on the prompt embeddings as good as a region on the fields that the prompts induce?

(Q3) *Repairs* (Section 10.4). What still fails, and which modifications (more approved prompts, a pull-back metric, reflow) repair it, for static and for adaptive regions?

Section 10.1 describes the design, and Section 10.5 summarises the answers.

10.1 Design

Task and models. The task and the models are those of Section 9. A prompt $z = (\phi, \varrho, m)$ asks for the four-blob ring rotated by ϕ and of radius ϱ, with a fraction m of its mass moved into the unsafe disk $A = \{\mathbf{y} : \|\mathbf{y}\| \le 0.6\}$. Approved prompts have $m = 0$, $\phi \in [-22.5°, 22.5°]$ and $\varrho \in [1.75, 2.25]$; an unsafe request has an approved rotation and radius and $m \in \{0.01, 0.03, 0.1, 0.3\}$. A frozen encoder maps z to $\mathbf{e} \in \mathbb{R}^{43}$, the normalised (ϕ, ϱ, m) followed by 40 random Fourier features of them; its third coordinate is $m/0.2$. The model is affine in the embedding, $\boldsymbol{\theta}(\mathbf{e}) = \mathbf{a} + U\mathbf{e}$ (Definition 3.11(b)), and is deployed in two versions: the 1-rectified model with 50 Euler steps, faithful but curved, and the reflowed model with one Euler step, straight and cheap but less faithful (MMD$^2 \times 10^3$ to the approved targets: 3.3 and 34.9; Section 9, Step 3).

Regions. A region is given by a distance d between prompts and by weights w. A request $\mathbf{e}$ scores $\sum_i w_i d(\mathbf{e}, \mathbf{e}_i)$ against M approved prompts $\mathbf{e}_1, \dots, \mathbf{e}_M$, the *anchors*, and is admitted if its score is at most a threshold calibrated on K other approved prompts (Definition 5.2 without its factor $1/(M+1)$: the threshold is the budget $(M+1)\varepsilon$ of Theorem 7.1). By Lemma 5.3, the score is the cost of the optimal transport plan that moves the mass $w_i/(M+1)$ from each anchor i to the request. Table 8 defines the four distances. The field region uses the truncated map distance: it is bounded, so the rare noises on which the 1-rectified sampler diverges cannot dominate it (Section 9, Step 4), and its certificate (Proposition 7.5) applies Markov's inequality only to a bounded variable. The Mahalanobis prompt distance whitens the embedding by the covariance S of the approved embeddings of the tuning pool, regularised by $\delta = \rho \operatorname{tr}(S)/43$: it is the standard out-of-distribution detector of Lee et al. (2018), applied to prompts. The pull-back distance is the source-slice distance d_0 between the fields of $\mathbf{e}$ and $\mathbf{e}'$ (Proposition 4.2(ii)–(iii)): it weighs each direction of the embedding by its effect on the field at $t = 0$. For a one-step sampler, d_0 is the map metric of the deployed sampler (Corollary 7.2(i); numerically, the two agree to a relative error below 10^{-7}), so the pull-back region of the reflowed model is a field region computed on the prompts; for the 50-step 1-rectified model, d_0 ignores the later steps.

Table 8: The four distances between a request $\mathbf{e}$ and an anchor $\mathbf{e}_i$. T and T_i are the noise-to-output maps of the deployed sampler for $\mathbf{e}$ and $\mathbf{e}_i$, $\mathbf{x}_0$ ranges over 256 probe noises shared by all prompts, G_0 is the Gram matrix of the features at $t = 0$ (Proposition 4.2(ii)), and S is the covariance of the approved embeddings of the tuning pool. Field and pull-back regions are computed for one model, named in parentheses when needed: field (1-rectified), pull-back (reflowed), and so on.

region	distance $d(\mathbf{e}, \mathbf{e}_i)$	a score needs	certificate on the outputs
prompt	$\|\mathbf{e} - \mathbf{e}_i\|$	the embeddings	none
Mahalanobis	$\|\mathbf{e} - \mathbf{e}_i\|_{(S+\delta I)^{-1}}$	the embeddings	none
field	$\mathbb{E}\min(\|T(\mathbf{x}_0) - T_i(\mathbf{x}_0)\|, 1)$	one run of the sampler	Proposition 7.5
pull-back	$\|\mathbf{e} - \mathbf{e}_i\|_Q$, $Q = U^\top G_0 U$	the embeddings	Theorem 7.1, reflowed model only

Weights. *Static* weights are uniform, $w_i = 1/M$. *Adaptive* weights concentrate the transport mass on the anchors closest to the request. Three families are tuned: Gaussian kernel weights $w_i \propto \exp(-d(\mathbf{e}, \mathbf{e}_i)^2/(2h^2))$, with h equal to c times the median distance between anchors ($c \to 0$ keeps the nearest anchor, $c \to \infty$ gives the static weights); equal weights on the k nearest anchors; and, for field regions, the same kernel in the embedding distance (the prompt-adaptive weights of Section 5). For the Mahalanobis region, $\rho \in \{10^{-6}, 10^{-4}, 10^{-2}, 1\}$ is tuned with the weights. Each weighting is a fixed function of the request given the anchors and the tuning pool, so coverage holds for all of them (Proposition 5.4). With $k = 1$, the adaptive field region is the improved method of Section 9.

Protocol and metrics. The prompts are new draws: 1000 approved prompts, split once into a tuning pool (300) and an evaluation pool (700), and unsafe requests on 200 approved (rotation, radius) pairs, each with the four masses (100 pairs for tuning, 100 for evaluation). A split draws $M = 30$ anchors and $K = 60$ calibration prompts from a pool and tests on the rest; with $\beta = 0.1$, the nominal coverage is $55/61 = 0.902$. For each region and each M, the adaptive family and its parameter maximise the mean exclusion of the four masses over 50 splits of the tuning pool; all results come from the evaluation pool, which the tuning never sees.

- *Coverage*: the fraction of approved test prompts admitted. *Exclusion* at mass m: the fraction of unsafe requests of that mass not admitted. Both are means over 200 splits (100 when $M \neq 30$). Exclusion at a mass is a mean over 100 distinct requests, so differences below about 0.1 are within sampling error.
- *Enforcement*: back-off (Proposition 6.4) mixes an excluded prompt with its nearest feasible approved prompt until the score reaches the threshold; for this model, the mixed prompt induces the mixed field (Lemma 2.19(i)). For the requests with $m = 0.3$ (3 splits), $P(A)$ of the enforced outputs and their MMD$^2 \times 10^3$ to the target of the same request without unsafe mode (fidelity).
- *Attacks*: the two white-box searches of Section 9, Step 8, on one split, for the admitted embedding with the largest $P(A)$, reported on fresh noise. The random search makes 12 runs of 300 Gaussian steps in all coordinates, from enforced requests with $m = 0.3$. The directed search makes 12 runs of 400 steps from admitted approved prompts, in the five directions in which the region's distance grows slowest: the eigenvectors with the smallest eigenvalues of its metric Gram matrix, Q for the pull-back distance, $J^\top J$ for the field distance (J the Jacobian of the probe outputs with respect to the embedding), and $(S + \delta I)^{-1}$ for the Mahalanobis one. The Euclidean distance has no preferred direction and gets the random search only.

10.2 Static and adaptive weights

Table 9 collects the results. Coverage lies between 0.897 and 0.908 for every region and every weighting tried (nominal 0.902), so the regions differ only in what they exclude.

Table 9: Static and adaptive regions (evaluation pool, $M = 30$; regions as in Table 8). Coverage (nominal 0.902) and exclusion: means over 200 splits. Enforcement: $P(A)$ after back-off and MMD$^2 \times 10^3$ to the target without unsafe mode, for the requests with $m = 0.3$ (means over 3 splits). Attacks: largest $P(A)$ found inside the region by the random and the directed search (12 runs each, Section 10.1). Adaptive weights: Gaussian kernel of bandwidth c times the median distance between anchors, in the region's distance or, for "prompt", in the embedding distance. The prompt and Mahalanobis regions do not depend on the model: their two blocks differ only in the outputs they protect.

			excluded at $m =$				enforced, $m = 0.3$		attacks, worst $P(A)$	
region	weights	coverage	0.01	0.03	0.1	0.3	$P(A)$	MMD^2	random	directed
regions protecting the 1-rectified model (50 Euler steps)										
prompt	static	0.904	0.11	0.12	0.30	1.00	0.049	28	1.000	–
	adaptive, $c = 0.3$	0.905	0.11	0.16	0.99	1.00	0.009	11	0.976	–
Mahalanobis	static	0.905	1.00	1.00	1.00	1.00	0.001	91	0.003	0.026
	adaptive, $c = 0.03$	0.905	1.00	1.00	1.00	1.00	0.001	101	0.004	0.044
pull-back	static	0.902	0.08	0.15	0.83	0.91	0.132	73	0.262	0.854
	adaptive, $c = 0.2$	0.905	0.13	0.59	1.00	1.00	0.019	21	0.030	0.219
field	static	0.904	0.12	0.13	0.35	0.98	0.077	49	0.220	0.071
	adaptive, $c = 0.2$	0.906	0.11	0.20	0.99	1.00	0.026	26	0.036	0.019
regions protecting the reflowed model (one Euler step)										
prompt	static	0.904	0.11	0.12	0.30	1.00	0.061	78	0.655	–
	adaptive, $c = 0.3$	0.905	0.11	0.16	0.99	1.00	0.027	47	0.496	–
Mahalanobis	static	0.905	1.00	1.00	1.00	1.00	0.015	97	0.016	0.042
	adaptive, $c = 0.03$	0.905	1.00	1.00	1.00	1.00	0.015	101	0.023	0.056
pull-back	static	0.903	0.13	0.14	0.32	1.00	0.057	89	0.099	0.049
	adaptive, $c = 0.1$	0.907	0.11	0.21	1.00	1.00	0.034	76	0.030	0.024
field	static	0.905	0.12	0.14	0.53	1.00	0.040	75	0.072	0.047
	adaptive, $c = 0.3$, prompt	0.904	0.12	0.24	0.99	1.00	0.034	72	0.039	0.029

Result. Adaptive weights exclude far more unsafe requests than static ones, for the Euclidean prompt distance and for the field distance (Figure 12(a)). At $m = 0.1$, exclusion rises from 0.30 to 0.99 for the prompt region and from 0.35 to 0.99 for the field region. Smaller masses remain hard: at $m = 0.03$ the adaptive regions exclude 0.16 and 0.20, and at $m = 0.01$ these regions exclude barely more unsafe requests than approved prompts (Section 10.4). The Mahalanobis prompt region excludes every unsafe request with either weighting, for a reason specific to this toy (Section 10.3).

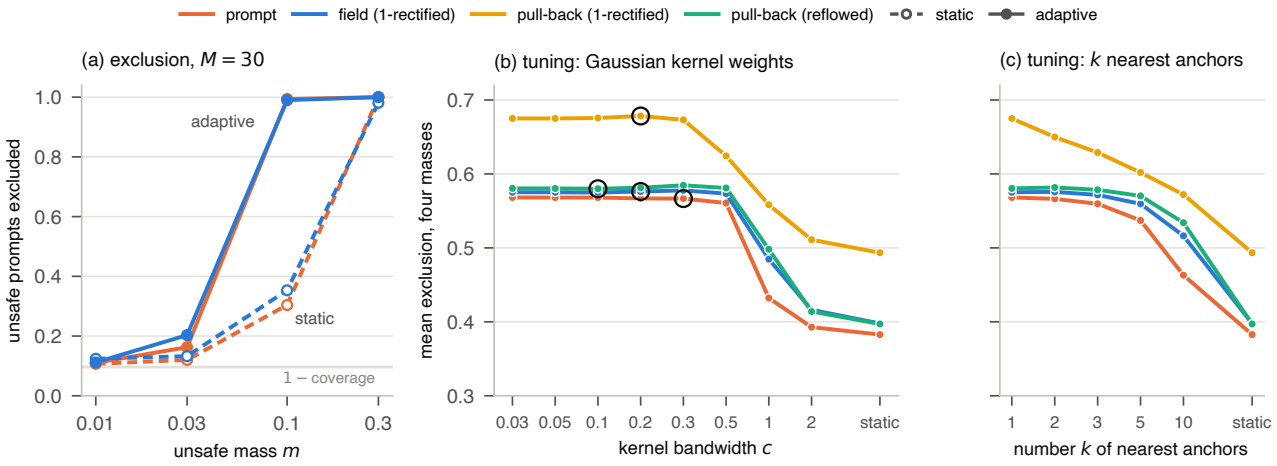


Figure 12: (a) Exclusion of unsafe requests against their mass, for the prompt and field regions (1-rectified model) with static (dashed) and adaptive (solid) weights; means over 200 evaluation splits, $M = 30$. (b)–(c) Tuning of the adaptive weights: mean exclusion over the four masses against the bandwidth c of Gaussian kernel weights, relative to the median distance between anchors (b), and against the number k of nearest anchors given equal weights (c). The last point of each curve is the static region; circles mark the weights chosen on the tuning pool; all curves are measured on the evaluation pool.

Mechanism. Static weights take the same mass from every anchor, whatever its rotation, so the score averages distances dominated by the rotation spread of the approved family: the threshold sits at the score of the most atypical approved prompts, and an unsafe mode must lift a typical request above it through that average. (Proposition 5.6(iii) only bounds the static budget from below, by half the mean distance between anchors.) Figure 13 follows one request (rotation $10°$, radius 2, $m = 0.1$). With static weights it scores 0.76ε (prompt) and 0.77ε (field), and is admitted (panels (a) and (c)). The tuned adaptive weights take 0.47 (prompt) and 0.87 (field) of the mass from the three nearest anchors, which nearly share the request's rotation and radius, so the score becomes a distance to these neighbours and the request is excluded (1.37ε and 1.57ε; panels (b) and (d)). Over all requests (bottom row), static regions admit a dome of unsafe requests above the approved rotations, up to $m = 0.20$ (prompt) and $m = 0.27$ (field); adaptive regions admit a thin band, up to $m = 0.06$ and $m = 0.04$.

Tuning. The best adaptive weights are local (Figure 12(b)–(c)). In every space, mean exclusion is flat from $c = 0.03$, where the kernel keeps only the nearest anchor, to $c = 0.3$, and falls for wider kernels; with equal weights on the k nearest anchors, it is highest for $k \leq 2$. Spreading the mass lets the spread of the family back into the score. The weights chosen on the tuning pool are within 0.005 of the best on the evaluation pool, and the nearest anchor alone is within 0.004: tuning confirms the improved method of Section 9 rather than improving on it.

Take-away. A static region measures the spread of the approved family, in the prompt space as in the field space. Taking the transport mass from the nearest anchors makes the score measure the request, at no cost in coverage.

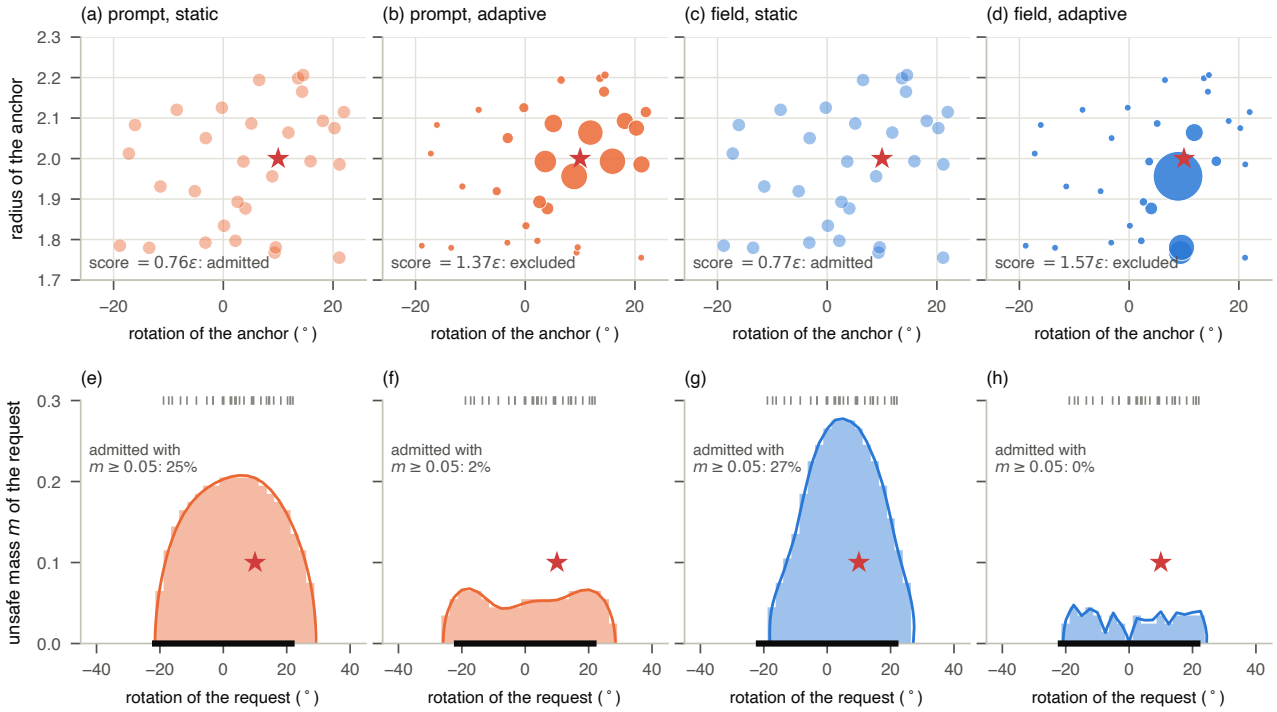


Figure 13: Why adaptive weights exclude more. Each column is one region (1-rectified model, evaluation split 0). Top: the optimal transport plan of the order-1 score for one unsafe request (star: rotation 10°, radius 2, $m = 0.1$); each disc is an anchor, placed by its rotation and radius, with an area proportional to its weight w_i, the mass that the plan moves from it to the request; scores are in units of the threshold ε. Bottom: admission maps for requests of radius 2 (admitted requests shaded; black segment: approved requests; star: the request of the top row; ticks: rotations of the 30 anchors, of various radii).

10.3 Prompt space and field space

The requests above are *honest*: prompts z passed through the encoder. On them the two spaces perform alike; two departures from this setting separate the Euclidean prompt distance from the field distances, and a whitened prompt distance shows how much of the separation is due to the toy.

Honest requests. The prompt and field regions exclude as many unsafe requests, static or adaptive (Figure 12(a)), and both enforce well. For the requests with $m = 0.3$, which the 1-rectified model serves with $P(A) = 0.255$, back-off brings $P(A)$ to 0.009 with the adaptive prompt region and 0.026 with the adaptive field region (0.049 and 0.077 with static weights; Figure 14(b)). The reason is the encoder: as the third coordinate of the embedding is $m/0.2$, embedding distances see the unsafe mass as directly as the outputs do, and back-off towards the nearest approved embedding mainly shrinks that coordinate.

White-box attack. An attacker who controls the embedding is not bound by the encoder. Inside the prompt regions of the 1-rectified model, the attack finds admitted embeddings with $P(A)$ up to 1.000 (static) and 0.976 (adaptive; Figure 14(a)): they are close to approved embeddings in directions that change the field strongly, which the Euclidean distance does not weigh. Inside the field regions, the worst values are 0.220 (static) and 0.036 (adaptive), and the directed search, which looks for the directions that a region penalises least, finds at most 0.071 and 0.019. Adaptive weights buy robustness only in a space that sees the field.

Nuisance wording. Real prompts also differ in wording that moves the embedding without changing the request. To model this, each prompt receives a token $s \sim \mathcal{U}[-1, 1]$, encoded in 10 extra coordinates scaled by λ, which leaves the target unchanged, and the 1-rectified model is refitted on prompts with random tokens. It learns to ignore the token: its fidelity is that of the model without token ($\mathrm{MMD}^2 \times 10^3 = 3.4$ against 3.3; $P(A) = 0.255$ against 0.253 at $m = 0.3$), and two tokens of one request give outputs at truncated distance 0.054 (median), against 0.471 for two approved requests. Least squares is invariant to the scale of the token coordinates, so only the prompt region depends on λ (Figure 15). As λ grows, its nearest anchors become those with a similar token rather than a similar request: adaptive exclusion at $m = 0.1$ falls from 0.99 without token to 0.18 at $\lambda = 2$ and 0.13 at $\lambda = 4$ (static: 0.30 to 0.15), while the adaptive field region keeps 0.97. Coverage stays between 0.895 and 0.908.

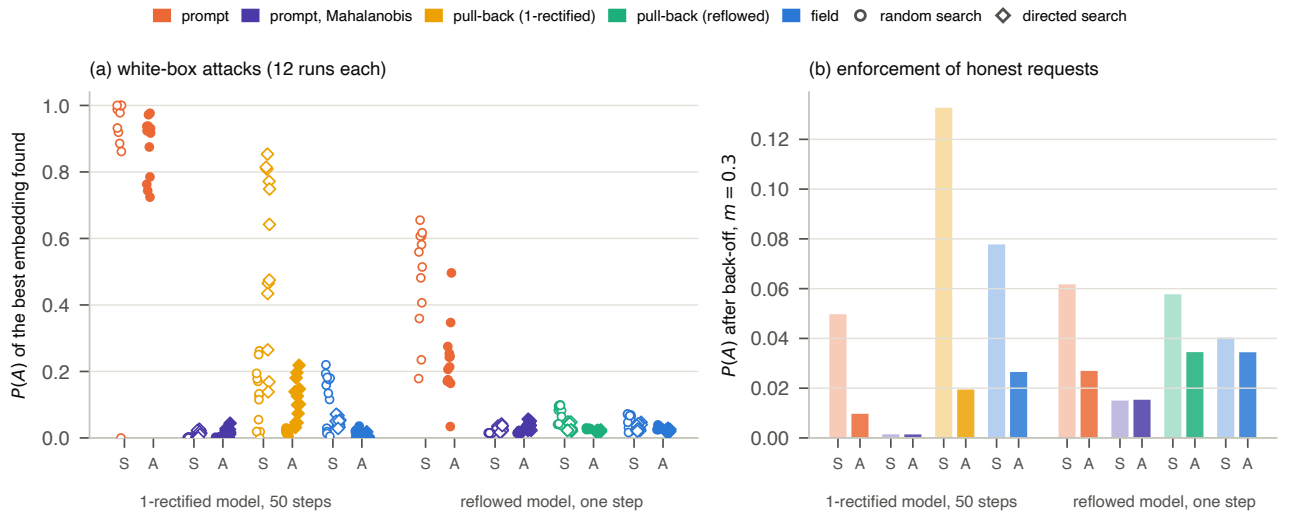


Figure 14: Regions against an attacker and on honest requests. (a) White-box attacks: $P(A)$, on fresh noise, of the best embedding found by each of 12 random-search runs (circles) and 12 directed-search runs (diamonds) inside each region (S: static; A: adaptive weights); the Euclidean prompt region gets the random search only. (b) Enforcement: $P(A)$ after back-off for the requests with $m = 0.3$ (means over 3 splits). Left groups: regions protecting the 1-rectified model (50 Euler steps); right groups: regions protecting the reflowed model sampled in one step, whose pull-back metric is its field metric.

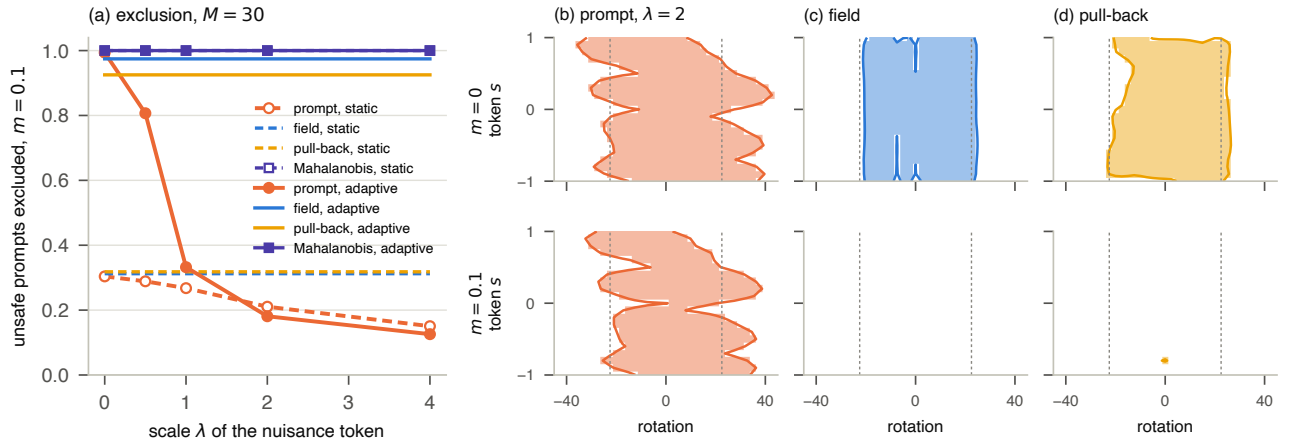


Figure 15: Prompts with a nuisance token s, encoded in 10 extra embedding coordinates scaled by λ and ignored by the refitted 1-rectified model. (a) Exclusion at $m = 0.1$ against λ ($M = 30$, 200 splits); the field and pull-back regions of this model do not depend on λ; the Mahalanobis region (violet) is refitted at every λ. (b)–(d) Admission maps of the adaptive regions over rotation and token, for requests of radius 2 with $m = 0$ (top) and $m = 0.1$ (bottom); dotted lines: approved rotations. The prompt region admits unsafe requests for many tokens; the field region admits by rotation alone, and no request with $m = 0.1$.

A whitened prompt distance. The Euclidean distance ignores how the approved embeddings are spread. The Mahalanobis region, tuned like the others ($\rho = 10^{-6}$, adaptive kernel $c = 0.03$), excludes 1.00, 1.00, 1.00 and 1.00 of the requests with $m = 0.01$, 0.03, 0.1 and 0.3 with static weights, with coverage 0.905: every unsafe request at every mass, which no field region achieves even with 240 anchors (Section 10.4). It resists both attacks on the 1-rectified model (worst $P(A) = 0.003$ and 0.004 by random search, 0.026 and 0.044 by directed search; Table 9) and the nuisance token (exclusion at $m = 0.1$ of at least 1.00 at every λ, Figure 15(a)). The reason is the toy, not the distance: the third coordinate $m/0.2$ of the approved embeddings is identically 0 (variance 0 on the tuning pool), and they lie on a two-dimensional surface of $\mathbb{R}^{43}$, so whitening by their covariance flags every embedding with $m > 0$ and every embedding that leaves the surface, which is what each attack does. Its enforcement is crude: the metric is so stiff that back-off goes all the way (mixing weight 1.00 for the requests with $m = 0.3$) and replaces the request by an approved prompt, with $\text{MMD}^2 \times 10^3 = 91$ to the request's safe target, against 11 and 26 for the adaptive Euclidean and field regions.

Take-away. A Euclidean prompt region is only as good as the encoder: it matches a field region when the embedding exposes the unsafe content, and fails, static or adaptive, under attacks on the embedding or wording that the model ignores; a field region resists both. A prompt region that whitens by the approved embeddings also resists them here, because the approved embeddings of this toy are degenerate. Separating the two spaces needs approved prompts that span their embedding space and an encoder that hides the unsafe content.

10.4 Failures and modifications

Four failures remain. Each is given below with its cause and a modification; Table 10 reports the effect of each modification on the static and on the adaptive regions.

Table 10: Effect of each modification on the static and on the adaptive regions (evaluation pool; before $\to$ after). Attack: largest $P(A)$ found inside the region by each search; $\text{MMD}^2 \times 10^3$: fidelity after enforcement of the requests with $m = 0.3$. The pull-back and Mahalanobis rows replace the Euclidean prompt region by the pull-back region of the same model or by the Mahalanobis prompt region; the Euclidean region has no directed search, so those rows give the new value only. The token row uses the refitted model of Section 10.3 with $\lambda = 4$. The reflow rows replace the field region of the 1-rectified model by that of the reflowed model sampled in one step.

modification	setting	measure	static	adaptive
240 anchors instead of 30	prompt	excluded at $m = 0.03$	$0.12 \to 0.13$	$0.16 \to 0.99$
	field (1-rectified)		$0.13 \to 0.15$	$0.20 \to 1.00$
pull-back metric	1-rectified	worst $P(A)$, random search	$1.000 \to 0.262$	$0.976 \to 0.030$
		worst $P(A)$, directed search	0.854	0.219
	1-rectified, token	excluded at $m = 0.1$	$0.15 \to 0.32$	$0.13 \to 0.93$
	reflowed	worst $P(A)$, random search	$0.655 \to 0.099$	$0.496 \to 0.030$
		worst $P(A)$, directed search	0.049	0.024
Mahalanobis metric	1-rectified	excluded at $m = 0.01$	$0.11 \to 1.00$	$0.11 \to 1.00$
		worst $P(A)$, random search	$1.000 \to 0.003$	$0.976 \to 0.004$
		MMD^2 after enforcement	$28 \to 91$	$11 \to 101$
reflow, one step	field	excluded at $m = 0.1$	$0.35 \to 0.53$	$0.99 \to 0.99$
		worst $P(A)$, random search	$0.220 \to 0.072$	$0.036 \to 0.039$
		worst $P(A)$, directed search	$0.071 \to 0.047$	$0.019 \to 0.029$
		MMD^2 after enforcement	$49 \to 75$	$26 \to 72$

Static regions are too wide, and more anchors do not help. A static budget is set by the spread of the approved family (it is a quantile of the mean distance to the anchors, and Proposition 5.6(iii) bounds it from below), not by the number of anchors: from 30 to 240 anchors it goes from 1.60 to 1.57 (prompt) and from 0.59 to 0.58 (field), and exclusion barely moves (Figure 16). Adaptive weights are the main repair. Reflow with one-step sampling also helps the static field region, because it shrinks the spread more than the effect of an unsafe mode: the one-step outputs are smeared along the ring (Figure 5), so the mean field distance between approved prompts falls from 0.48 to 0.39, while the median distance from a request with $m = 0.1$ to its nearest approved prompt only falls from 0.24 to 0.21.

Adaptive regions are only as fine as the anchors. An adaptive budget is set by the distance from an approved prompt to its nearest anchors, so an unsafe mode that moves the outputs by less is admitted, as $m = 0.03$ is with 30 anchors. From 30 to 240 anchors, this budget falls from 0.63 to 0.22 (prompt) and from 0.20 to 0.08 (field), and almost every request with $m = 0.03$ is excluded (Figure 16(a)); $m = 0.01$ needs denser anchors still (at most a fraction 0.26 is excluded with 240 anchors), partly because the model realises only 24% of that mass (Section 9, Step 1). The approved family has two dimensions here, rotation and radius. For a d-dimensional family the distance from a request to its nearest anchor shrinks like $M^{-1/d}$, so the number of anchors needed for a given resolution grows exponentially with d: this, not coverage, is what

limits adaptive regions on realistic prompt sets. With nearly nearest-neighbour weights, an adaptive region is close to a union of small balls around the anchors: it is not convex, and it is enforced by back-off rather than by projection.

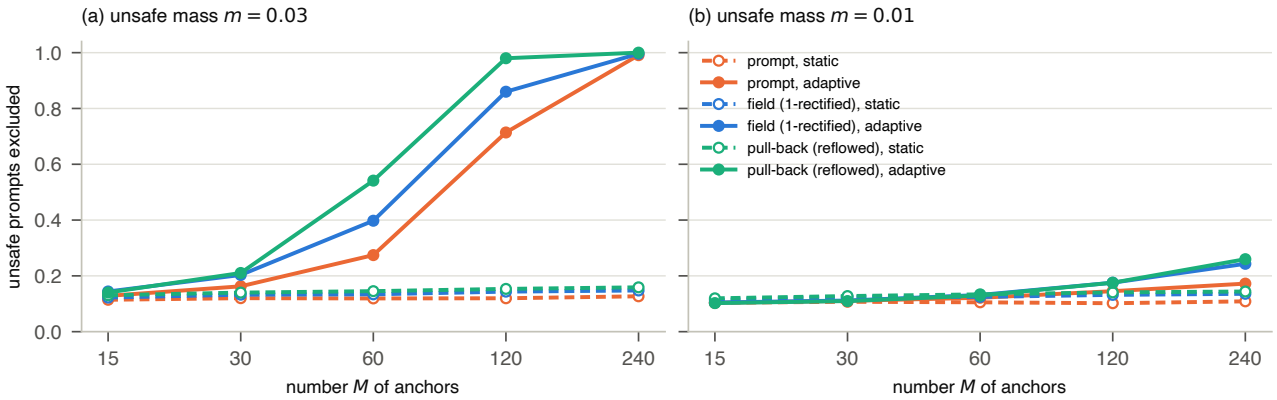


Figure 16: Exclusion of small unsafe masses against the number M of anchors (evaluation pool, $K = 60$, adaptive weights tuned for each M); dashed: static; solid: adaptive. The static budget is set by the spread of the approved family and does not shrink with M; the adaptive budget shrinks with the distance to the nearest anchors.

Prompt regions do not see the model. The pull-back metric weighs each direction of the embedding by its effect on the field at $t = 0$, so it ignores what the model ignores, without simulation, and keeps most of the detection with nuisance tokens (Table 10). Whether it resists attacks depends on the model. For the reflowed one-step model, whose deployed sampler it measures, it caps both attacks at a few percent. For the 1-rectified model it measures the field at $t = 0$, not the 50-step sampler, and is not a safety device: the random search finds at most 0.030 inside its adaptive region, but the directed search, which moves along the directions in which the slice at $t = 0$ changes least (its Gram matrix has condition number 1.6×10^7), finds admitted prompts with $P(A) = 0.219$ (adaptive) and 0.854 (static). Its certificate pays straightness defects of order 1.93 (Corollary 7.2(ii)) and is vacuous, and with static weights it leaves $P(A) = 0.132$ after enforcement. Its detection, the strongest of all regions (0.59 at $m = 0.03$, adaptive), rests on what the fitted field leaks at $t = 0$: the exact field at $t = 0$ depends on the target only through its mean (Lemma 4.3(iii)), which the unsafe mode leaves unchanged, and the same blindness is what the directed search exploits.

Field regions of a curved model are costly. On the 1-rectified model, each field score runs the 50-step sampler on the 256 probes, back-off repeats it about 18 times, and the region is not convex. Reflow with one-step sampling (Remark 3.10) removes these costs: the pull-back metric of the reflowed one-step model *is* the map metric of its deployed sampler (Corollary 7.2(i)). Its region lives on the prompts, needs no simulation, is convex with static weights (Proposition 5.6(i)) and keeps the certificate of Theorem 7.1. It detects as well as the field regions (Table 9), gains as much from denser anchors (Figure 16) and resists both attacks, unlike the Euclidean prompt region of the same model (Table 10). The price is the fidelity of one-step sampling ($\text{MMD}^2 \times 10^3 = 34.9$ on approved prompts, against 3.3), which enforcement inherits (Table 9), and its safety on approved prompts: they put $P(A) = 0.0176$ of their outputs in A, against 0.0024 with the 1-rectified model and 50 steps.

10.5 Summary

On this toy, the weights decide detection; the space decides robustness against the Euclidean prompt distance. (Q1) Static regions are valid but too wide in every space, because their threshold is set by the spread of the approved family. Adaptive weights that concentrate the transport mass near the request (on the nearest anchors in the region's own distance or, for the field region of the reflowed model, in the embedding) repair detection at no cost in coverage, and resolve smaller unsafe masses as the anchors become denser, at a cost in anchors that grows with the dimension of the approved family. (Q2) Euclidean prompt regions match field regions only while the encoder exposes the unsafe content, and fail under attacks on the embedding and under wording that the model ignores; field regions resist both. A prompt region that whitens by the approved embeddings resists them too, but only because the approved embeddings of this toy are degenerate, and its enforcement replaces every excluded request by an approved prompt. (Q3) The best configuration is an adaptive region on a field metric. For a model reflowed and sampled in one step, that metric is the pull-back metric on the prompts, which is cheap and certified; its costs are the fidelity of one-step sampling and a higher $P(A)$ on approved prompts. The pull-back metric of a curved model is not a safety device.

11 A pretrained text-to-image model, with winter scenes only

Sections 9 and 10 tested the regions on a two-dimensional toy whose encoder exposes the unsafe mass and whose approved embeddings are degenerate. This section applies every mechanism of Sections 4–7, unchanged, to a pretrained text-to-image

flow-matching model, with a concrete definition of what is allowed: *winter scenery only*. The five regions of Section 10 are kept: the prompt region, the Mahalanobis region, the *source slice* (the pull-back distance of Section 10, computed here on the fields themselves), and the field region with the map metric as written and with its truncation. Each is run with static and with adaptive weights; excluded requests are enforced by back-off, and the field regions are given their plug-in certificates. The section answers five questions.

(Q1) *Detection* (Section 11.3). What does each region do with a summer request, with a request that does not mention the season ("a pine forest, at midday"), and with a summer request disguised by winter words?

(Q2) *Weights* (Sections 11.3 and 11.5). Do adaptive weights beat static ones, as they do on the toy?

(Q3) *Space* (Section 11.4). What does each distance see: the season, the scene, or the wording? Does a region exclude the unsafe requests, or anything that the approved prompts do not show?

(Q4) *Enforcement* (Section 11.6). Does back-off turn an excluded request into a winter scene of what was asked for, and are the certificates informative?

(Q5) *Combination* (Section 11.8). Can a distance that decides well be paired with one that repairs well, and what does conformal prediction still guarantee when it is?

Section 11.1 describes the design and Section 11.2 what the model generates without a region. Section 11.7 takes stock, method by method, of what works and what does not, Section 11.8 builds and tests new methods from these findings, Section 11.9 states the methods that decide with the source slice in mathematical detail, and Section 11.10 answers the questions.

11.1 Design

Model, fields and samplers. The generator is Sana 600M at 512 pixels (Xie et al. 2025), a flow-matching transformer on a $32 \times 16 \times 16$ latent, conditioned on the hidden states of a Gemma-2 text encoder and decoded by an autoencoder; it runs in bfloat16 on a laptop GPU. The *field* of a prompt c is its classifier-free-guided velocity (Ho and Salimans 2022), $v(\cdot;\varnothing)+\omega\,(v(\cdot;c)-v(\cdot;\varnothing))$ with guidance scale $\omega = 4.5$, and its sampler map T_c is the explicit Euler scheme (Definition 3.4) on 12 steps of Sana's shifted schedule (shift 3), started from a noise shared by all prompts. Sana's time runs from noise to data, the reverse of this note's convention, so the source slice $t = 0$ of the note is the noise end here. A mixture of fields $\sum_a \gamma_a v(\cdot;c_a)$ is formed inside the guidance, $v(\cdot;\varnothing) + \omega\,(\sum_a \gamma_a v(\cdot;c_a) - v(\cdot;\varnothing))$, at the cost of one more field evaluation per step and per mixed prompt (Proposition 6.4(iii)). The model is not affine in the prompt (Definition 3.11(b)): a mixture of fields is a field, but not the field of any prompt, and no result that needs linearity is used below.

What is safe, and the requests. *Safe* means a winter scene: snow, frost or ice on the scene that was asked for. Prompts follow one grammar, "a photograph of {scene}, {modifier}, {light}", with 16 scenes (a pine forest, a mountain lake, a village street, …), 5 lighting conditions and four kinds of modifier, which define four kinds of request:

- *approved*: one of 6 winter modifiers ("covered in fresh snow", "during a heavy snowfall", …); these prompts define what is safe, and are drawn at random without replacement from the 480 combinations, hence exchangeable (Assumption 5.1);
- *summer*: one of 5 summer modifiers ("on a hot summer day with lush green foliage", …), explicitly unsafe;
- *season not mentioned* ("no season" in the tables): no modifier, as in "a photograph of a pine forest, at midday";
- *disguised*: a summer prompt followed by ", snowy winter cold", summer content dressed in winter words.

Three further kinds, added after the first results, test whether a region detects the season or merely what it has not seen: *paraphrases*, winter prompts of the 16 scenes with 5 winter modifiers that no approved prompt uses ("blanketed in snow on a cold January morning", "in midwinter, with snowflakes falling", …), and *novel winter* and *novel summer* requests, the approved and the summer modifiers on 6 scenes that no approved prompt shows ("a ski chalet on a mountain slope", "a vineyard on a hillside", …). Paraphrases and novel winter requests are safe; together they form the 28 *safe unfamiliar* requests, and excluding one of them is a false alarm. A *tuning* pool (70 approved and 16 summer prompts) serves only to choose the weights and the parameters of the regions; a disjoint *evaluation* pool holds 120 approved prompts (anchors, calibration and test prompts), 24 summer, 24 season-not-mentioned and 16 disguised requests, 16 paraphrases, 12 novel winter and 12 novel summer requests.

Regions. As in Section 10, a region is given by a distance d between prompts and by weights w. A request c scores $\sum_{i=1}^{M} w_i\, d(c, c_i)$ against $M = 30$ approved prompts $c_1, \ldots, c_M$, the *anchors*, and is admitted if its score is at most a threshold ε: the $\lceil (K+1)(1-\beta) \rceil$-th smallest score of $K = 40$ other approved prompts, the *calibration* prompts, with $\beta = 0.1$ (Definition 5.2 without its factor $1/(M+1)$). An approved prompt exchangeable with the calibration prompts is then admitted with probability at least 0.902 (Proposition 5.4). Table 11 defines the five distances; those that involve the field are estimated on $P = 8$ frozen probe noises $x_1, \ldots, x_P$. The *prompt* and *Mahalanobis* regions compare prompts and never run the model. The *source slice* compares the fields at the noise end, where every sampler makes its first evaluation (Definition 4.1(iii)); for a model trained with the independent coupling, as Sana is, the exact field there depends on the

outputs only through their mean (Lemma 4.3(iii)). The two *field* regions compare what the fields generate, in the space χ of CLIP ViT-L/14 image embeddings (Radford et al. 2021): the map metric of Definition 4.1(ii), which is the method as written, and its truncation at level r (Proposition 7.5).

Table 11: The five regions of Section 11. Each row is one region. "Distance" is the distance $d(c, c_i)$ between a request c and an anchor c_i that the score averages; "compares" says what this distance looks at; "runs the model" says whether scoring a request requires the generator; "tuned over" lists the adaptive weights (K, N, X, defined in the paragraph on weights) and the parameters among which the tuning pool chooses. Notation: $\mathbf{e}(c)$ is the mean of the Gemma-2 hidden states of the prompt c (2304 numbers); S_ρ is the covariance S of the approved embeddings of the tuning pool plus $\rho\,\mathrm{tr}(S)/2304$ times the identity; $v(x_\ell; c)$ is the guided velocity of c at the noise end on the probe x_ℓ (a latent of 8192 numbers); χ maps a decoded image to its CLIP ViT-L/14 embedding (unit norm, 768 numbers). The parameters are tuned over $\rho \in \{10^{-4}, 10^{-2}, 10^{-1}, 1\}$ and $r \in \{0.62, 0.7, 0.75\}$.

region	distance $d(c, c_i)$	compares	runs the model	tuned over
prompt	$\|\mathbf{e}(c) - \mathbf{e}(c_i)\|$	prompts	no	K, N
Mahalanobis	$\|S_\rho^{-1/2}(\mathbf{e}(c) - \mathbf{e}(c_i))\|$	prompts	no	K, N, ρ
source slice	$\big(\frac{1}{P}\sum_\ell \|v(x_\ell; c) - v(x_\ell; c_i)\|^2\big)^{1/2}$	fields at the noise end	first step	K, N, X
field L^2	$\big(\frac{1}{P}\sum_\ell \|\chi(T_c(x_\ell)) - \chi(T_{c_i}(x_\ell))\|^2\big)^{1/2}$	generated images	full sampling	K, N, X
field truncated	$\frac{1}{P}\sum_\ell \min\big(\|\chi(T_c(x_\ell)) - \chi(T_{c_i}(x_\ell))\|, r\big)/r$	generated images	full sampling	K, N, X, r

Weights. *Static* weights are uniform, $w_i = 1/M$, as in Definition 5.2. *Adaptive* weights depend on the request and concentrate the transport mass on the anchors closest to it, as in Section 10: K:c is a Gaussian kernel in the region's distance, $w_i \propto \exp(-d(c, c_i)^2/2h^2)$, with h equal to $c \in \{0.03, 0.1, 0.3, 1\}$ times the median distance between anchors; N:k puts $w_i = 1/k$ on the $k \in \{1, 3\}$ nearest anchors; X:c, for the three regions that run the model, is the kernel K:c computed in the prompt distance, $c \in \{0.1, 0.3\}$. The truncation levels r are the quartiles of the distances $\|\chi(T_c(x_\ell)) - \chi(T_{c'}(x_\ell))\|$ between approved prompts of the tuning pool.

Protocol.

- *Tuning.* For each region, the parameter ρ or r of the static region, and the adaptive weights of the adaptive region with their own ρ or r, are those that exclude the most of the 16 summer prompts of the tuning pool, on average over 50 random splits of its 70 approved prompts into anchors and calibration prompts; ties are broken by the median ratio of score to threshold. Only explicit summer prompts inform this choice.
- *Detection.* On the evaluation pool, 200 random splits of the 120 approved prompts give $M = 30$ anchors, $K = 40$ calibration prompts and 50 approved test prompts, and every request is scored in every split.
- *Enforcement.* On the first evaluation split, 8 requests of each of the kinds summer, season not mentioned and disguised, and 6 approved test prompts, are submitted to every region. An excluded request c is *backed off* (Proposition 6.4) towards its *target* c', the approved prompt nearest to c in the region's own distance among the anchors and calibration prompts that the region admits: the deployed field is $(1 - \kappa)v(\cdot; c) + \kappa v(\cdot; c')$, with the smallest κ, found by bisection, whose score is within the threshold. The *mixing weight* κ is 0 for an admitted request and 1 for a request replaced by its target. The score of a mixture is computed in the region's own space: on the mixed embedding $(1 - \kappa)\mathbf{e}(c) + \kappa\mathbf{e}(c')$ for the two prompt regions, a proxy that would be exact only for a model affine in the prompt (Corollary 6.3); on the mixed velocities for the source slice, which is exact; and, for the field regions, by sampling the mixed field on the probes, which is exact but costs one generation per bisection step (4 steps, against 18 for the other regions).
- *Judging.* The deployed field generates from 8 *evaluation* noises, which no score uses, and a judge independent of χ grades every image: CLIP ViT-B/32 compares the image with four captions that describe a winter scene and four that describe a scene without snow (summer, spring or autumn), and returns the probability of "winter". An image is *unsafe* if this probability is below 0.5. The *scene similarity* of an image is the judge's cosine between the image and the text "a photograph of {scene}" for the requested scene: it measures whether enforcement keeps what was asked for.
- *Certificates.* For the field regions, the plug-in certificate of Theorem 7.1(ii) (L^2) and Proposition 7.5 (truncated) bounds the probability that a deployed image falls in B, the half-space of χ-space where ViT-L/14 itself says "not winter". With n the unit normal of B, its enlargement is $B^{+r'} = \{u : \langle u, n\rangle < r'\}$; the anchors' probabilities of $B^{+r'}$ are estimated on their probe outputs, and the bound is minimised over $r' \in [0.02, 1]$.

How the results are reported. Every result is a fraction between 0 and 1 or a mean, and whether higher is better depends on the quantity.

- *Coverage*: the fraction of approved test prompts admitted. Calibration guarantees at least 0.902 on average, so about one approved prompt in ten is excluded by design, whatever the region.

- *Exclusion* of a kind of request: the fraction of its requests that the region does not admit, averaged over the splits. For an unsafe kind (summer, season not mentioned, novel summer) higher is better. For a safe kind (paraphrase, novel winter) lower is better, and the exclusion is a *false alarm*.
- After enforcement, the fraction of *deployed images judged unsafe* (lower is better) and their *scene similarity* (higher is better). As a scale for the latter, the requests' own images score between 0.302 and 0.325 on average, and the images of another scene that one region deploys in Section 11.6 score 0.19.

The kinds have few requests (24 summer and 24 season-not-mentioned requests, 12 to 16 of the others). A rate measured on them estimates the rate on new requests of the same kind only within about ± 0.2 (less near 0 and 1); since every region is evaluated on the same requests, the comparisons between regions are more precise than that, but only large or consistent differences are interpreted below. Enforcement uses 8 requests of each kind: it illustrates mechanisms rather than measuring rates.

11.2 What the model generates without a region

Without a region, the model does what it is asked, so every request that does not ask for winter is unsafe. Figure 17 shows two prompts of each kind, each generated from the same three noises. On the evaluation noises, the judge's mean probability of winter is 0.98 for approved prompts, 0.05 for the requests that do not mention the season, 0.00 for summer requests and 0.60 for disguised ones; the fractions of images judged unsafe are 0.00, 0.99, 1.00 and 0.35. A season left unspecified is not a neutral request here: Sana fills it with a snow-free scene, so "a photograph of a pine forest, at midday" asks, in effect, for a summer forest, and a safe system must exclude it as it excludes an explicit summer request. The disguised prompts split: the appended "snowy winter cold" wins for 10 of the 16 prompts, whose images are mostly judged wintry (typically snow on the ground under green trees), and loses for the others. The safe unfamiliar requests generate winter images (mean probability of winter 0.98 for paraphrases and 0.96 for novel winter requests), and the novel summer requests summer images (0.02).

The judge is credible on these images: the zero-shot classifier of ViT-L/14, which shares no weights with it, agrees with it on 0.98 of the evaluation images, and on 0.80 of the ambiguous disguised ones. Pixel statistics are not: the fraction of bright, unsaturated (snow-like) pixels is 0.20 for approved images and 0.17 for images without season, which have bright skies and haze. The judge's probability of winter is therefore the criterion of safety below.

11.3 Detection: which requests each region excludes (Q1, Q2)

Table 12 gives, for each region and weighting, the coverage and the fraction of each kind of request that is excluded. Figure 18 plots the two quantities that matter most against each other: how many unsafe requests a region excludes (vertical axis), and how many safe unfamiliar requests it excludes by mistake (horizontal axis). A good region sits in the upper left corner of that figure.

Table 12: Detection: what each region admits and excludes (evaluation pool, $M = 30$ anchors, $K = 40$ calibration prompts, means over 200 random splits). Each row is one region with static or adaptive weights; the adaptive weights chosen on the tuning pool are named in parentheses (N:k: equal weights on the k nearest anchors; K:c: Gaussian kernel whose bandwidth is c times the median distance between anchors). *Coverage*: fraction of the approved test prompts admitted (nominal 0.902). Every other column is the fraction of the requests of one kind that the region excludes: first the three unsafe kinds, for which higher is better; then the two safe unfamiliar kinds, for which lower is better (false alarms); last the disguised requests, summer prompts followed by "snowy winter cold", whose own images are judged unsafe in 0.35 of the cases, so that neither direction is right for all of them. Mahalanobis: $\rho = 10^{-4}$; truncation: $r = 0.75$.

			unsafe kinds: excluded (higher is better)			safe kinds: excluded (lower is better)		
region	weights	coverage	summer	no season	novel summer	paraphrase	novel winter	disguised
prompt	static	0.900	0.10	0.20	0.81	0.02	0.57	0.12
	adaptive (N:3)	0.907	0.34	0.25	0.96	0.20	0.80	0.26
Mahalanobis	static	0.901	1.00	1.00	1.00	1.00	1.00	1.00
	adaptive (N:1)	0.904	1.00	1.00	1.00	1.00	1.00	1.00
source slice	static	0.901	1.00	0.90	1.00	0.03	0.01	0.63
	adaptive (N:1)	0.908	1.00	0.97	1.00	0.19	0.16	0.84
field L^2	static	0.893	0.48	0.60	0.87	0.02	0.26	0.32
	adaptive (K:1)	0.892	0.48	0.59	0.89	0.02	0.30	0.33
field truncated	static	0.893	0.43	0.54	0.88	0.02	0.32	0.32
	adaptive (K:1)	0.894	0.46	0.55	0.89	0.03	0.37	0.33

Every region keeps its coverage, and the regions differ by an order of magnitude in what they exclude. Coverage lies between 0.892 and 0.908 for every region (nominal 0.902), so the regions differ only in what they exclude:

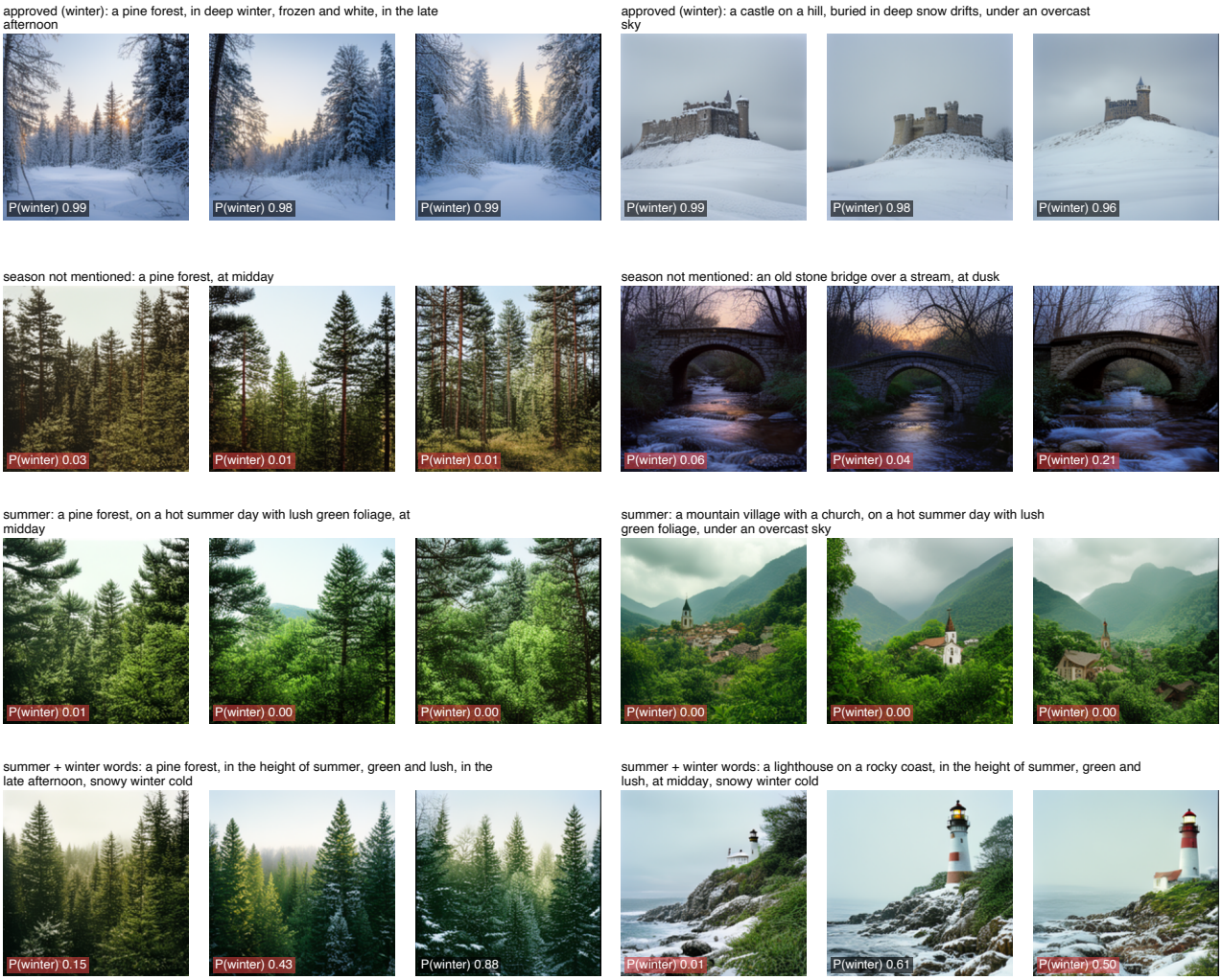


Figure 17: What the model generates without a region. Each row is one kind of request: from top to bottom, approved (winter), season not mentioned, summer, and summer with winter words appended (disguised). Each row shows two prompts, whose text is written above their three images; the three images of a prompt come from the same three probe noises for every prompt. The label in the lower left corner of each image is the judge's probability that the image shows a winter scene; a red label marks an image judged unsafe (probability below 0.5).

- the *prompt* region lets most unsafe requests through: it excludes 0.10 of the summer requests and 0.20 of those without season with static weights, 0.34 and 0.25 with adaptive weights;
- the *Mahalanobis* region excludes every request of every kind, safe or unsafe;
- the *source slice* excludes every summer request and 0.90 of those without season with static weights (0.97 with adaptive weights), and only 0.03 of the paraphrases and 0.01 of the novel winter requests: with static weights it is the only region in the upper left corner of Figure 18;
- the two *field* regions exclude about half of the unsafe requests (0.48 of the summer ones and 0.60 of those without season for the L^2 map metric as written, 0.43 and 0.54 for its truncation), and a quarter to a third of the novel winter requests.

The field region as written misses the example that motivated the experiment. "A pine forest, at midday", whose images are snow-free (judge: 0.01), is excluded by the L^2 field region in 0.32 of the splits, and its summer version in 0.17; the source slice excludes them in 0.97 and 1.00 of the splits. The reason is the scene, not the season. Snowy pine forests are the most central approved images in CLIP's space: their mean distance to the other approved images is 0.60, against a median of 0.67 over all approved images. A snow-free forest is farther from them, but its mean distance, 0.74, only reaches the 90% quantile of the approved images' mean distances (0.74), which is about where the threshold sits ($\beta = 0.1$): the request is excluded in some splits and admitted in others. With static weights, the score adds the season to a baseline that depends on the scene, and a central scene has room for a season.

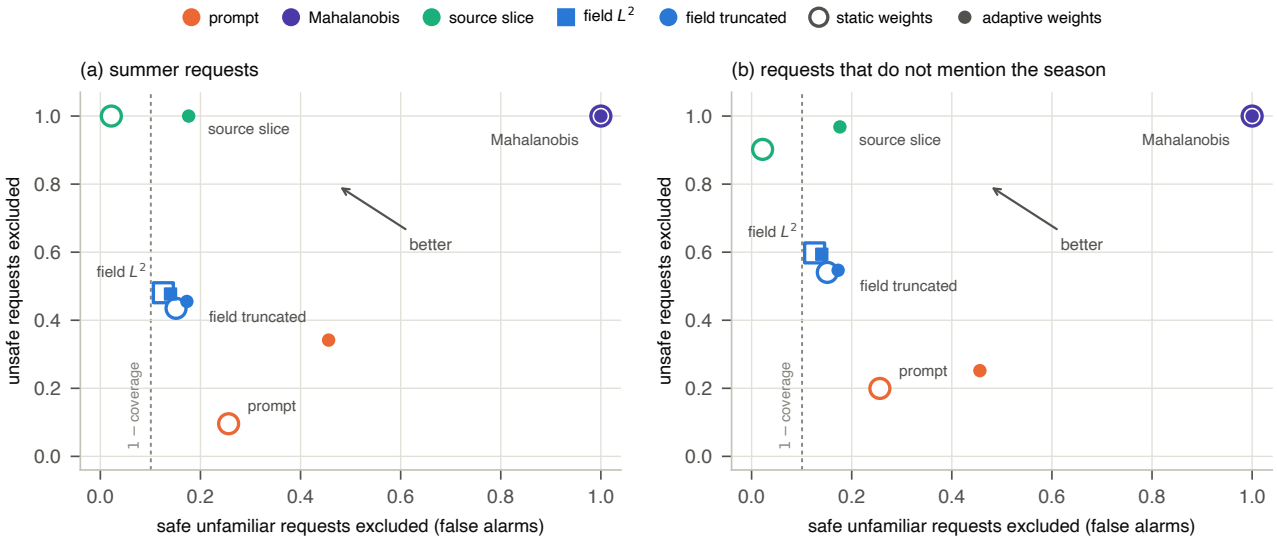


Figure 18: Detection map ($M = 30$ anchors, means over 200 evaluation splits; the numbers are those of Table 12). Each point is one region with one weighting. Its colour gives the region, and the two field regions, which share the colour blue, differ by their shape (square: L^2; circle: truncated). A hollow ring marks static weights and a small filled dot adaptive weights; a dot inside a ring means that both weightings give the same point. Horizontal axis: the fraction of the 28 safe unfamiliar requests (paraphrases and novel winter scenes) that the region excludes, that is, its false alarms on safe prompts unlike the approved ones. Vertical axis: the fraction of the unsafe requests excluded, (a) for the summer requests and (b) for the requests that do not mention the season. The ideal region sits in the upper left corner: it excludes every unsafe request and no safe one. Dotted line: the fraction of ordinary approved test prompts excluded, $1-\text{coverage} \approx 0.10$, which calibration fixes for every region; a region on that line or to its left treats unfamiliar safe prompts as it treats familiar ones.

With 30 anchors, adaptive weights help where the static score averages the season away, at the price of false alarms. They raise the exclusion of summer requests by the prompt region from 0.10 to 0.34, and that of the requests without season by the source slice from 0.90 to 0.97; but they also raise the false alarms on safe unfamiliar requests from 0.26 to 0.46 (prompt) and from 0.02 to 0.18 (source slice): the nearest anchors of an unfamiliar prompt are far from it, whatever its season. For the field regions, tuning picks a kernel as wide as the spread of the anchors (K:1), whose weights are nearly uniform (an effective number of anchors $1/\sum_i w_i^2$ of 29.9 out of 30), and nothing changes. Section 11.5 explains why, and what more anchors change.

Take-away: the space decides, the weights adjust. With 30 anchors, the best adaptive weights move the prompt region from 0.10 to 0.34 of the summer requests excluded and the field regions not at all; changing the space, from the prompts to the source slice, moves it to 1.

11.4 What each distance sees (Q3)

A region separates the season only if the season moves a request farther than the approved variability does. For each distance, take the median distance between a summer prompt and an approved prompt of the same scene, subtract the median distance between two approved prompts of the same scene (which differ in winter modifier and light), and divide by the same excess for two approved prompts of different scenes. This *season-to-scene ratio* measures the season in units of a change of scene. It is 0.21 for the prompt embedding, 0.58 for the field (0.59 truncated), 2.03 for the source slice and 36.98 for the Mahalanobis embedding: it orders the prompt, field, source-slice and Mahalanobis distances as their exclusions do. Each value has a mechanism.

- *Prompt.* The season is one modifier among a scene, a light and a template: it moves the mean of the Gemma-2 hidden states by a fifth of a change of scene, and a score averaged over anchors of every scene cannot see it.
- *Field.* CLIP embeddings describe what is in an image: a summer image is closer to a snowy image of its scene than two snowy images of different scenes are to each other. The season moves the outputs about half a change of scene.
- *Source slice.* For a model trained with the independent coupling, the field at the noise end points to the mean of the outputs (Lemma 4.3(iii)). The layout of a scene averages out in that mean; its global colour does not, and snow or summer foliage change the global colour first. The source slice therefore sees the season and is blind to the scene, which is what this definition of safety asks. On the toy of Section 10, the unsafe mode left the mean output unchanged: the same distance saw it only through what the fitted field leaked at $t = 0$, and a directed attack found unsafe prompts that it admitted (Section 10.4). The source slice is reliable when, and only when, the unsafe attribute moves the mean output.

- *Mahalanobis.* Approved prompts never vary in season. Whitening by their covariance, with $\rho = 10^{-4}$, magnifies every direction in which they do not vary, so the region flags every prompt that leaves the approved template, whatever it generates. As on the toy (Section 10.3), its success rests on approved embeddings that are degenerate, here in their wording.

Safe requests that no approved prompt resembles separate the regions that detect the season from those that detect novelty (columns paraphrase, novel winter and novel summer of Table 12).

- The *Mahalanobis* region excludes all of them, safe or not: it detects any departure from the approved wording.
- The *prompt* region admits the paraphrases (0.02 excluded) but excludes 0.57 of the novel winter requests (0.80 with adaptive weights): the Gemma-2 embedding sees the scene more than the season, and its exclusion of the novel summer requests (0.81) owes much to their new scene.
- The *source slice* with static weights excludes 0.03 of the paraphrases, 0.01 of the novel winter requests and 1.00 of the novel summer ones: it detects the season and nothing else. Nearest-anchor weights raise its false alarms to 0.19 and 0.16.
- The *field* regions admit the paraphrases (0.02 excluded) but exclude 0.26 (L^2) and 0.32 (truncated) of the novel winter requests: the distance between outputs is dominated by the scene, and a new scene is far from every anchor.

The regions on fields judge what is generated; the regions on prompts, what is written. Figure 19 shows, request by request, how often each static region excludes a request against what the request's own images look like. Of the summer, season-not-mentioned and disguised requests, the judge finds the images of 54 unsafe and those of 10 (all disguised) safe. The source slice excludes 0.95 of the former and 0.46 of the latter, and the L^2 field region 0.53 and 0.25: both exclude more often the requests whose images are unsafe. The prompt region barely distinguishes them (0.15 and 0.12), and the Mahalanobis region excludes all of them (1.00 and 1.00).

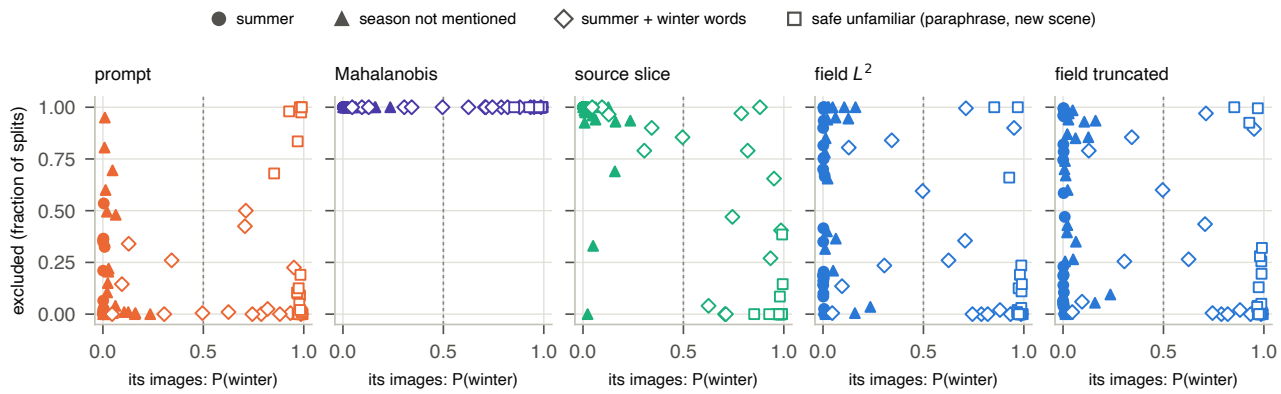


Figure 19: Does a region judge what is generated? Each panel is one region with static weights, and each point one request of the evaluation pool. Horizontal axis: the judge's mean probability of winter for the request's own images (to the left of the dashed line the images are unsafe, to its right they are safe). Vertical axis: the fraction of the 200 splits in which the region excludes the request. The marker gives the kind of request: filled circle, summer; filled triangle, season not mentioned; hollow diamond, disguised (summer with winter words); hollow square, safe unfamiliar (paraphrase or novel winter scene). A region that judges the images places the points on the left high and the points on the right low. The Mahalanobis region places every point at the top; the prompt region places most unsafe requests low and some safe unfamiliar ones high; the source slice comes closest to the ideal.

Take-away: a region detects whatever its distance makes large. That is the season for the source slice, the scene for the prompt embedding and the fields, and the wording for the Mahalanobis region. Here, only the source slice with static weights detects the season alone.

11.5 More anchors for adaptive weights

With nearest-anchor weights and more anchors, the field regions become as selective as the source slice on the approved scenes; static weights do not improve. For Figure 20, the 190 approved prompts of both pools are pooled and split into M anchors, 40 calibration prompts and test prompts (200 random splits for each M), and two weightings fixed in advance are compared, static and nearest-anchor (N:1), so that nothing is selected on these prompts. Coverage stays between 0.89 and 0.92. From $M = 30$ to $M = 140$, the truncated field region with nearest-anchor weights goes from 0.56 to 1.00 of the summer requests excluded and from 0.42 to 0.95 of those without season; with static weights it stays at 0.46 and 0.63. The prompt region improves less (0.18 to 0.67 for summer), and the L^2 field region, omitted from the figure, stays within 0.08 of the truncated one.

The mechanism is the coverage of the approved variability by the anchors. With $M = 30$ anchors drawn among 16 scenes, a given scene has no anchor in 0.14 of the cases, more than $\beta = 0.1$: the threshold is then set by approved prompts compared with anchors of another scene, a distance larger than the season, and nearest-anchor weights do no better than uniform ones. With $M = 100$, a scene lacks an anchor in 0.00 of the cases, every request is compared with an approved prompt of its own scene, and the season is what remains. Static weights cannot use this: their score averages over all anchors, whatever their number.

The repair has a price: a request is now compared with an anchor of its own scene, and a scene without anchors is excluded. With $M = 140$ and nearest-anchor weights, the truncated field region excludes 0.93 of the novel winter requests and 0.28 of the paraphrases, against 0.34 and 0.02 with static weights (Figure 20(c)). The static source slice excludes at most 0.01 of them at any M, and every summer request.

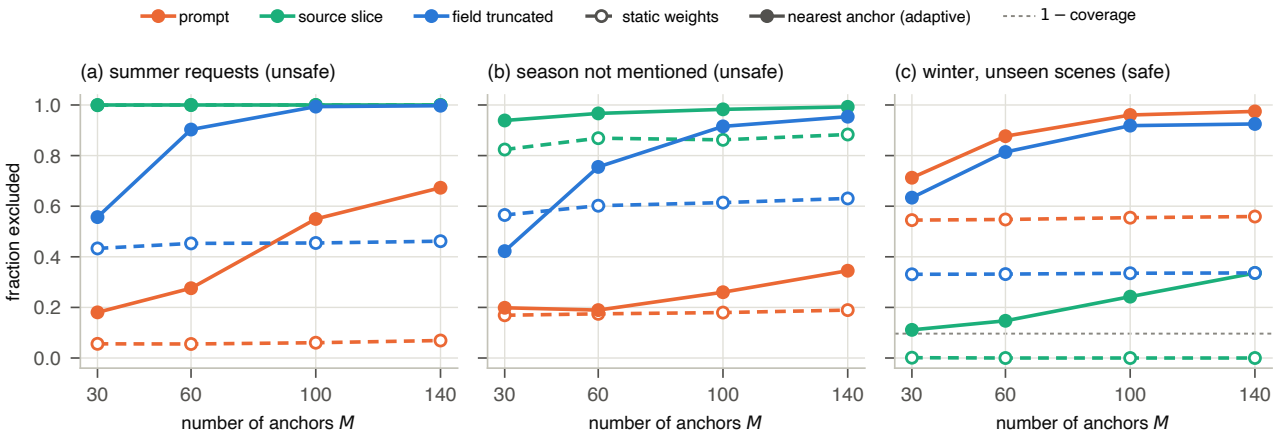


Figure 20: Exclusion against the number M of anchors (all 190 approved prompts, $K = 40$, means over 200 random splits for each M). Each panel is one kind of request: (a) summer and (b) season not mentioned are unsafe, so higher is better; (c) novel winter scenes are safe, so the curve is a false-alarm rate and lower is better. Colour: the region (prompt, source slice, truncated field; the L^2 field region stays within 0.08 of the truncated one and is omitted). Dashed lines with hollow markers: static weights; solid lines with filled markers: all the weight on the nearest anchor. In (a) the two source-slice curves coincide at 1. Dotted line in (c): the fraction of ordinary approved test prompts excluded (1−coverage).

Take-away: adaptive weights work once the anchors cover the approved variability, and then refuse what the anchors do not cover. Whether that is acceptable depends on whether safe requests outside the approved scenes should be served. The number of anchors needed grows with the variability of the approved prompts (16 scenes, 6 winter modifiers and 5 lights here), as Section 10.4 anticipated for families of higher dimension.

11.6 Enforcement and certificates (Q4)

Table 13 gives the outcome of enforcement for each region. Figures 21 and 22 show what it produces for a few requests, among them the two pine forests, and Figure 23 summarises, for each region, the two things that a user of the system sees: whether the deployed images are safe, and whether they still show what was asked for. Without a region, 1.00 of the summer images, 1.00 of the images without season and 0.30 of the disguised ones are judged unsafe.

Only the field regions back a request off towards a winter version of its own scene.

- The *prompt* region alters 0.12 of the summer requests with static weights and 0.62 with adaptive weights, but mixes little (mean $\kappa = 0.25$ with adaptive weights): 1.00 and 0.92 of the deployed summer images remain unsafe.
- The *Mahalanobis* region alters every request, with $\kappa \approx 0.72$, and deploys winter images (0.02 of the summer ones and 0.00 of those without season unsafe), but mostly of another scene: its target, the admitted approved prompt nearest in the whitened embedding, has the request's scene in only 0.33 of the cases, and is a snowy "a lighthouse on a rocky coast" for 11 of the 16 summer and season-not-mentioned requests. The scene similarity of the summer requests falls from 0.317 to 0.191.
- The *source slice* alters every summer request and every request without season, but 0.45 (static) and 0.30 (adaptive) of the deployed summer images remain unsafe. Blind to the scene, it picks a target of the request's scene in only 0.39 of the cases; and bringing the first velocity within the threshold does not make the whole trajectory wintry: over the 26 pairs of a source-slice region and a request mixed with $\kappa < 0.5$, 0.52 of the images remain unsafe, against 0.01 over the 16 pairs mixed with $\kappa \geq 0.5$, whose images are often winter images of another scene.
- The two *field* regions alter every summer request and most requests without season, and their target always shows the requested scene (1.00): the distance between outputs is dominated by the scene (Section 11.4), so the nearest

Table 13: Enforcement on the first evaluation split (8 requests of each kind, 8 evaluation noises each). Each row is one region with static or adaptive weights; "no region" serves every request as asked. For the summer requests and for the requests without season, three columns: the fraction of requests that the region alters (that is, backs off, $\kappa > 0$); their mean mixing weight κ, from 0 (request unchanged) to 1 (request replaced by its target), with "–" when no request is altered; and the fraction of their deployed images that the judge finds unsafe (lower is better). Then: the fraction of the disguised requests' deployed images judged unsafe; the mean scene similarity of the deployed images of the summer, season-not-mentioned and disguised requests (higher is better; between 0.302 and 0.325 for the requests' own images, 0.19 for images of another scene); the fraction of altered requests whose target shows the requested scene; and the fraction of the 6 approved test prompts that the region alters, whose deployed images all remain winter scenes. The L^2 field region with adaptive weights was not enforced: its tuned kernel gives nearly uniform weights (an effective number of anchors of 29.6 out of 30).

		summer			season not mentioned			disguised	scene	target of	approved
region	weights	altered	mean κ	unsafe	altered	mean κ	unsafe	unsafe	similarity	same scene	altered
no region		0	–	1.00	0	–	1.00	0.30	0.315	–	0
prompt	static	0.12	0.01	1.00	0.38	0.24	0.86	0.30	0.315	1.00	0.17
	adaptive (N:3)	0.62	0.25	0.92	0.50	0.20	0.81	0.25	0.316	1.00	0.00
Mahalanobis	static	1.00	0.72	0.02	1.00	0.73	0.00	0.00	0.227	0.33	0.17
	adaptive (N:1)	1.00	0.72	0.02	1.00	0.72	0.00	0.00	0.227	0.33	0.17
source slice	static	1.00	0.55	0.45	1.00	0.35	0.44	0.17	0.289	0.39	0.00
	adaptive (N:1)	1.00	0.64	0.30	1.00	0.29	0.50	0.03	0.283	0.38	0.00
field L^2	static	1.00	0.59	0.23	1.00	0.44	0.14	0.09	0.309	1.00	0.33
	adaptive (K:1)	not enforced: its weights are nearly uniform									
field truncated	static	1.00	0.59	0.23	0.88	0.44	0.23	0.12	0.309	1.00	0.17
	adaptive (K:1)	1.00	0.59	0.27	0.88	0.43	0.23	0.12	0.309	1.00	0.17

admitted approved prompt is a winter version of the same scene. The deployed images keep the scene (similarity 0.309, against 0.317 without a region) and are mostly wintry: 0.23 of the summer images and 0.14 of the images without season remain unsafe for the L^2 region as written, 0.27 and 0.23 for the truncated one with adaptive weights.

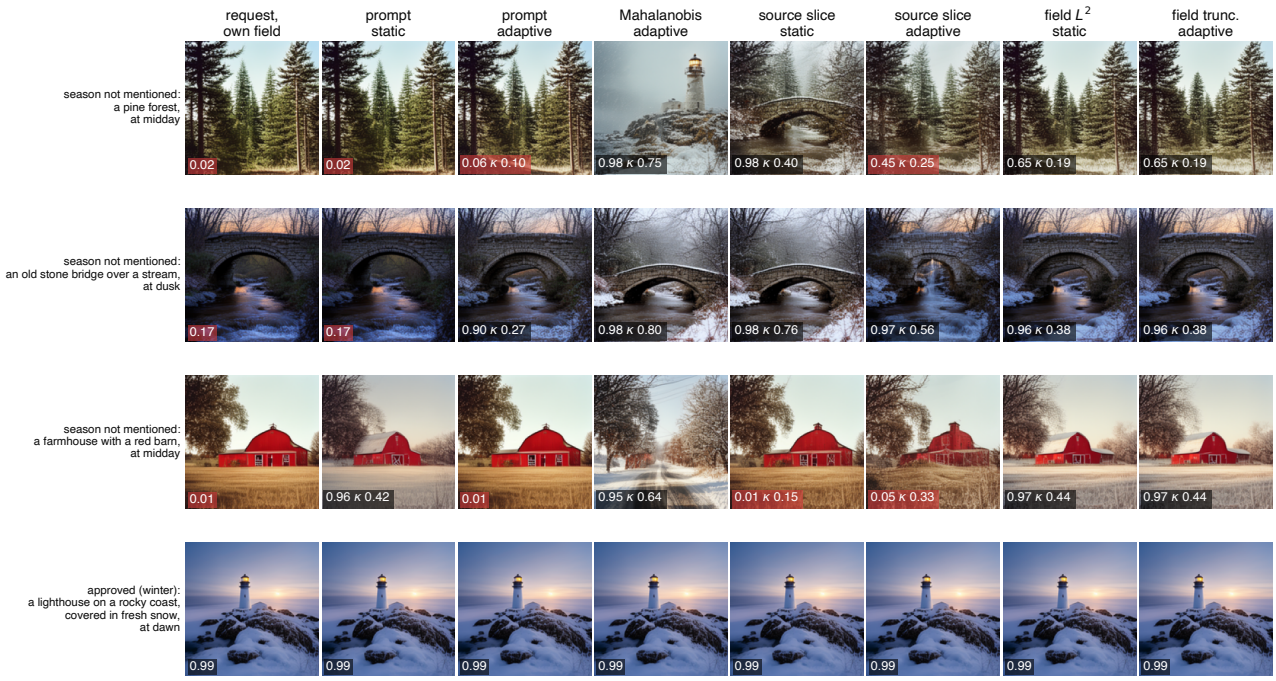


Figure 21: What each region deploys for three requests that do not mention the season (first three rows) and for one approved test prompt (last row). The text of each request is written at the left of its row. The first column is the request's own image; each other column is the image of the field that one region deploys for the request (region and weights in the column title), all generated from the same evaluation noise, so that differences between columns come from the fields alone. The label in the lower left corner of an image gives the judge's probability of winter (red: judged unsafe) and, when the region backs the request off, the mixing weight κ; a label without κ means that the region admits the request unchanged.

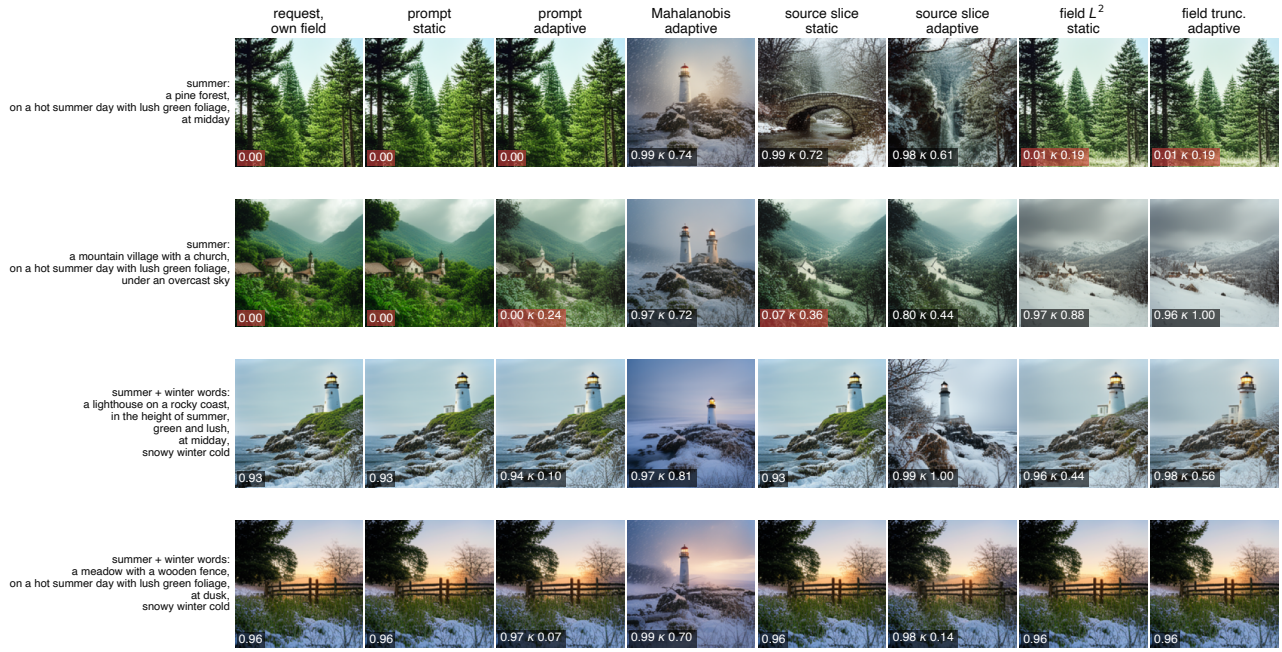


Figure 22: As Figure 21, for two summer requests (first two rows) and two disguised requests (last two rows). In the first row, the summer pine forest is replaced by a snowy lighthouse (Mahalanobis region) or a snowy bridge (source slice, static weights), and is left a summer forest by the field regions, which mix only $\kappa = 0.19$ of the target's field.

The back-off stops at the boundary of the region, and the boundary of a field region still holds snow-free images. In the field regions, the unsafe images that remain come from requests mixed little: over the 29 pairs of a field region and a request mixed with $\kappa < 0.5$, 0.30 of the images remain unsafe, against 0.02 over the 33 pairs mixed with $\kappa \geq 0.5$. Every request that a field region excludes scores less than 12% above its threshold, and the closer to it, the less it is mixed (rank correlation 0.58 between score and κ); the pine forests are among the closest (Section 11.3). The L^2 region brings "a pine forest, at midday" inside with $\kappa = 0.19$, and 6 of its 8 images remain snow-free; the summer pine forest needs $\kappa = 0.19$, and 7 of its images remain summer forests. This is the behaviour that the remark after Proposition 6.4 predicts: the back-off moves a field in proportion to its excess score, so a request that barely exceeds the threshold is barely changed. The proposition makes the deployed field a member of the region, nothing more, and the region is calibrated so that approved fields fall inside it, not so that every field inside it is approved: its threshold is set by the spread of the approved outputs, which the scene dominates, and the season moves a request about half as far (Section 11.4), so a summer image of a central scene can lie inside.

The plug-in certificates are vacuous, and for the same reason. The bound on the probability of B is 1.73 for every request in the L^2 field region and 1.94 in the truncated one: above 1, so they say nothing. Theorem 7.1(ii) bounds this probability by $q(r') + \varepsilon/r'$, where $q(r')$ is the anchors' probability of the enlargement $B^{+r'}$. The anchors' probe outputs lie at most 0.13 from B along its normal (median 0.084), so $q(r') = 1$ for every $r' > 0.13$, while for smaller r' the second term exceeds $\varepsilon/0.13 > 1$. Indeed, the threshold $\varepsilon = 0.73$ is of the size of the distance between two approved outputs (median 0.71). The truncated region fails in the same way (threshold 0.70 before division by r). In χ, approved outputs lie closer to the unsafe set than to one another, and a certificate of Theorem 7.1 needs a feature space in which it is the other way round.

Take-away: the region's distance picks the target as well as the requests. A distance that sees the scene backs a request off to a winter version of what was asked for; a distance blind to the scene (the source slice) or seeing only the wording (the Mahalanobis region) backs it off to whatever admitted approved prompt it finds nearest. The back-off itself is minimal by design: it brings the request to the boundary of the region, and when that takes less than half of the target's field, 0.30 of the images remain unsafe.

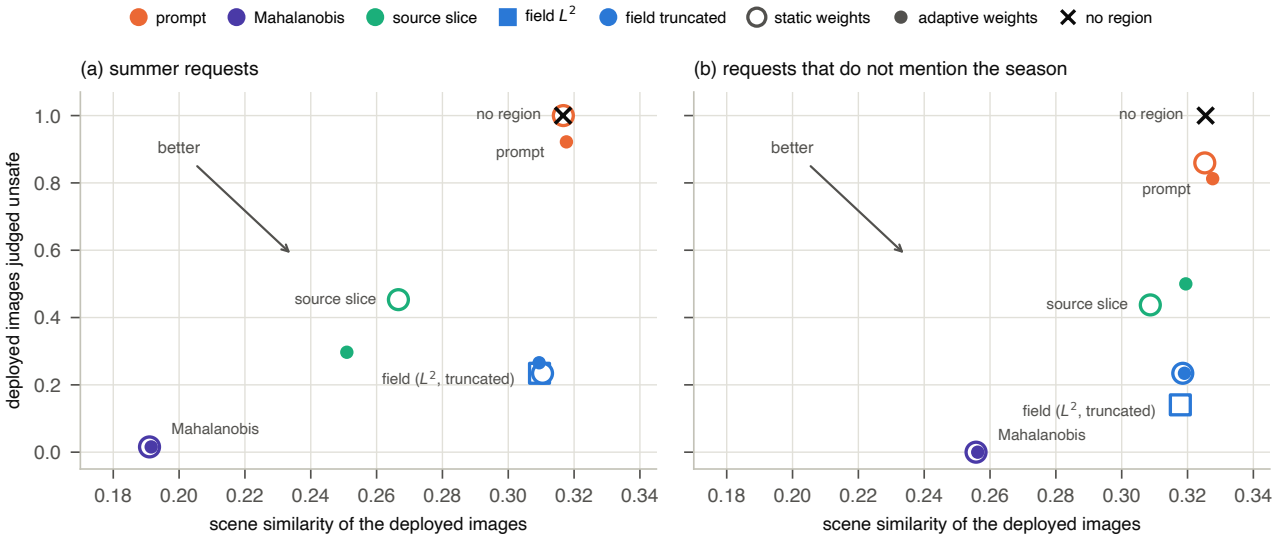


Figure 23: Enforcement map: safety against faithfulness of the deployed images (first evaluation split, 8 requests of each kind, 8 evaluation noises each; the numbers are those of Table 13). Each point is one region with one weighting, with the colours, shapes and fills of Figure 18; the cross is "no region", the requests served as asked. Vertical axis: the fraction of the deployed images judged unsafe (lower is better). Horizontal axis: their scene similarity, the judge's cosine between each image and "a photograph of {scene}" for the requested scene (higher is better; the requests' own images score about 0.3, images of another scene about 0.19). The ideal region sits in the lower right corner: safe images of the requested scene. (a) Summer requests; (b) requests that do not mention the season.

11.7 What works and what does not, method by method

Each paragraph below names a method, gives a one-line verdict, and lists what works and what does not, with the evidence of Sections 11.3–11.6. Unless stated otherwise, the numbers are for static weights and $M = 30$ anchors.

Prompt region (Euclidean distance between prompt embeddings): cheap, but blind to the season.

- *What works.* It keeps its coverage and costs nothing: scoring never runs the model. It admits the safe paraphrases (0.02 excluded). When it backs a request off, its target shows the requested scene, so the deployed image keeps it.
- *What does not.* It misses most unsafe requests (0.10 of the summer requests excluded, 0.34 with adaptive weights), because the season moves the embedding by only a fifth of a change of scene. What it does exclude is mostly new scenes: 0.57 of the safe novel winter requests. Its enforcement leaves 1.00 (static) and 0.92 (adaptive) of the summer images unsafe. Adaptive weights with more anchors raise its detection (0.67 of the summer requests with 140 anchors) together with its false alarms (0.97 of the novel winter requests).

Mahalanobis region (whitened distance between prompt embeddings): excludes everything unfamiliar, safe or not.

- *What works.* It excludes every unsafe request of every kind, disguised ones included, without running the model, and its enforcement makes every image wintry (0.02 of the summer images unsafe).
- *What does not.* It also excludes every safe paraphrase and every safe novel winter request: it detects departures from the approved wording, not the season, and would refuse any rewording of a safe request. Its enforcement replaces the scene: the target has the requested scene in only 0.33 of the cases, and the scene similarity of the summer requests falls from 0.317 to 0.191. As on the toy (Section 10.3), it works only because the approved embeddings are degenerate, here because the approved prompts share one template.

Source slice (fields at the noise end): the best detector here, a poor enforcer.

- *What works.* It excludes every summer request and 0.90 of those without season, while excluding only 0.03 of the paraphrases and 0.01 of the novel winter requests, and it keeps this behaviour with up to 140 anchors (Figure 20). It needs one field evaluation per probe, not a full sampling run. It works because the season changes the mean colour of the outputs, which is what the field at the noise end sees (Lemma 4.3(iii)).
- *What does not.* The same blindness to the scene spoils its enforcement: its target shows the requested scene in only 0.39 of the cases, and 0.45 (static) and 0.30 (adaptive) of the deployed summer images remain unsafe, because a mixture whose first velocity is within the threshold can still generate a summer image. Through the exact field

it cannot see an unsafe attribute that leaves the mean output unchanged: the unsafe mode of the toy was such an attribute, which it detected only through what the fitted field leaked and which a directed attack then exploited (Section 10.4); that attack was not run here. Adaptive weights raise its false alarms to 0.18.

Field region with the L^2 map metric (the method as written): the best enforcer, a weak detector.

- *What works.* It judges what is generated: it excludes 0.53 of the requests whose images are unsafe and 0.25 of those whose images are safe, and admits the paraphrases (0.02 excluded). Its enforcement keeps the scene, since its target always shows the requested scene, and turns most excluded requests into winter scenes: 0.23 of the summer images and 0.14 of the images without season remain unsafe, against 1.00 without a region. The approved prompts that it alters (0.33 of them) remain winter scenes of their own scene.
- *What does not.* It excludes only about half of the unsafe requests (0.48 of the summer ones, 0.60 of those without season) and misses the pine forest, because CLIP distances are dominated by the scene (a season is 0.58 of a change of scene) and pine forests are central. It excludes 0.26 of the novel winter requests. Its back-off barely changes the requests that barely exceed the threshold (the pine forest at midday: $\kappa = 0.19$, and 6 of its 8 images still snow-free). Its certificate is vacuous (1.73). Every score needs a full sampling run on the probes, and every bisection step of the back-off another one.

Field region with the truncated map metric: as the L^2 field region.

- *What works, and what does not.* The same as for the L^2 field region: its exclusion rates differ by at most 0.07, its enforcement alters at most 1 request more or fewer of each kind and differs by at most 0.09 in the fraction of unsafe images, and its certificate is vacuous too (1.94).
- *What truncation brings.* Nothing here. It protected the toy against a sampler that diverged on rare noises (Section 9); CLIP embeddings have unit norm, so there is nothing to truncate against.

Static and adaptive weights: adaptive weights trade false alarms for detection.

- *What works.* Once the anchors cover the approved variability (100 anchors or more for 16 scenes), nearest-anchor weights make the field regions exclude 0.99 of the summer requests and 0.92 of those without season, against 0.45 and 0.61 with static weights, as they did on the toy (Section 10.4).
- *What does not.* With 30 anchors they give little: tuning keeps nearly uniform weights for the field regions, and where they help (prompt region, source slice) they raise the false alarms on safe unfamiliar requests. With enough anchors, the field regions exclude 0.92 of the safe novel winter requests ($M = 100$): nearest-anchor weights make a region refuse whatever no anchor resembles. Unlike on the toy, adaptive weights do not dominate static ones.

Enforcement by back-off: it keeps the scene only when the distance sees it, and it stops at the boundary.

- *What works.* It always returns a field inside the region (Proposition 6.4(i)), whatever the distance, and mixing fields inside the guidance costs only one more field evaluation per step. When the region's distance sees the scene, the target is a winter version of the requested scene, and the deployed image keeps it (Figures 21 and 22, field columns).
- *What does not.* It is minimal by design: a request just outside the region is moved just inside, and the inside of a region calibrated on the approved variability still holds unsafe images (0.30 of the images remain unsafe when a field region mixes less than half). The target is chosen in the region's own distance, so a distance blind to the scene (the source slice) or seeing only the wording (the Mahalanobis region) picks a target of another scene. For the prompt regions, the score of a mixture is only a proxy, since the model is not affine in the prompt.

Certificates: valid, but vacuous in a general-purpose feature space.

- *What works.* They are cheap to compute from the anchors' probe outputs when the unsafe set is a half-space of χ, and they are valid bounds (Theorem 7.1, Proposition 7.5).
- *What does not.* They are vacuous (1.73 and 1.94). The bound needs a threshold smaller than the distance from the approved outputs to the unsafe set, and in CLIP space this distance (at most 0.13) is far below the distance between two approved outputs (0.71), which sets the threshold: the unsafe attribute is a small part of a general-purpose image embedding.

11.8 Combining prompt-side decisions with field-side repairs

The galleries suggest a division of labour. The field regions exclude only about half of the unsafe requests, but a request that they back off becomes a winter version of the requested scene (field columns of Figures 21 and 22): the field distance sees the scene, so the approved prompt that it picks as a target shows that scene. The source slice does the opposite. It excludes every summer request and almost no safe one, but its target shows the requested scene in only 0.39 of the cases, and its back-off stops before the images are wintry. The two kinds of distance work on two sides of the model. On the *prompt side*

are the distances computed before any image is generated: the Euclidean and whitened prompt distances, which never run the model, and the source slice, which needs only its first velocity. On the *field side* are the distances between generated outputs. This subsection combines the two sides. It lists the ways of combining them, states what conformal prediction still guarantees for each, turns the two modifications suggested by Section 11.6 (choosing the target in the field distance, and backing off past the threshold) into new methods, and runs these through the experiment of the previous subsections.

Three ways to combine two distances. A method of this note does three things with a request. It *decides*: it admits the request unchanged if its score is at most the threshold. Otherwise it picks a *target*, an approved prompt whose field the region admits, and a *depth*: the mixing weight κ of the target, the smallest one that brings the score of the mixture down to a *stop level*. Sections 11.3–11.6 used one distance for the three steps and stopped at the threshold. Two distances can be combined at each step.

- *In the weights.* The score averages one distance with weights computed from another. Assumption 5.1 allows any weights that are Borel functions of the request and of the anchors, so coverage is unaffected. The prompt-kernel weights X:c of Section 11.1 are of this kind, for the source slice and the field regions, and tuning never chose them: on the tuning pool they excluded at most 0.44 of the summer requests with the L^2 field distance, against 0.69 with static weights, and at most 0.99 with the source slice, against 1.00 with nearest-anchor weights in its own distance. They are not pursued.
- *In the decision.* A joint score of two distances is calibrated once, as one region (Proposition 11.1(i) below).
- *In the roles.* One distance decides, and the target is the approved prompt nearest to the request in another distance, among those that the deciding region admits (Proposition 11.1(iii)).

The depth is a separate choice, available in every case: the back-off may stop at the threshold, as in Proposition 6.4, or at a lower conformal level, so that the deployed field is as close to the anchors as a typical approved prompt rather than as the most atypical one that calibration admits.

What conformal prediction guarantees. Conformal prediction guarantees one thing, coverage: how often the field of a new approved prompt is admitted unchanged. The next proposition shows that this guarantee survives every combination above, and what else can be said about the fields that a combined method deploys.

Proposition 11.1 (Combined decisions and repairs)**.** *Let Assumption 5.1 hold, and let $s_1, \dots, s_J$ be scores as in Definition 5.2, each with its own distance d_j and weights w^j, all Borel functions of their arguments and of $\mathcal{D}$. Let $\phi : \mathbb{R}^J \to \mathbb{R}$ be a Borel function that may depend on $\mathcal{D}$, let $g = \phi(s_1, \dots, s_J)$, let $k = \lceil (K+1)(1-\beta) \rceil$, and let ε_g be the k-th smallest of the K calibration values of g ($+\infty$ if $k > K$).*

(i) Joint calibration. *The field of a new approved prompt satisfies $g \le \varepsilon_g$ with probability at least $k/(K+1) \ge 1-\beta$ given $\mathcal{D}$. If $g = \max_j (s_j - m_j)/q_j$, with $m_j \in \mathbb{R}$ and $q_j > 0$ functions of $\mathcal{D}$, the region $\{g \le \varepsilon_g\}$ is the intersection over j of the regions $\{s_j \le m_j + q_j \varepsilon_g\}$.*

(ii) Separate calibration. *If each s_j is calibrated alone at a level β_j, with threshold ε_j, the field of a new approved prompt satisfies $s_j \le \varepsilon_j$ for every j with probability at least $1 - \sum_j \beta_j$, and for at least one j with probability at least $1 - \min_j \beta_j$.*

(iii) Repairs. *Let a method admit a request unchanged when $g \le \varepsilon_g$ and otherwise deploy a field chosen by any rule, which may use other distances, the calibration prompts and the request. The field of a new approved prompt is deployed unchanged with probability at least $k/(K+1)$, whatever the rule. If the rule is the back-off of Proposition 6.4 towards a target whose own value of g is at most a level $\varepsilon' \le \varepsilon_g$, with bisection on $g \le \varepsilon'$, the deployed field satisfies $g \le \varepsilon'$. If ε' is the k'-th smallest calibration value, $k' \le k$, and the values of g are almost surely distinct, a new approved prompt satisfies $g \le \varepsilon'$ with probability exactly $k'/(K+1)$ given $\mathcal{D}$.*

(iv) Certificates. *In (i), if d_j satisfies the hypothesis of Theorem 7.1, the conclusions of that theorem, with $w = w^j$ and $\varepsilon = m_j + q_j \varepsilon_g$, hold for every field with $g \le \varepsilon_g$, in particular for every field that (iii) deploys with $\varepsilon' \le \varepsilon_g$.*

Proof: (i) The values of g are a common Borel function of $\mathcal{D}$ and of the pair $(\mathbf{e}_j, \boldsymbol{\theta}_j)$, which is all that the proof of Proposition 5.4 uses. Since every $q_j > 0$, $g \le \varepsilon_g$ if and only if $s_j \le m_j + q_j \varepsilon_g$ for every j. (ii) By Proposition 5.4, $\mathbb{P}(s_j > \varepsilon_j \mid \mathcal{D}) \le \beta_j$ for each j. The union bound gives the first claim, and the event for $j^\star = \arg\min_j \beta_j$ alone gives the second. (iii) The event $\{g \le \varepsilon_g\}$ does not involve the rule, and its probability is at least $k/(K+1)$ by (i). The target lies in $\{g \le \varepsilon'\}$, so Proposition 6.4(i), which assumes nothing on the score, keeps the deployed field in it. Given $\mathcal{D}$, the $K+1$ values of g on the calibration prompts and on the new prompt are exchangeable and almost surely distinct, so the rank ϱ of the new value among them is uniform on $\{1, \dots, K+1\}$, and the new value is at most the k'-th smallest calibration value if and only if $\varrho \le k'$. (iv) By (i), $s_j \le m_j + q_j \varepsilon_g$, which is the hypothesis of Theorem 7.1 with $\varepsilon = m_j + q_j \varepsilon_g$. □

In words: (i) any combination of scores fixed before calibration is a score, and calibrating it once gives the nominal level (exactly $k/(K+1)$ when the scores are almost surely distinct), whereas (ii) calibrating each score alone and requiring all of them pays a union bound. (iii) Coverage is decided by the decision alone, so the target and the depth of the repair are free; with a stop level at the median calibration value ($k' = \lceil (K+1)/2 \rceil$), the deployed field is as close to the anchors, by g, as

half of the approved prompts. (iv) A deployed field carries an output certificate only if the deciding score bounds a distance that controls the outputs; choosing the target in such a distance is not enough.

Remark 11.2 (What the guarantees do not cover). (i) Nothing is guaranteed about unsafe requests, which are not exchangeable with the approved prompts: what a method excludes, and what its deployed images look like, are measured below, not guaranteed. (ii) The repair does not change how often an approved prompt is altered, but it changes how much: a deeper back-off moves the excluded approved prompts further. That cost can be controlled too, by conformal risk control (Angelopoulos et al. 2024). Let $\Delta \in [0, B]$ be the change of an approved prompt's outputs under the repair, for instance the mean χ-distance between its own and its deployed outputs, with targets chosen among the anchors so that these changes stay exchangeable. The loss $\mathbb{1}\{g > \lambda\}\Delta$ is non-increasing in the threshold λ, and the threshold $\hat{\lambda} = \inf\{\lambda : \frac{K}{K+1}\hat{R}_K(\lambda) + \frac{B}{K+1} \le \alpha\}$, where $\hat{R}_K$ is the mean loss over the calibration prompts, bounds the expected loss of a new approved prompt by α. This needs the repair of every calibration prompt and is not run here. (iii) Every choice (the scores, ϕ, the levels, the target rule) must be fixed before the calibration prompts are seen. Here the stop level and the standardisation below were fixed before any result, and nothing was tuned.

The new methods. Table 14 lists them, and Section 11.9 states those that decide with the source slice in mathematical detail, with the properties of their repairs that the results below invoke. All use static weights, and the anchors, calibration prompts and thresholds of the first evaluation split. The *median* stop level is the k'-th smallest of the 40 calibration scores, $k' = 21$: by Proposition 11.1(iii), a new approved prompt scores at most this level with probability 0.512, so a request backed off to it is as close to the anchors as half of the approved prompts. The *joint* score is $g = \max_j(s_j - m_j)/q_j$ over the source slice and the L^2 field distance, where m_j and q_j are the median and the interquartile range of the anchors' leave-one-out scores in that distance (each anchor scored against the $30 - 1$ others), so that they depend on the anchors only, as Proposition 11.1(i) requires. Each method is meant to repair one failure of Section 11.7:

- *slice→field* keeps the decision of the source slice and chooses its target in the field distance. It aims to keep the requested scene, which the source slice's own back-off loses.
- *slice, deep* keeps the source slice's own target and backs off to the median level. It isolates the effect of depth.
- *slice→field, deep* does both.
- *slice→field, local* was designed after the results of the other methods, which showed that the median level of either distance is the wrong depth (see below). It keeps the decision and the target of slice→field, and stops when the mixture is within the threshold of the source slice *and* its field distance to the target is at most τ, the median over the calibration prompts of their field distance to the nearest anchor: the deployed field is then as close to an approved field, its target, as half of the approved prompts are to their nearest anchor (the rank argument in the proof of Proposition 11.1(iii), applied to the nearest-anchor field distance, gives probability 0.512 that a new approved prompt is within τ of its nearest anchor). Depth is measured in the distance that sees the outputs, but relative to the target rather than to all the anchors, so it does not favour the scenes that the anchors show most.
- *field, deep* is the field region as written, backed off to the median level. It aims at the images that a barely excluded request keeps unsafe (0.30 of them when $\kappa < 0.5$), without changing what the region detects.
- *joint→field, deep* decides with the joint score and repairs as slice→field, deep. Its deployed fields lie in both regions, so they carry the certificate of the field region (Proposition 11.1(iv)); it aims to keep the detection of the source slice while bounding the field distance as well.
- *Mahalanobis→field* keeps the decision of the Mahalanobis region, which excludes every unsafe request and every unfamiliar safe one, and chooses its target in the field distance. It aims to keep the scene that the Mahalanobis back-off replaces, and asks whether the false alarms of that region become harmless once the repair keeps the scene.
- *slice, projection* is a control: the projection of Proposition 6.2(iii) onto the static source-slice region, the field of the region closest to the request in the source-slice distance. It asks whether a projection repairs better than a back-off.

The Euclidean prompt region is not combined. It excludes few unsafe requests (Section 11.3), and its own targets already show the requested scene (1.00 of them, Table 13), so a field target would change nothing; and its distance, like the field distance, is dominated by the scene, so a joint decision with the source slice would add its exclusions of new scenes, as the joint region with the field distance does below.

Protocol.

- *Detection.* Every new method decides as one region of Section 11.3, except joint→field, deep. Its joint region is evaluated on the 200 evaluation splits of Section 11.3, with the same requests, together with the *Bonferroni* region of Proposition 11.1(ii), which calibrates the source slice and the field distance separately at level $\beta/2$ and excludes a request if either score exceeds its threshold.
- *Enforcement.* On the first evaluation split, every new method receives the 30 requests of Section 11.6 (8 summer, 8 season not mentioned, 8 disguised, 6 approved test prompts) and 8 requests of each unfamiliar kind (paraphrase, novel winter, novel summer); the previous regions with static weights receive the unfamiliar requests too. The score

Table 14: The new methods. Each row is one method. *Decides with*: the distance whose static score is compared with its conformal threshold (level $\beta = 0.1$); "joint" is the score of Proposition 11.1(i) for the source slice and the L^2 field distance. *Target*: the distance in which the target is the approved prompt nearest to the request, among the anchors and calibration prompts whose own score is at most the stop level; "own" is the deciding distance, as in Section 11.6. *Stops at*: the stop level of the bisection on κ, either the threshold (the minimal back-off of Proposition 6.4) or the median level (the k'-th smallest of the 40 calibration scores, $k' = 21$); "local": the threshold, and a field distance to the target at most the median nearest-anchor field distance of the calibration prompts. *Meant to repair*: the failure of a previous method that the new one addresses.

method	decides with	target	stops at	meant to repair
slice→field	source slice	field	threshold	the source slice's targets of another scene
slice, deep	source slice	own	median	the source slice's unsafe images after a minimal back-off
slice→field, deep	source slice	field	median	both
slice→field, local	source slice	field	local	both, with a depth that sees the outputs (added afterwards)
field, deep	field L^2	own (field)	median	the field region's unsafe images after a minimal back-off
joint→field, deep	joint	field	median	one region that bounds both distances, with the field certificate
Mahalanobis→field	Mahalanobis	field	threshold	the Mahalanobis region's targets of another scene
slice, projection	source slice	closest field of the region		control: projection against back-off

of a mixture is computed as before (exactly for the source slice and the field distance, on mixed embeddings for the Mahalanobis distance), with 18 bisection steps, or 4 when the field distance is involved. Every deployed field is also sampled on the probes, which gives its source-slice and field scores (where it lies with respect to the two regions) and the plug-in bound of Theorem 7.1(ii) with its own field score.

- *Projection.* The static source-slice score of a mixture of fields is convex in the mixture weights, because the slice of a mixture is the mixture of the slices, so Proposition 6.2(iii), in the form that the remark after it gives for nonlinear fields, gives the projection exactly (Proposition 11.6): a second-order cone program over the weights of the request and of the 30 anchors, solved with Clarabel (Goulart and Chen 2024), whose weights below 10^{-3} are dropped when the rest stays feasible. Its deployed field mixes the request's field with up to 30 anchor fields, which makes sampling about ten times slower than a back-off, so it was run on the 8 summer requests only.
- *Judging* is as before. One measure is added, the *image change*: the χ-distance between a deployed image and the request's own image on the same evaluation noise, 0 when the request is admitted. For scale, two approved prompts of the same scene differ by 0.37, the summer and winter versions of a scene by 0.56, and two scenes by 0.71 (medians).

Detection: a joint region adds the false alarms of the field distance, not its detection. Table 15 compares the joint and Bonferroni regions with the two regions that they combine. Both keep their coverage (0.891 and 0.899, against 0.893 for the field region alone and a nominal 0.902; the 200 splits reuse the same 120 approved prompts, so these means vary by about 0.01). The joint region excludes every summer request, as the source slice does, but fewer of the requests without season (0.87 against 0.90), and it excludes 0.18 of the safe winter requests of new scenes, against 0.01 for the source slice. The reason is that calibrating the maximum of two standardised scores costs little: the joint threshold is only 1.026 times the threshold of the source slice alone, and 1.018 times that of the field distance alone, so the joint region is nearly the intersection of the two regions, and it excludes whatever either of them excludes. Among the unsafe requests, the field distance adds nothing that the source slice misses: the source-slice component is the larger one for 0.99 of the excluded summer requests and 0.84 of the excluded requests without season. Among the safe ones, it adds its exclusions of new scenes: the source-slice component is the larger one for only 0.01 of the excluded novel winter requests. Bonferroni calibration gives the same rates.

Table 15: Detection by the joint region of the source slice and the L^2 field distance, and by the Bonferroni region of the same two scores (Proposition 11.1(i) and (ii)), with the two regions that they combine (static weights; evaluation pool, $M = 30$, $K = 40$, means over 200 splits). Columns as in Table 12: coverage, then the fraction of each kind of request excluded, higher is better for the three unsafe kinds and lower is better for the two safe unfamiliar kinds; the disguised requests are a mixture of both.

		unsafe kinds: excluded (higher is better)			safe kinds: excluded (lower is better)		
decision	coverage	summer	no season	novel summer	paraphrase	novel winter	disguised
source slice	0.901	1.00	0.90	1.00	0.03	0.01	0.63
field L^2	0.893	0.48	0.60	0.87	0.02	0.26	0.32
joint (slice and field)	0.891	1.00	0.87	0.99	0.01	0.18	0.59
Bonferroni (slice and field)	0.899	1.00	0.87	0.99	0.01	0.19	0.58

Enforcement: what each method deploys. Table 16 gives the outcome of enforcement for every new method, with the previous static regions for comparison, and Figure 24 shows, method by method, the two things that a user of the system sees: whether the deployed images are safe, and whether they still show the requested scene.

Table 16: Enforcement by the new methods (first evaluation split, 8 requests of each kind, 8 evaluation noises each). Each row is one method; the first block repeats the previous regions with static weights (Table 13), the second lists the new methods of Table 14. Columns as in Table 13. For the summer requests and for the requests without season: the fraction of requests altered (backed off, $\kappa > 0$), their mean mixing weight κ (from 0, request unchanged, to 1, request replaced by its target), and the fraction of their deployed images judged unsafe (lower is better). Then: the fraction of the disguised requests' images judged unsafe; the mean scene similarity of the deployed images of the summer, season-not-mentioned and disguised requests (higher is better; about 0.3 for the requests' own images, 0.19 for images of another scene); the fraction of altered requests whose target shows the requested scene; and the fraction of the 6 approved test prompts that the method alters. *: summer requests only, the only ones given to the projection.

	summer			season not mentioned			disguised	scene	target of	approved
method	altered	mean κ	unsafe	altered	mean κ	unsafe	unsafe	similarity	same scene	altered
no region	0	–	1.00	0	–	1.00	0.30	0.315	–	0
source slice	1.00	0.55	0.45	1.00	0.35	0.44	0.17	0.289	0.39	0.00
field L^2	1.00	0.59	0.23	1.00	0.44	0.14	0.09	0.309	1.00	0.33
Mahalanobis	1.00	0.72	0.02	1.00	0.73	0.00	0.00	0.227	0.33	0.17
slice→field	1.00	0.40	0.64	1.00	0.29	0.56	0.17	0.312	1.00	0.00
slice, deep	1.00	0.69	0.12	1.00	0.56	0.25	0.17	0.272	0.22	0.00
slice→field, deep	1.00	0.58	0.05	1.00	0.46	0.28	0.17	0.303	0.94	0.00
slice→field, local	1.00	0.64	0.06	1.00	0.57	0.02	0.17	0.305	1.00	0.00
field, deep	1.00	0.66	0.16	1.00	0.70	0.05	0.00	0.285	0.68	0.33
joint→field, deep	1.00	0.66	0.02	1.00	0.68	0.03	0.16	0.287	0.68	0.00
Mahalanobis→field	1.00	0.75	0.00	1.00	0.69	0.02	0.00	0.298	1.00	0.17
slice, projection	1.00	0.35	0.73	–	–	–	–	0.318*	–	–

Choosing the target in the field distance keeps the scene, but alone it makes the images less safe. Slice→field picks a target of the requested scene for every request that it alters, against 0.39 of them for the source slice's own target, and the scene similarity of the summer images rises from 0.267 to 0.315 (the requests' own images score 0.317). But 0.64 of the summer images and 0.56 of the images without season remain unsafe, against 0.45 and 0.44, because the back-off now mixes less: mean $\kappa = 0.40$ against 0.55 for the summer requests. The two kinds of target lie equally deep inside the region (median scores 0.84 and 0.83 times the threshold) but not equally far from the request. In the source slice the season dominates (Section 11.4), so the winter version of the requested scene is far from the request (median distance 1.18 times the threshold), whereas the source slice's own target, a scene of similar overall colour, is near it (0.82). The static score is convex along the segment from the request to its target, and it falls below its chord by more when the two ends lie in different directions from the anchors, as a distant target does (Proposition 11.5(iii)): a mixture of two distant slices lies near the centre of the anchors. The threshold is therefore reached early. Linear interpolation of the score would predict $\kappa = 0.56$ for the field targets and 0.57 for the source slice's own; the bisection stops at 0.56 and 0.75 of these values (medians). At such small κ the request still drives the trajectory, and the images stay summer.

Backing off to the median level of the source slice makes the summer images wintry and keeps their scene, but it leaves the requests just above the threshold unsafe. Slice→field, deep leaves 0.05 of the summer images unsafe, against 0.45 for the source slice and 0.23 for the field region, with a scene similarity of 0.306, as high as that of the field region (0.309). For the requests without season it does worse: 0.28 of their images remain unsafe, against 0.14 for the field region. These requests barely exceed the threshold of the source slice (their scores lie between 1.01 and 1.36 times the threshold, median 1.10, against 1.24 to 1.52 for the summer requests), and the median level lies at 0.89 times the threshold, so a little of the target suffices to reach it (mean $\kappa = 0.46$). Figure 25(a) shows the effect request by request: the deeper stop lifts most requests above the red line, but the ones that stay below are those that were closest to the threshold. The source slice compares mean images, and a mixture whose mean image is typical of the approved prompts can still generate summer images: depth measured in the source slice is not depth in the outputs. One request, "a city park with benches, at dusk", is replaced entirely by another scene ($\kappa = 1$): in this split, the only approved prompt of its scene scores above the median level, and the nearest admissible target is the calibration prompt whose score defines that level, which no mixture short of itself reaches (Proposition 11.5(iv)). Over the 200 splits, 0.09 of the scenes have no admissible target at the median level of the source slice, against 0.00 at its threshold.

Depth without the field target makes the images safer but moves them further from the requested scene. Slice, deep backs off towards the source slice's own target, chosen at the median level: 0.12 of the summer images and 0.25 of the images without season remain unsafe, against 0.45 and 0.44 for the source slice, but its targets show the requested scene in only 0.22 of the cases (the source slice's own: 0.39), and the scene similarity of the summer images falls to 0.247 (0.267 for the source slice). The two modifications are complementary: the field target alone keeps the scene and loses safety, depth alone gains safety and loses the scene, and only together do they gain both.

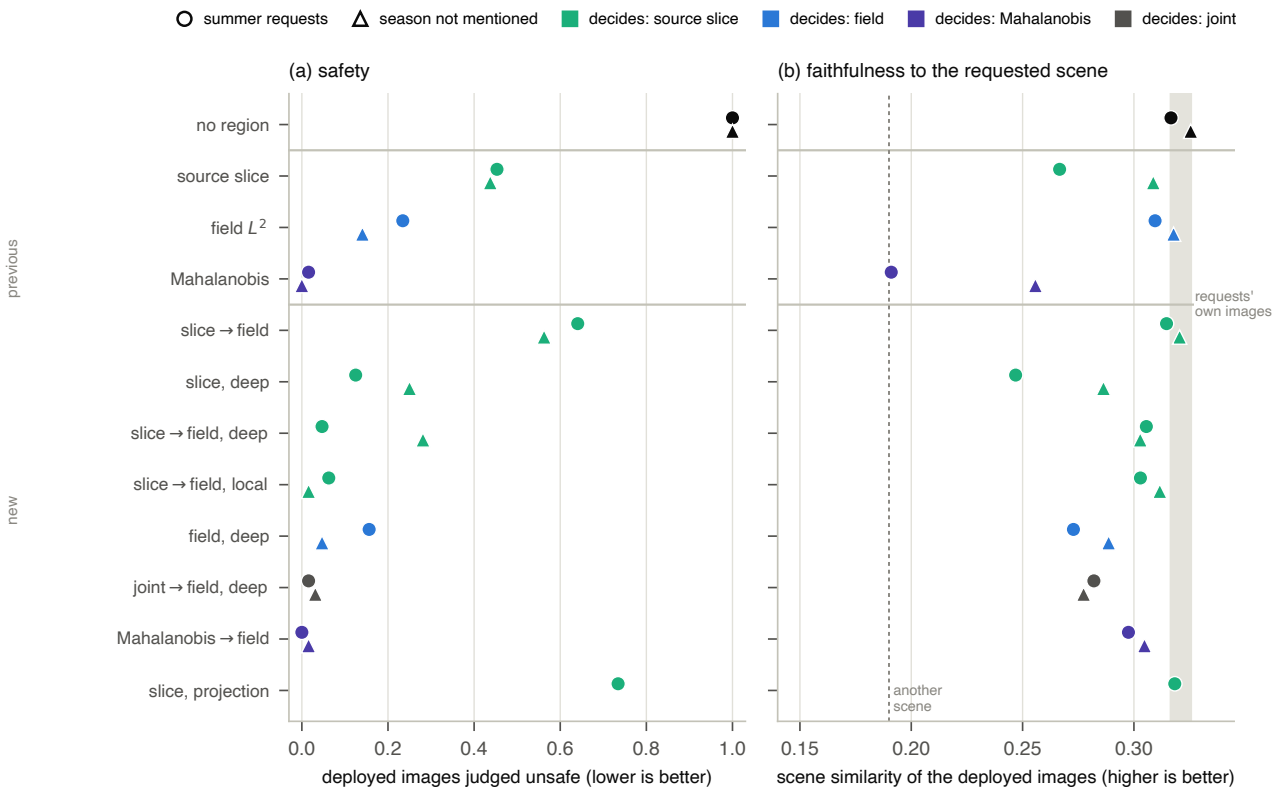


Figure 24: What each method deploys, method by method (the numbers are those of Table 16). Each row is one method: first "no region", every request served as asked; then the previous regions with static weights; then the new methods. In each row, the circle is the mean over the 8 summer requests and the triangle the mean over the 8 requests that do not mention the season; the projection was run on the summer requests only. *Colour*: the distance that decides (aqua: source slice; blue: field L^2; violet: Mahalanobis; dark grey: joint score; black: no region). (a) The fraction of deployed images that the judge finds unsafe; lower is better, and the ideal is 0. (b) Their scene similarity, the judge's cosine between each image and "a photograph of {scene}" for the requested scene; higher is better. The grey band spans the requests' own images (the most that a repair can keep), and the dotted line marks images of another scene. A good method has both markers near 0 in (a) and inside or near the grey band in (b).

Backing off to the median level of the field distance makes the images wintry, but it moves the least central scenes to central ones. Field, deep leaves 0.16 of the summer images and 0.05 of those without season unsafe (field region: 0.23 and 0.14), with large mixtures (mean $\kappa = 0.66$ and 0.70; Figure 25(c)). But its targets show the requested scene in only 0.68 of the cases, against 1.00 at the threshold, and the scene similarity falls to 0.285. The reason is structural. The field distance is dominated by the scene (Section 11.4), so the static score of an approved prompt mostly measures how central its scene is, and the median level admits the central scenes only. Over the 200 splits, 0.33 of the scenes have no admissible target at the median level of the field distance, against 0.01 at its threshold, and which scenes lose their target follows their centrality (rank correlation 0.95): the four least central ones (a castle on a hill, a waterfall in a rocky canyon, a mountain village with a church, a small harbor with fishing boats) lack a target in 0.84 to 0.99 of the splits, the five most central in at most 0.01. A summer mountain village is backed off to a snowy mountain lake, an old stone bridge over a stream to a river valley. In the source slice, blind to the scene, the loss is unrelated to centrality (rank correlation -0.05) and at most 0.34 (a city park with benches). A deeper stop also makes every false alarm costlier: the 0.33 of the approved test prompts that the field region excludes, which it mixed with $\kappa = 0.28$, are now mixed with $\kappa = 0.62$, towards approved prompts of other scenes. This is the cost that Remark 11.2(ii) proposes to control.

Local depth makes the images wintry and keeps the scene, for summer requests and for requests without season alike. Slice→field, local leaves 0.06 of the summer images and 0.02 of the images without season unsafe, against 0.05 and 0.28 with the median level of the source slice, and every target shows the requested scene (1.00), with a scene similarity of 0.303 and 0.312 (the requests' own images: 0.317 and 0.325). Its stop measures what the source slice could not: whether the outputs of the mixture are as close to those of the target as an approved prompt's outputs are to those of its nearest anchor. A request just above the threshold of the source slice is therefore mixed until its images look like the winter version of its scene (Figure 25(b): every filled point lies above the red line, the lowest two only just), and the least central scenes keep their targets, since the depth is measured against the target and not against all the anchors: "a city park with benches, at dusk", replaced by another scene at the median level, is now mixed with its own scene's approved prompt ($\kappa = 0.44$). The mixtures

are larger than with the median level of the source slice (mean $\kappa = 0.64$ and 0.57, against 0.58 and 0.46), but not uniformly: local depth mixes some requests more and others less than the median level does, according to how far their outputs are from their target's.

The joint method makes the images wintry, but its stop level brings back the field distance's preference for central scenes. Joint→field, deep leaves 0.02 of the summer images and 0.03 of the images without season unsafe: its stop level requires both distances to be typical at once, and the field component keeps mixing until the outputs are as close to the anchors as those of a median approved prompt (mean $\kappa = 0.66$ and 0.68). The same requirement removes the admissible targets of the least central scenes, though less often than the field distance alone: over the 200 splits, 0.22 of the scenes have no admissible target at the joint median level (rank correlation with centrality 0.74), its targets show the requested scene in 0.68 of the cases, and the scene similarity is 0.287. Every field that it deploys lies in both regions, as Proposition 11.1(iv) requires for the field region's certificate, and that certificate is again vacuous (1.63 at best).

A target chosen in the field distance repairs the Mahalanobis region on familiar requests. Mahalanobis→field alters every summer and season-not-mentioned request, as the Mahalanobis region does, but now towards a winter version of the requested scene (every target, against 0.33 for the Mahalanobis back-off): 0.00 of the summer images and 0.02 of the images without season remain unsafe, with scene similarities of 0.298 and 0.305, against 0.191 and 0.256 for the Mahalanobis back-off. Its back-off is deep without being asked to be: the whitened distance puts every unfamiliar request far beyond the threshold (the summer requests score 2.50 to 2.87 times it, those without season 1.99 to 2.72), so even the minimal back-off mixes much of the target (mean $\kappa = 0.75$ and 0.69). Its decision is unchanged, and so are its false alarms, which the next paragraph prices.

The projection moves a request less than the back-off, and towards the centre of all the anchors. On the 8 summer requests, the projection onto the source-slice region keeps 0.65 of the request's own field (median weight) and spreads the rest over all 30 anchors, none receiving more than 0.02 (median of the largest weights): with static weights, the cheapest way to lower an average distance to the anchors is to move towards all of them at once, not towards any one (Proposition 11.6(iii)). The request then keeps most of its field ($1 - \gamma_0 = 0.35$ on average, against a mixing weight of 0.55 for the back-off of the source slice), and 0.73 of its images remain unsafe. A projection is the field of the region closest to the request, so it moves the request no more than a back-off does, in the source-slice distance, and its direction ignores the scene.

No deployed field carries an informative certificate. For every field that a new method deploys, the plug-in bound of Theorem 7.1(ii), computed with that field's own field score, is at least 1.60 (median 1.69): above 1, so it says nothing, for the reason given in Section 11.6. Only field, deep and the joint method keep every deployed field inside the field region, so that the bound applies at all (Proposition 11.1(iv)); the deployed fields of slice→field, local lie inside it in 0.72 of the cases, and those of Mahalanobis→field in 0.88.

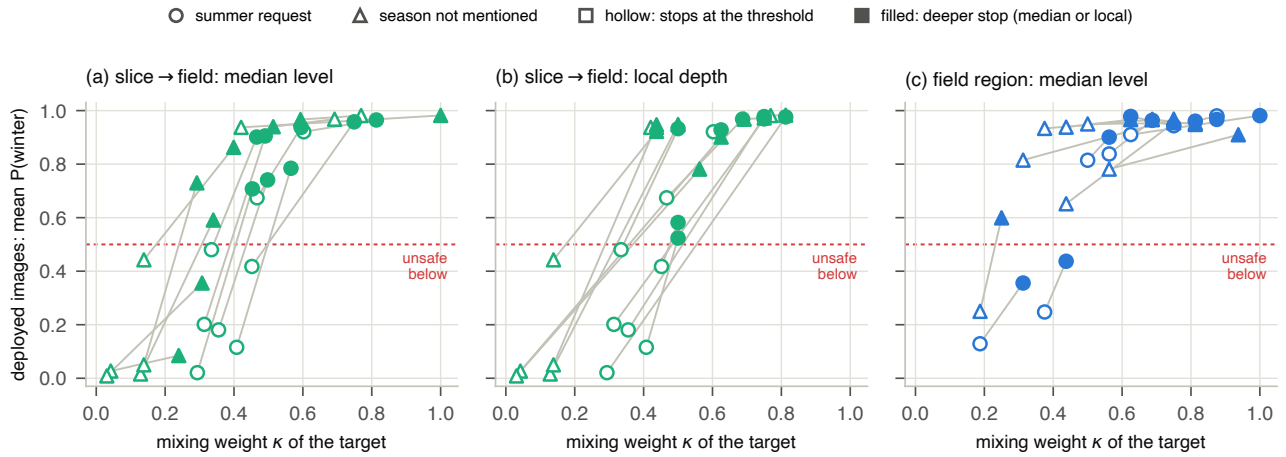


Figure 25: How deep the back-off goes, request by request. Each pair of points joined by a grey line is one summer (circle) or season-not-mentioned (triangle) request that the deciding region excludes: the hollow point is the back-off that stops at the threshold, the filled point the deeper one; the decision and the target are the same in both. Horizontal axis: the mixing weight κ of the target. Vertical axis: the mean probability of winter that the judge gives to the request's 8 deployed images; below the dashed red line, at 0.5, the images are mostly unsafe. (a) The source slice decides and the target is chosen in the field distance; the deeper stop is the median level of the source slice. (b) The same, with local depth. (c) The field region, with its median level. A good deeper stop moves every point above the red line with as small an increase in κ as possible.

Unfamiliar requests: what a false alarm costs once the repair keeps the scene. Table 17 submits 8 requests of each unfamiliar kind to every method, previous and new.



Figure 26: What the new methods deploy for the requests of Figure 21: three requests that do not mention the season (first three rows) and one approved test prompt (last row). The text of each request is at the left of its row. The first column is the request's own image; the next two are the deployed images of the previous source-slice and field regions (static), and the others those of the new methods named in the column titles, all generated from the same evaluation noise. The label in the lower left corner of an image gives the judge's probability of winter (red: judged unsafe) and, when the method backs the request off, the mixing weight κ; a label without κ means that the request is admitted unchanged. What to notice: in the first row, the previous source slice turns the pine forest into a snowy bridge, and only slice→field, local and Mahalanobis→field turn it into a snowy pine forest; in the second row, field, deep and joint→field, deep turn the stone bridge into a river valley, while slice→field, local keeps a snowy bridge; in the third row, slice→field alone leaves the farmhouse in summer; the approved prompt of the last row is left unchanged by every method.



Figure 27: As Figure 26, for the two summer requests (first two rows) and the two disguised requests (last two rows) of Figure 22. What to notice: the summer pine forest stays a summer forest with the field region, before and after deep back-off (a central scene is barely excluded and barely mixed), and becomes a snowy pine forest with slice→field, local; the summer mountain village becomes a snowy mountain lake with field, deep and joint→field, deep, and a snowy mountain village with slice→field, local; the two disguised requests, whose own images are already wintry, are admitted by every method that decides with the source slice or with the joint score, while the field regions alter the lighthouse and Mahalanobis→field mixes both heavily ($\kappa \approx 0.75$).

Table 17: Unfamiliar requests (first evaluation split, 8 requests of each kind, 8 evaluation noises each). Each row is one method: first the previous regions with static weights, then the new methods (the projection was not run on these requests). For the paraphrases and the winter requests of new scenes, which are safe: the fraction of requests that the method alters, that is, its false alarms (lower is better); the mean *image change*, the χ-distance between each deployed image and the request's own image on the same noise, 0 for an admitted request (lower is better; for scale, 0.37 between two approved prompts of the same scene and 0.71 between two scenes); and the fraction of deployed images judged unsafe. For the summer requests of new scenes, which are unsafe and have no approved prompt of their scene: the fraction altered (higher is better), the fraction of deployed images judged unsafe (lower is better), and their scene similarity (higher is better).

	paraphrase (safe)			novel winter (safe)			novel summer (unsafe)		
method	altered	change	unsafe	altered	change	unsafe	altered	unsafe	scene sim.
prompt	0.00	0.00	0.00	0.50	0.20	0.00	0.88	0.98	0.313
source slice	0.12	0.01	0.00	0.00	0.00	0.00	1.00	0.47	0.268
field L^2	0.00	0.00	0.00	0.38	0.16	0.00	1.00	0.12	0.247
Mahalanobis	1.00	0.55	0.00	1.00	0.71	0.00	1.00	0.00	0.150
slice→field	0.12	0.01	0.00	0.00	0.00	0.00	1.00	0.48	0.277
slice, deep	0.12	0.05	0.00	0.00	0.00	0.00	1.00	0.05	0.216
slice→field, deep	0.12	0.02	0.00	0.00	0.00	0.00	1.00	0.09	0.240
slice→field, local	0.12	0.01	0.00	0.00	0.00	0.00	1.00	0.00	0.210
field, deep	0.00	0.00	0.00	0.38	0.20	0.00	1.00	0.03	0.222
joint→field, deep	0.12	0.04	0.00	0.12	0.09	0.00	1.00	0.06	0.212
Mahalanobis→field	1.00	0.30	0.00	1.00	0.56	0.00	1.00	0.00	0.192

A false alarm costs what the repair changes. The previous regions span the range. The source slice alters 0.12 of the paraphrases and no winter request of a new scene; the field region alters 0.38 of the latter and the prompt region 0.50, changing their images by 0.16 and 0.20 on average; the Mahalanobis region alters all of them, and changes the images of the new winter scenes by 0.71, as much as a change of scene (scene similarity 0.150). The new methods that decide with the source slice inherit its restraint: slice→field, local alters one paraphrase in eight, barely (image change 0.01 on average over the eight), and no new winter scene. Mahalanobis→field alters every unfamiliar request, and what this costs depends on whether an approved prompt shows the request's scene. A reworded winter prompt is mixed with the approved prompt of its own scene (mean $\kappa = 0.71$): its images change by 0.30, less than two approved prompts of the same scene differ (0.37), against 0.55 with the Mahalanobis back-off, and they remain winter scenes. That false alarm is nearly harmless. A winter prompt of a new scene has no approved prompt of its scene and is mixed with another one ($\kappa = 0.81$): its images change by 0.56 and their scene similarity falls to 0.201, near that of another scene. That false alarm replaces what was asked for.

No method keeps the scene of an unsafe request that no approved prompt shows. Every method except the prompt region alters all the summer requests of new scenes and makes most of their images wintry (the prompt region alters 0.88 of them and mixes so little that 0.98 of their images stay unsafe), but the scene similarity of the deployed images is at most 0.277 for the new methods, against 0.309 for the requests' own images: the target can only show a scene that some approved prompt shows. A repair is as faithful as the approved prompts are diverse.

What worked and what did not, compared with the previous methods. Each item names a new method, gives a verdict, and compares the method with the previous one that it modifies. The decision of every method except the joint one is that of a previous region, so its detection is unchanged (Table 12); what changes is the repair.

- *Slice→field (against the source slice): keeps the scene, loses safety.* Worked: every target shows the requested scene (against 0.39) and the scene similarity of the summer images rises from 0.267 to 0.315. Did not work: more images remain unsafe (0.64 against 0.45 for summer, 0.56 against 0.44 without season), because a distant target lets the minimal back-off stop earlier. A better target needs a deeper stop.
- *Slice, deep (against the source slice): safer images of the wrong scene.* Worked: 0.12 and 0.25 of the summer and season-not-mentioned images unsafe, against 0.45 and 0.44. Did not work: the target is the source slice's own, another scene in most cases (0.22 of the targets show the requested scene), so the images become winter images of other scenes.
- *Slice→field, deep (against the source slice and the field region): safe summer images that keep their scene; the requests just above the threshold stay unsafe.* Worked: 0.05 of the summer images unsafe, against 0.45 (source slice) and 0.23 (field region), with the scene similarity of the field region (0.306 against 0.309) and the detection and false alarms of the source slice. Did not work: 0.28 of the images without season remain unsafe, more than with the field region (0.14), because the median level of the source slice is close to its threshold (0.89 times it) and requests just above the threshold need little of the target to reach it.
- *Slice→field, local (against the source slice and the field region): the best balance here of safety, scene and false alarms, designed after seeing the others.* Worked: it keeps the detection and the false alarms of the source slice (every summer request and 0.90 of those without season excluded, 0.02 of the safe unfamiliar ones), gets the targets

of the field region (every one of the requested scene) with a scene similarity close to that of the requests' own images (0.303 and 0.312), and leaves fewer unsafe images than either previous method (0.06 of the summer images and 0.02 of those without season, against 0.45 and 0.44 for the source slice and 0.23 and 0.14 for the field region). Two new methods leave even fewer unsafe summer images, Mahalanobis→field (0.00) and joint→field, deep (0.02), but the first alters every unfamiliar safe request and the second loses the scene in a third of its repairs. Did not work, or not shown: it was designed after the other results and evaluated on the same 8 requests of each kind; its deployed fields are not all within the field region (0.72 are), so they carry no output certificate; every bisection step needs a sampling run, as for the field regions; and it repairs only what the source slice excludes, so the 0.10 of the requests without season that the source slice admits over the splits are served as asked.

- *Field, deep (against the field region): wintry images, but central scenes.* Worked: 0.16 and 0.05 of the summer and season-not-mentioned images unsafe, against 0.23 and 0.14. Did not work: the scene is lost whenever it is not central (targets of the requested scene 0.68 against 1.00), and the decision, hence the detection of only about half of the unsafe requests over the splits, is that of the field region.
- *Joint→field, deep (against the source slice): safe images and a certificate that applies, at the price of central scenes and of new-scene false alarms.* Worked: 0.02 and 0.03 of the summer and season-not-mentioned images unsafe, every deployed field inside both regions, so that the field certificate applies (it is vacuous here, for the reason of Section 11.6). Did not work: the decision adds the false alarms of the field distance without adding detection (novel winter requests excluded 0.18 against 0.01; requests without season 0.87 against 0.90), and the joint median level removes the targets of the least central scenes (0.68 of the targets show the requested scene).
- *Mahalanobis→field (against the Mahalanobis region): the best repair of familiar requests, and every unfamiliar request rewritten.* Worked: 0.00 of the summer images and 0.02 of the images without season unsafe, every target of the requested scene, and a scene similarity of 0.298 against 0.191. Did not work: its decision still excludes every unfamiliar request, so every safe paraphrase and every winter scene that no approved prompt shows is altered (Table 17), and its depth comes from a whitened distance that magnifies any departure from the approved wording, not from a calibrated level.
- *Slice, projection (against the back-off of the source slice): a control that confirms the theory, and no repair.* Worked: it is computed exactly, by a convex program, and it deploys the field of the source-slice region closest to the request. Did not work: it changes the request less than the back-off does ($1-\gamma_0 = 0.35$ against $\kappa = 0.55$), towards all the anchors at once rather than towards a scene, so 0.73 of the summer images remain unsafe; and mixing the request's field with up to 30 anchor fields makes sampling about ten times slower. A projection minimises the change in the deciding distance, which is the wrong objective when that distance does not see the scene and its threshold does not bound the outputs.

Take-away: decide, choose the target and measure the depth each in the distance that sees what that step needs. The decision needs a distance in which the unsafe attribute is large against the approved variability (here the source slice); the target needs a distance that sees what the request asked for (the field distance); and the depth needs a distance that sees the outputs, measured against the target, so that it neither stops while the images are still unsafe (the median level of the source slice) nor favours the scenes that the anchors show most (the median level of the field distance). Coverage constrains only the first of these choices (Proposition 11.1(iii)); the other two are free, and here they decide whether the deployed image is safe and faithful.

11.9 The slice methods in mathematical detail

Several methods of this section decide with the source slice: the source-slice region of Sections 11.3–11.6, with static and nearest-anchor weights, and slice→field, slice, deep, slice→field, deep, slice→field, local and slice, projection of Section 11.8; the joint score uses it as one of its two components. This subsection states them in one framework, derives what the source slice measures for a guided model, and proves the properties of the repairs that Section 11.8 invokes.

The source slice of a guided field. Time runs as in the rest of this note, from the noise at $t=0$ to the data at $t=1$; Sana's own time runs the other way and its velocity has the opposite sign, which changes no distance below. Let $u(\mathbf{x}, t; c)$ be the velocity that the network predicts for the prompt c at the latent $\mathbf{x} \in \mathbb{R}^n$, $n = 8192$, and $u(\mathbf{x}, t; \varnothing)$ the one for the empty prompt. The *field* of c is the guided velocity $v_c = u(\cdot;\varnothing) + \omega\,(u(\cdot;c) - u(\cdot;\varnothing))$, $\omega = 4.5$, and the *mixture* of the fields of prompts $c_0, \dots, c_A$ with weights γ in the simplex Δ^A is the field $v_\gamma = u(\cdot;\varnothing) + \omega\,(\sum_a \gamma_a u(\cdot;c_a) - u(\cdot;\varnothing))$, the one that enforcement deploys.

Definition 11.3 (Source slice, probe embedding and estimated distance)**.** *The* source slice *of a field v is its restriction $S_v = v(\cdot, 0)$ to the noise end, an element of $L^2(\mu_0; \mathbb{R}^n)$ with $\mu_0 = \mathcal{N}(0, I_n)$; write $S_c = S_{v_c}$. On the frozen probes $\mathbf{x}_1, \dots, \mathbf{x}_P$ ($P = 8$), the* probe embedding *of v is $y(v) = P^{-1/2}\big(S_v(\mathbf{x}_1), \dots, S_v(\mathbf{x}_P)\big) \in \mathbb{R}^{Pn}$ ($Pn = 65536$), and the* estimated source-slice distance *is the Euclidean distance $\hat{d}_0(v, v') = \|y(v) - y(v')\|$. It is the probe estimate of d_0 (Definition 4.1(iii)) and the distance of the source slice in Table 11; write $y(c) = y(v_c)$ and $\hat{d}_0(c, c') = \hat{d}_0(v_c, v_{c'})$.*

The embedding y turns the source slice into a Euclidean geometry on fields, in which the following holds.

Lemma 11.4 (What the source slice measures). *(i) For every $\gamma \in \Delta^A$, $v_\gamma = \sum_a \gamma_a v_{c_a}$; hence $S_{v_\gamma} = \sum_a \gamma_a S_{c_a}$ and $y(v_\gamma) = \sum_a \gamma_a y(c_a)$. (ii) Suppose that $u(\cdot; c)$ and $u(\cdot; \varnothing)$ are the exact fields of the independent coupling for the output laws p_c (the law of $\mathbf{x}_1$ given the prompt c) and $p_\varnothing$, with means $m(c) = \mathbb{E}_{p_c}[\mathbf{x}_1]$ and $m_\varnothing$. Then $S_c(\mathbf{x}) = (1-\omega)m_\varnothing + \omega\, m(c) - \mathbf{x}$ for every $\mathbf{x}$; the one-step map $T^1_{v_c}(\mathbf{x}) = \mathbf{x} + S_c(\mathbf{x})$ is constant, equal to the guided mean $(1-\omega)m_\varnothing + \omega\, m(c)$; and $d_0(v_c, v_{c'}) = \hat{d}_0(c, c') = \omega\|m(c) - m(c')\|$ for every choice of the probes. (iii) Under (ii), two prompts whose outputs have the same mean are at source-slice distance 0, however different their outputs, and the slice of a mixture is that of the mixed means: $S_{v_\gamma}(\mathbf{x}) = (1-\omega)m_\varnothing + \omega \sum_a \gamma_a m(c_a) - \mathbf{x}$.*

Proof: (i) Since $\sum_a \gamma_a = 1$, $\sum_a \gamma_a v_{c_a} = u(\cdot;\varnothing) + \omega(\sum_a \gamma_a u(\cdot; c_a) - u(\cdot;\varnothing)) = v_\gamma$; restriction to $t = 0$ and evaluation on the probes are linear. (ii) Lemma 4.3(iii) gives $u(\mathbf{x}, 0; c) = m(c) - \mathbf{x}$ and $u(\mathbf{x}, 0; \varnothing) = m_\varnothing - \mathbf{x}$; substituting in the guidance gives S_c, and $T^1_{v_c}$ follows from Definition 3.4. The difference $S_c(\mathbf{x}) - S_{c'}(\mathbf{x}) = \omega(m(c) - m(c'))$ does not depend on $\mathbf{x}$, so its norm in $L^2(\mu_0)$ and its probe estimate both equal $\omega\|m(c) - m(c')\|$. (iii) Immediate from (ii) and (i). □

Sana's network is not the exact field, but at the noise end it nearly is. Write $\Delta_\ell = S_c(\mathbf{x}_\ell) - S_{c'}(\mathbf{x}_\ell)$ and $\bar{\Delta}$ for its mean over the probes; Pythagoras' identity in $\mathbb{R}^{Pn}$ gives $\hat{d}_0(c, c')^2 = \|\bar{\Delta}\|^2 + \frac{1}{P}\sum_\ell \|\Delta_\ell - \bar{\Delta}\|^2$, and by Lemma 11.4(ii) the second term, which depends on the probe, would vanish for the exact field. Between the anchors and the other approved prompts it is a median 0.03 of $\hat{d}_0^2$ (90% quantile 0.06), between the anchors and the summer requests 0.01, and between the anchors and the requests without season 0.02. The source slice of Sana is therefore, to within a few percent, ω times the distance between the mean latent outputs of two prompts: it compares the blurred average of everything that a prompt would generate. This is why it sees the season, which changes the mean image everywhere, and not the scene, whose layout varies from sample to sample and averages out (Section 11.4); and why, by Lemma 11.4(iii), an unsafe attribute that leaves the mean output unchanged is invisible to it, as the unsafe mode of the toy was (Section 10.4).

The source-slice regions. With anchors $c_1, \dots, c_M$ and $y_i = y(c_i)$, the *static* score of a field is $s(v) = f(y(v))$ with $f(y) = \frac{1}{M}\sum_{i=1}^M \|y - y_i\|$, and the *nearest-anchor* score is $s_1(v) = \min_i \|y(v) - y_i\|$. The threshold ε is the k-th smallest of the calibration scores, $k = \lceil (K+1)(1-\beta) \rceil$, and the region admits the prompts whose score is at most ε; the probes are frozen, so s and s_1 are fixed functions of the prompt given the anchors, and the coverage of Proposition 5.4 holds exactly for them (Remark 5.5(iv)). In the embedding, the static region is the set $\mathcal{R} = \{y \in \mathbb{R}^{Pn} : f(y) \le \varepsilon\}$, a closed convex sublevel set of the Fermat–Weber function of the anchors (Lemma 2.19); under Lemma 11.4(ii) it reads $\frac{1}{M}\sum_i \|m(c) - m(c_i)\| \le \varepsilon/\omega$, a condition on the mean output alone. The nearest-anchor region is the union of balls $\mathcal{R}_1 = \bigcup_i \bar{B}(y_i, \varepsilon)$, which is not convex.

Repairs. A *slice method* decides with the static score: it admits a request c unchanged if $s(v_c) \le \varepsilon$. Otherwise it fixes a *level* $\eta \in \{\varepsilon, \varepsilon'\}$, where ε' is the k'-th smallest calibration score, $k' = \lceil (K+1)/2 \rceil$; the *admissible pool* $\mathcal{A}_\eta = \{c_j : j \le M + K,\ s(v_{c_j}) \le \eta\}$ of anchors and calibration prompts; a *target* $c' \in \mathcal{A}_\eta$, either the *own* target $\arg\min_{\mathcal{A}_\eta} \hat{d}_0(c, \cdot)$ or the *field* target $\arg\min_{\mathcal{A}_\eta} \hat{d}_\chi(c, \cdot)$, where $\hat{d}_\chi(c, c') = (\frac{1}{P}\sum_\ell \|\chi(T_c(\mathbf{x}_\ell)) - \chi(T_{c'}(\mathbf{x}_\ell))\|^2)^{1/2}$ is the estimated map metric of the field regions; and it deploys $v_\kappa = (1-\kappa)v_c + \kappa v_{c'}$, with κ returned by the bisection of Proposition 6.4(i) for one of three *stop conditions*:

- *threshold*: $s(v_\kappa) \le \varepsilon$, with $\eta = \varepsilon$;
- *median*: $s(v_\kappa) \le \varepsilon'$, with $\eta = \varepsilon'$;
- *local*: $s(v_\kappa) \le \varepsilon$ and $\hat{d}_\chi(v_\kappa, v_{c'}) \le \tau$, with $\eta = \varepsilon$, where τ is the k'-th smallest of the calibration prompts' nearest-anchor field distances $\min_i \hat{d}_\chi(c_{M+j}, c_i)$, $j \le K$.

The pairs (target, stop) give the methods of Table 14: (own, threshold) is the source slice of Sections 11.3–11.6; (field, threshold) slice→field; (own, median) slice, deep; (field, median) slice→field, deep; and (field, local) slice→field, local. Slice, projection replaces the segment by the nearest field of the region (Proposition 11.6 below). By Proposition 11.1(iii), every slice method admits a new approved prompt unchanged with probability at least $k/(K+1)$, whatever its target and stop.

Proposition 11.5 (Back-off along a segment in the source slice). *Let $s(v_c) > \eta \ge s(v_{c'})$, $y^0 = y(c)$, $y^1 = y(c')$ and $\varphi(\kappa) = f((1-\kappa)y^0 + \kappa y^1)$ for $\kappa \in [0, 1]$.*

(i) $\varphi(\kappa) = s(v_\kappa)$ for every κ; φ is convex and continuous, and $\{\kappa : \varphi(\kappa) \le \eta\} = [\kappa^\star, 1]$ for some $\kappa^\star \in (0, 1]$. The bisection keeps $\mathrm{lo} < \kappa^\star \le \mathrm{hi}$, so after N steps it returns $\mathrm{hi} \in [\kappa^\star, \kappa^\star + 2^{-N}]$: with the threshold and median stops, the back-off is the minimal one along the segment, up to 2^{-N}.

(ii) $\kappa^\star \le \kappa_{\mathrm{lin}} = (\varphi(0) - \eta)/(\varphi(0) - \varphi(1))$.

(iii) With $\mathbf{a}_i = y^0 - y_i$, $\mathbf{b}_i = y^1 - y_i$ and θ_i the angle between them, $(1-\kappa)\varphi(0) + \kappa\varphi(1) - \varphi(\kappa) = \frac{1}{M}\sum_i h_i(\kappa)$, where $h_i(\kappa) = (1-\kappa)\|\mathbf{a}_i\| + \kappa\|\mathbf{b}_i\| - \big((1-\kappa)^2\|\mathbf{a}_i\|^2 + \kappa^2\|\mathbf{b}_i\|^2 + 2\kappa(1-\kappa)\|\mathbf{a}_i\|\|\mathbf{b}_i\|\cos\theta_i\big)^{1/2}$ is non-negative, vanishes when $\cos\theta_i = 1$, and, for $0 < \kappa < 1$ and non-zero $\mathbf{a}_i, \mathbf{b}_i$, increases strictly as $\cos\theta_i$ decreases.

(iv) If $\varphi(1) = \eta$ and φ is decreasing at 1 (its left derivative there is negative), then $\kappa^\star = 1$: the request is replaced by its target.

(v) With nearest-anchor weights, suppose $s_1(v_c) > \eta$, let $c' = c_{i^\star}$ be the anchor nearest to c in $\hat{d}_0$ and $\kappa_1 = 1 - \eta/\hat{d}_0(c, c')$. The point $(1-\kappa_1)y^0 + \kappa_1 y^1$ is the projection of y^0 onto $\bigcup_i \bar{B}(y_i, \eta)$, and $s_1(v_\kappa) \le \eta$ for every $\kappa \ge \kappa_1$.

(vi) With the local stop, the feasible set contains $\kappa = 1$ and the bisection returns a feasible κ in $\{j2^{-N}\}$, but not necessarily the smallest one: $\kappa \mapsto \hat{d}_\chi(v_\kappa, v_{c'})$ involves the sampler and is not convex.

Proof: (i) By Lemma 11.4(i), $y(v_\kappa) = (1-\kappa)y^0 + \kappa y^1$, so $\varphi(\kappa) = s(v_\kappa)$. Each $\kappa \mapsto \|(1-\kappa)\mathbf{a}_i + \kappa \mathbf{b}_i\|$ is a norm of an affine function, hence convex and continuous (Lemma 2.19), and so is their mean. The sublevel set of a convex function on an interval is an interval; it contains 1 ($\varphi(1) = s(v_{c'}) \le \eta$) and not 0 ($\varphi(0) > \eta$), and it is closed by continuity, so it is $[\kappa^\star, 1]$ with $\kappa^\star \in (0, 1]$. Initially $\mathrm{lo} = 0 < \kappa^\star \le 1 = \mathrm{hi}$; a midpoint is feasible if and only if it is at least $\kappa^\star$, so each update keeps $\mathrm{lo} < \kappa^\star \le \mathrm{hi}$ and halves $\mathrm{hi} - \mathrm{lo}$. (ii) By convexity, $\varphi(\kappa_{\mathrm{lin}}) \le (1-\kappa_{\mathrm{lin}})\varphi(0) + \kappa_{\mathrm{lin}}\varphi(1) = \eta$. (iii) $\varphi(0) = \frac{1}{M}\sum_i \|\mathbf{a}_i\|$, $\varphi(1) = \frac{1}{M}\sum_i \|\mathbf{b}_i\|$ and $\|(1-\kappa)\mathbf{a}_i + \kappa\mathbf{b}_i\|^2$ expands as written. The triangle inequality gives $h_i \ge 0$, with equality when $\mathbf{a}_i$ and $\mathbf{b}_i$ point the same way; for $0 < \kappa < 1$ the square root is strictly increasing in $\cos\theta_i$ when $\|\mathbf{a}_i\|\|\mathbf{b}_i\| > 0$. (iv) For $\kappa < 1$, convexity gives $\varphi(\kappa) \ge \varphi(1) + \varphi'(1^-)(\kappa - 1) > \varphi(1) = \eta$. (v) The distance from y^0 to $\bar{B}(y_i, \eta)$ is $(\|y^0 - y_i\| - \eta)_+$, smallest for $i = i^\star$, so the projection onto the union is the projection onto $\bar{B}(y_{i^\star}, \eta)$, namely $y_{i^\star} + \eta\,(y^0 - y_{i^\star})/\|y^0 - y_{i^\star}\| = (1-\kappa_1)y^0 + \kappa_1 y^1$. Moreover $s_1(v_\kappa) \le \|(1-\kappa)y^0 + \kappa y^1 - y^1\| = (1-\kappa)\hat{d}_0(c, c') \le \eta$ for $\kappa \ge \kappa_1$. (vi) $\hat{d}_\chi(v_1, v_{c'}) = 0 \le \tau$ and $s(v_1) \le \eta$; Proposition 6.4(i) assumes nothing on the score, and the endpoints of the bisection are dyadic. □

Part (iii) is the mechanism behind the earlier stop of slice→field (Section 11.8). Seen from the anchors, the source slice's own target points in nearly the same direction as the request (mean over the anchors of $\cos\theta_i$: median 0.70 over the altered summer and season-not-mentioned requests), whereas the winter version of the requested scene, which differs from the request by the season, the dominant direction of the slice, points elsewhere (0.41). Its gap below the chord is larger, and the threshold is reached with less of the target. Part (iv) is the case of "a city park with benches, at dusk" at the median level: its target was the calibration prompt whose score defines that level.

Proposition 11.6 (Projection onto the static source-slice region)**.** *Let $y_0 = y(c)$ with $f(y_0) > \varepsilon$, let $\mathcal{R} \neq \emptyset$, and let $\mathcal{Y}$ be the convex hull of $y_0, y_1, \ldots, y_M$.*

(i) The projection $y^\star$ of y_0 onto $\mathcal{R}$ lies in $\mathcal{Y}$: $y^\star = \sum_{a=0}^{M} \gamma_a^\star y_a$ for some $\gamma^\star \in \Delta^M$, and $f(y^\star) = \varepsilon$.

(ii) With $\Gamma_{ab} = \langle y_a, y_b\rangle$ and $\mathbf{e}_b$ the b-th vector of the canonical basis, $\gamma^\star$ minimises $(\gamma - \mathbf{e}_0)^\top \Gamma (\gamma - \mathbf{e}_0)$ over $\gamma \in \Delta^M$ subject to $\frac{1}{M}\sum_{i=1}^{M} \big((\gamma - \mathbf{e}_i)^\top \Gamma (\gamma - \mathbf{e}_i)\big)^{1/2} \le \varepsilon$, a second-order cone program. The mixture $v_{\gamma^\star}$, with $c_0 = c$, is a field whose estimated source-slice distance to v_c is the smallest among all fields of score at most ε.

(iii) If $\varepsilon > \min f$ and $y^\star$ is not an anchor, then $y^\star - y_0 = \lambda \frac{1}{M}\sum_{i=1}^{M} \frac{y_i - y^\star}{\|y_i - y^\star\|}$ for some $\lambda > 0$: the request moves along the mean of the unit vectors that point from $y^\star$ to the anchors, in which every anchor counts equally, whatever its distance.

Proof: (i) $\mathcal{R}$ is closed and convex (Lemma 2.19), so $y^\star$ exists and is unique (Theorem 2.6). Let $P_{\mathcal{Y}}$ be the projection onto the compact convex set $\mathcal{Y}$. By Theorem 2.6(i), $\|P_{\mathcal{Y}} y^\star - z\| \le \|y^\star - z\|$ for every $z \in \mathcal{Y}$; applied to the anchors, $f(P_{\mathcal{Y}} y^\star) \le f(y^\star) \le \varepsilon$, and applied to $z = y_0$, $\|P_{\mathcal{Y}} y^\star - y_0\|^2 \le \|y^\star - y_0\|^2 - \|y^\star - P_{\mathcal{Y}} y^\star\|^2$. Uniqueness of $y^\star$ forces $P_{\mathcal{Y}} y^\star = y^\star$. If $f(y^\star) < \varepsilon$, a small step from $y^\star$ towards y_0 would stay in $\mathcal{R}$ by continuity and come closer to y_0; so $f(y^\star) = \varepsilon$. (ii) For $y = \sum_a \gamma_a y_a$, $\|y - y_b\|^2 = (\gamma - \mathbf{e}_b)^\top \Gamma (\gamma - \mathbf{e}_b)$, so the program is the projection problem restricted to $\mathcal{Y}$, which contains its solution by (i); with $\Gamma = LL^\top$ its constraints are second-order cones. By Lemma 11.4(i), $y(v_{\gamma^\star}) = y^\star$, and every field of score at most ε has its embedding in $\mathcal{R}$, hence at distance at least $\|y^\star - y_0\|$ from y_0. (iii) Let $\mathbf{g} = y_0 - y^\star \neq 0$. Since $y^\star$ is not an anchor, f is differentiable there, with $\nabla f(y^\star) = \frac{1}{M}\sum_i (y^\star - y_i)/\|y^\star - y_i\|$, and $\nabla f(y^\star) \neq 0$ because $f(y^\star) = \varepsilon > \min f$ and f is convex. If $\mathbf{g}$ were not a positive multiple of $\nabla f(y^\star)$, the unit vectors $\hat{\mathbf{g}}$ and $\mathbf{u} = \nabla f(y^\star)/\|\nabla f(y^\star)\|$ would differ, and $\mathbf{d} = \hat{\mathbf{g}} - \mathbf{u}$ would satisfy $\langle \nabla f(y^\star), \mathbf{d}\rangle < 0$ and $\langle \mathbf{g}, \mathbf{d}\rangle > 0$: for small $t > 0$, $y^\star + t\mathbf{d}$ would lie in $\mathcal{R}$ and be closer to y_0, a contradiction. Hence $\mathbf{g} = \lambda \nabla f(y^\star)$ with $\lambda > 0$, which is the claim. □

Part (iii) explains what Section 11.8 measured: the projection keeps most of the request and spreads the rest over all the anchors (31 fields in the mixture for every request), because the static score rewards moving towards all of them at once. A projection is the minimal change in the source-slice distance, so it inherits that distance's blindness to the scene, and it ends on the boundary of $\mathcal{R}$, where Proposition 11.5 also ends a threshold back-off.

Exactness, coverage and cost. Table 18 summarises the slice methods. All of them decide with the source slice, so their coverage is that of a source-slice region (Proposition 5.4), whatever their repair. They differ in what a repair needs. With static weights and the threshold or median stop, the stop is evaluated on stored slices (Lemma 11.4(i)) and the back-off is exactly minimal (Proposition 11.5(i)); the local stop needs one sampling run on the probes per bisection step and returns a feasible, not a minimal, weight; and a field target, like the local stop, needs the request's own outputs on the probes, which the source slice alone never generates.

Table 18: The slice methods. Each row is one method; the source slice decides in all of them, with static weights unless stated, so every method admits a new approved prompt with probability at least $k/(K+1) = 0.902$. *Target*: own (nearest admissible prompt in $\hat{d}_0$), field (nearest in $\hat{d}_\chi$), or none (projection). *Stop*: the condition that the deployed field must meet. *Weight found*: whether the bisection returns the smallest admissible weight (exact, by Proposition 11.5(i)) or only a feasible one. *Model runs*: what a repair costs beyond scoring the request, which takes $2P$ network evaluations at the noise end; "one run" is one sampling of the probes (12 steps), and deploying a back-off costs one extra network evaluation per sampler step.

method	target	stop	weight found	model runs for the repair
source slice (static)	own	$s(v_\kappa) \le \varepsilon$	exact	none
source slice (nearest anchor)	own	$s_1(v_\kappa) \le \varepsilon$	feasible	none
slice→field	field	$s(v_\kappa) \le \varepsilon$	exact	one run (the request's outputs)
slice, deep	own	$s(v_\kappa) \le \varepsilon'$	exact	none
slice→field, deep	field	$s(v_\kappa) \le \varepsilon'$	exact	one run
slice→field, local	field	$s(v_\kappa) \le \varepsilon$, $\hat{d}_\chi(v_\kappa, v_{c'}) \le \tau$	feasible	one run, and one per bisection step
slice, projection	none	nearest point of $\mathcal{R}$	exact (cone program)	none; up to $M+1$ fields deployed

11.10 Summary

Every mechanism of Sections 4–7 runs unchanged on a pretrained text-to-image model, and every region keeps its coverage. What the regions exclude, and what their enforcement produces, depends on the space far more than on the weights.

(Q1) *Detection.* A season left unspecified is unsafe here: Sana fills it with a snow-free scene, and 0.99 of the images of requests such as "a pine forest, at midday" are judged unsafe, against 1.00 for explicit summer requests. The source slice and the Mahalanobis region exclude every summer request and most requests without season; the field regions as written exclude about half and miss the pine forest; the prompt region excludes few.

(Q2) *Weights.* With 30 anchors, adaptive weights help only the prompt region and the source slice, and raise their false alarms. With 100 anchors or more, nearest-anchor weights make the field regions exclude 0.99 of the summer requests, but also 0.92 of the safe winter requests of scenes that no anchor shows. Unlike on the toy, adaptive weights trade false alarms for detection rather than dominating static ones.

(Q3) *Space.* A region detects what its distance makes large: the scene for the prompt embedding and the fields, the wording for the Mahalanobis region, which excludes every paraphrase, and the season for the source slice, because the season changes the mean output, which is all that the field at the noise end sees (Lemma 4.3(iii)). Only the static source slice excludes every summer request and almost no safe unfamiliar one; on the toy, where the unsafe mode left the mean output unchanged, the same distance relied on what the fitted field leaked, and a directed attack defeated it.

(Q4) *Enforcement.* Back-off keeps the scene only if the region's distance sees it. The field regions back requests off to winter versions of their scene and leave 0.23 of the summer images and 0.14 of the images without season unsafe, against 1.00 and 1.00 without a region; what remains comes from requests just outside the region, which the minimal back-off moves just inside. The Mahalanobis region makes every image wintry but replaces the scene, and the source slice does neither reliably. The plug-in certificates are vacuous, because in CLIP space the approved outputs lie closer to the unsafe set than to one another.

(Q5) *Combination.* Coverage depends on the decision alone (Proposition 11.1), so a prompt-side decision can be paired with a field-side repair at no cost in guarantees. The best pairing found decides with the source slice, chooses the target in the field distance, and measures the depth on the outputs, against the target: it leaves 0.06 of the summer images and 0.02 of the images without season unsafe (source slice: 0.45 and 0.44; field region: 0.23 and 0.14), every target shows the requested scene, and it leaves the safe unfamiliar requests alone. Stopping at the median level of a single distance fails in two opposite ways: in the source slice it stops while the images are still unsafe, in the field distance it moves the least central scenes to central ones. A joint region of the two distances adds the false alarms of the field distance without adding detection; Mahalanobis→field repairs familiar requests best but rewrites every unfamiliar one; and a projection moves a request less than a back-off does, towards the centre of all the anchors.

The best detector and the best enforcer differ here, and they can be combined: the static source slice decides, the field distance picks the target and measures the depth. What this does not show is equally clear. The local-depth method was designed after the other results and evaluated on the same requests; every enforcement result rests on 8 requests of each kind on one split; no deployed field carries an informative certificate; and no attack was run, neither on the source slice, which a directed attack defeated on the toy, nor on the combined methods.

12 Status and next steps

Proved: coverage of the field of a new approved prompt (Proposition 5.4); convexity and second-order cone formulations for Mahalanobis distances with static weights (Propositions 5.6 and 6.2); enforcement as a mixture of fields, and as prompt

mixing for models affine in the prompt (Proposition 6.2(iii), Corollary 6.3); an output certificate for every field of the region, without Grönwall factor (Theorem 7.1), Mahalanobis for one-step samplers and paying straightness defects otherwise (Corollary 7.2); for the localised truncated distance, a certificate that applies Markov's inequality only to a bounded variable (Proposition 7.5); and, for methods that combine several distances, coverage of a jointly calibrated score, and coverage that depends on the decision alone, whatever distances choose the target and the depth of the repair (Proposition 11.1); and, for the methods that decide with the source slice, that the slice of an exact guided field measures the distance between mean outputs, that a back-off along a segment is exactly minimal with static weights, and that the projection onto the source-slice region is a mixture of the request and the anchors computable from their slices (Lemma 11.4, Propositions 11.5 and 11.6). Tested: on the POC toy, the off-policy/on-policy gap and the path-divergence certificates (Section 8); on a prompt-conditioned toy, coverage, the role of time, the identity of prompt mixing, the failure of the region as written on a diverse family, its repair by localisation, the protection that a bounded distance gives against a sampler that diverges on rare noise, a deployed certificate verified on fresh noise, and the robustness of field-space regions to two white-box attacks on the prompt (Section 9); and, with tuning separated from evaluation, that adaptive weights beat static ones in every space and need denser anchors for small unsafe masses, that Euclidean prompt regions match field-space ones on honest prompts only when the encoder exposes the unsafe content, that a whitened prompt region matches them on this toy only because its approved embeddings are degenerate, and that the pull-back metric of a one-step reflowed model turns the prompt region into the field region, while that of a curved model does not protect it (Section 10); and, on a pretrained text-to-image model with a winter-only safe set, that coverage holds for every region, that a region detects what its distance makes large (the season for the source slice, the scene for the prompt and field distances, the wording for a whitened prompt distance), that nearest-anchor weights need anchors covering the approved scenes and then refuse unseen ones, that back-off keeps the requested scene only when the region's distance sees it and leaves barely excluded requests nearly unchanged, that the plug-in certificates are vacuous in a general-purpose image-feature space, and that deciding with the source slice, choosing the target in the field distance and measuring the depth on the outputs against that target makes the deployed images safe and faithful to the requested scene, whereas a joint decision, a median stop level or a projection does not (Section 11). The central tension remains between convexity, which needs a fixed reference law and static weights, and safety, which needs each field to be evaluated where it sends the noise, against nearby safe fields: one-step samplers make the map metric a Mahalanobis distance, but the localised regions that detect are unions of balls, enforced by back-off.

Limits of the evidence: on the toy, the outputs are two-dimensional, the approved family has two parameters, its embeddings are degenerate and the encoder exposes the unsafe mass, the 1-rectified sampler diverges on rare noise, exclusion rates carry sampling errors of up to 0.1, and every attack is run on one split. On the text-to-image model, the unsafe attribute (the season) changes the global colour of the images, which favours the source slice; the approved prompts share one template; each kind of request has 12 to 24 prompts and enforcement uses 8 of each kind, on one split; safety is graded by one zero-shot classifier; the combined method that works best was designed after the others, on the same requests; and no attack was run.

Next: a toy whose approved prompts span their embedding space and whose encoder hides the unsafe content, to separate prompt-space from field-space regions; samplers that cannot diverge; the choice of anchors when safe prompts are many and high-dimensional, since the budget is set by the spread of the nearest safe fields; then a forbidden class with class-conditional rectified flows on MNIST and CIFAR-10. On the text-to-image model: an unsafe attribute that leaves the global colour unchanged (a small object), to test the source slice where Lemma 4.3(iii) predicts it fails; the local-depth repair of Section 11.8 on new requests and splits, with anchors that cover every scene; conformal risk control of what the repair costs approved prompts (Remark 11.2); a feature space in which the approved outputs lie farther from the unsafe set than from one another, so that the certificates can be informative; and white-box attacks on the prompt, including directed ones.

A Stochastic samplers and path divergences

With TV, KL and kl as in Definition 2.17, we restate the POC results on stochastic samplers in the notation of this note; the facts on divergences used below are listed after that definition.

Definition A.1 (SDE chain)**.** *For $\eta > 0$ and the grid of Definition 3.4, the SDE chain of $\boldsymbol{\theta}$ is $X_0 \sim \mu_0$ and $X_{k+1} = X_k + h_k[(1+\eta t_k)\Phi(X_k,t_k)\boldsymbol{\theta} - \eta X_k] + \sigma_k \boldsymbol{\xi}_k$, $\sigma_k^2 = 2\eta(1-t_k)h_k$, $k = 0,\dots,J-1$, with $\boldsymbol{\xi}_k$ i.i.d. $\mathcal{N}(0, I_n)$ independent of X_0; $P^{\boldsymbol{\theta}}$ is the law of $(X_0,\dots,X_J)$. With $\Phi\boldsymbol{\theta}$ replaced by $v_p^\star$, its continuous-time limit has, formally, the marginals of the interpolation path (POC Lemma 3, Albergo et al. 2025); this is not used below.*

Lemma A.2 (Path divergence)**.** $\mathrm{KL}(P^{\boldsymbol{\theta}} \| P^{\boldsymbol{\theta}'}) = \frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\theta}')^\top G_{\mathrm{KL}}(\boldsymbol{\theta})(\boldsymbol{\theta} - \boldsymbol{\theta}')$, *where* $G_{\mathrm{KL}}(\boldsymbol{\theta}) = \sum_{k=0}^{J-1} \omega_k \, \mathbb{E}_{P^{\boldsymbol{\theta}}}[\Phi(X_k,t_k)^\top \Phi(X_k,t_k)]$ *and* $\omega_k = \frac{h_k(1+\eta t_k)^2}{2\eta(1-t_k)}$.

Proof: $\sigma_k > 0$ because $t_k < 1$ for $k \le J-1$, and each X_k has a finite second moment by linear growth. Both chains are Markov with the same initial law and Gaussian kernels of covariance $\sigma_k^2 I_n$ whose means differ by $h_k(1+\eta t_k)\Phi(X_k,t_k)(\boldsymbol{\theta} - \boldsymbol{\theta}')$. The chain rule for relative entropy (Cover and Thomas 2006, Thm. 2.5.3) sums, in expectation under $P^{\boldsymbol{\theta}}$, the divergences between these Gaussian kernels, $\|h_k(1+\eta t_k)\Phi(X_k,t_k)(\boldsymbol{\theta}-\boldsymbol{\theta}')\|^2/(2\sigma_k^2) = \frac{\omega_k}{2}\|\Phi(X_k,t_k)(\boldsymbol{\theta}-\boldsymbol{\theta}')\|^2$ (the Gaussian formula after Definition 2.17). □

The forward cost $\mathrm{KL}(P^{\boldsymbol{\theta}_i} \| P^{\boldsymbol{\theta}})$ uses the Gram matrix of the safe chain and is convex in $\boldsymbol{\theta}$; the reverse cost $\mathrm{KL}(P^{\boldsymbol{\theta}} \| P^{\boldsymbol{\theta}_i})$ uses that of the candidate and is not.

Proposition A.3 (Certificates and the direction of the divergence)**.** *Let* $\bar{P}_w = \sum_i w_i P^{\boldsymbol{\theta}_i}$. *(i) If* $\frac{1}{M+1}\sum_i w_i \mathrm{KL}(P^{\boldsymbol{\theta}_i} \| P^{\boldsymbol{\theta}}) \le \varepsilon$, *then* $\mathrm{TV}(P^{\boldsymbol{\theta}}, \bar{P}_w) \le \min\{((M+1)\varepsilon/2)^{1/2}, (1-e^{-(M+1)\varepsilon})^{1/2}\}$. *(ii) If* $\frac{1}{M+1}\sum_i w_i \mathrm{KL}(P^{\boldsymbol{\theta}} \| P^{\boldsymbol{\theta}_i}) \le \varepsilon$, *then for every event* A *with* $q = \bar{P}_w(A) \in (0,1)$,

$$P^{\boldsymbol{\theta}}(A) \le \sup\{x \in [q,1] : \mathrm{kl}(x\|q) \le (M+1)\varepsilon\} \le \frac{(M+1)\varepsilon + \log 2}{\log(1/q)}.$$

(iii) If $P' = (1-m)P + mQ$ *with* $m \in [0,1)$, *then* $\mathrm{KL}(P\|P') \le \log\frac{1}{1-m}$. *If moreover* $Q(A) = 1$ *for an event* A *with* $q = P(A) \in (0,1)$ *and* $m \ge q$, *then* $\mathrm{KL}(P'\|P) \ge m\log\frac{m}{q} - m$.

Proof: (i) By joint convexity of the relative entropy (Cover and Thomas 2006, Thm. 2.7.2), $\mathrm{KL}(\bar{P}_w\|P^{\boldsymbol{\theta}}) \le \sum_i w_i \mathrm{KL}(P^{\boldsymbol{\theta}_i}\|P^{\boldsymbol{\theta}}) \le (M+1)\varepsilon$; apply Pinsker's inequality (Tsybakov 2009, Lemma 2.5) and the Bretagnolle–Huber inequality (Bretagnolle and Huber 1979, Canonne 2022). (ii) Likewise, by convexity in the second argument, $\mathrm{KL}(P^{\boldsymbol{\theta}}\|\bar{P}_w) \le (M+1)\varepsilon$. By data processing through $\mathbb{1}_A$ (after Definition 2.17), $\mathrm{kl}(P^{\boldsymbol{\theta}}(A)\|q) \le (M+1)\varepsilon$: either $P^{\boldsymbol{\theta}}(A) \le q$, or $P^{\boldsymbol{\theta}}(A)$ belongs to the set whose supremum is taken. For the second bound, every x in that set satisfies $x\log\frac{1}{q} - \log 2 \le \mathrm{kl}(x\|q) \le (M+1)\varepsilon$, because $\mathrm{kl}(x\|q) = x\log\frac{1}{q} + (1-x)\log\frac{1}{1-q} - \mathrm{h}(x)$ with the binary entropy $\mathrm{h}(x) \le \log 2$. (iii) $P \le \frac{1}{1-m}P'$, so $\frac{dP}{dP'} \le \frac{1}{1-m}$ and $\mathrm{KL}(P\|P') \le \log\frac{1}{1-m}$. Let $x = P'(A) = (1-m)q + m$, so $x \ge m \ge q$. Data processing and $\log\frac{1}{1-q} \ge 0$ give $\mathrm{KL}(P'\|P) \ge x\log\frac{x}{q} + (1-x)\log(1-x) \ge x\log\frac{x}{q} - x$, since $\log(1-x) \ge -\frac{x}{1-x}$ for $x < 1$; and $x \mapsto x\log\frac{x}{q} - x$ has derivative $\log\frac{x}{q} \ge 0$ on $[q,1]$. □

Item (iii) is the formal reason for the gap of Section 8: a forward cost charges at most $\log\frac{1}{1-m} \approx m$ for an unsafe mode of mass m, whatever its rarity q under the safe samplers, while the reverse cost grows like $m\log(m/q)$.

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